

Fermi Notes on Physics

Volume Two

Introductory Undergraduate
Lectures on Thermal and
Statistical Physics

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Kinetic Theory and Transport in Gases

1.1 Thermal Physics at School

This section collects the thermal physics you learned at school and fixes the notation we shall use for it. You have likely seen these results at school, so we will not spend long on them, and nothing here is derived again. In the three sections that follow we take the picture of a gas as a crowd of moving atoms or molecules and work out from it the pressure, the distribution of molecular speeds, the rate at which molecules escape through a small hole, the distance a molecule travels between collisions, and the viscosity, thermal conductivity, and diffusion coefficient of a gas.

The Particle Model

Matter is made of atoms and molecules, and the three ordinary states differ in how those constituents are arranged and how freely they can move. In a solid the molecules are held close together in fixed neighbouring positions and vibrate about them. In a liquid they are still close together, so the density is not far from that of the solid, but they can move past one another, which is why a liquid flows and takes the shape of its container. In a gas the molecules

are far apart compared with their size, they move freely between occasional collisions, and the gas expands to fill whatever volume it is given. The density of a body of mass M occupying a volume V is $\rho = M/V$, and the density of a gas at ordinary pressures is typically smaller than that of the corresponding liquid by a factor of order a thousand.

The amount of a substance is measured in moles. One mole contains the Avogadro number N_A of molecules, so a sample of N molecules is N/N_A moles, and a sample of ν moles has a mass νM_m , where M_m is the molar mass.

Temperature and Thermal Equilibrium

When two bodies are placed in contact and left alone, their measurable properties, such as their temperatures and their volumes, eventually stop changing, as happens to a spoon left standing in a hot drink, and the two bodies are then said to be in thermal equilibrium with one another. Temperature is the quantity that two bodies in thermal equilibrium have in common, and energy passes from the hotter body to the colder one until the two temperatures are equal. Energy transferred because of a temperature difference is called heat.

Temperature is measured on the Celsius scale, whose degree was originally fixed by the freezing and boiling points of water, and on the absolute or kelvin scale. The two scales have the same size of degree and differ only in where the zero is placed, so that a temperature of θ degrees Celsius is $T = \theta + 273.15$ kelvin. Every formula in this course will use the absolute temperature.

The Gas Laws

For a fixed amount of gas at ordinary densities, experiment gives the following laws:

1. Boyle's law, that pV is constant at fixed temperature;
2. Charles's law, that V is proportional to T at fixed pressure;
3. the pressure law, that p is proportional to T at fixed volume.

Taken together with the observation that the volume at fixed p and T is proportional to the amount of gas, these three laws give the ideal gas equation,

$$pV = \nu RT = Nk_B T, \quad (1.1)$$

where ν is the amount of gas in moles, R is the gas constant, N is the number of molecules, and $k_B = R/N_A$ is Boltzmann's constant. A gas that obeys this equation is called ideal, and real gases obey it well when they are dilute and far from condensing.

Charles's law extrapolates to zero volume at $T = 0$, and the pressure law extrapolates to zero pressure at the same point. No gas actually reaches that state, since every gas liquefies first, but the two extrapolations agree, and the temperature at which they meet is the zero of the absolute scale. It is called absolute zero.

Solids and liquids expand a little when heated. Over a modest range of temperature the fractional change in the length of a solid rod is proportional to the change in temperature, $\Delta\ell = \alpha\ell\Delta T$, where α is the coefficient of linear expansion.

Heat Transfer

Heat travels from a hot place to a cold one in three ways. In conduction the energy is passed along through a material without the material itself moving, and the rate at which it is passed along is proportional to the area and to the temperature gradient. In convection the heated fluid itself moves, carrying its energy with it, usually because warm fluid is less dense than cold fluid and rises through it. In radiation the energy travels as electromagnetic waves, which need no material to cross, and the rate at which a body radiates rises very sharply with its temperature.

Kinetic Theory at School

The school model of a gas rests on a short list of assumptions. The gas contains a very large number of identical molecules in rapid random motion; the molecules occupy a negligible fraction of the volume of the container; they exert no forces on one another except

during collisions, which take a negligible time; and all collisions, with each other and with the walls, are elastic. The molecules bouncing off the walls deliver momentum to them, and the resulting force per unit area is the pressure. The model gives

$$pV = \frac{1}{3}Nmc^2, \quad (1.2)$$

where m is the mass of one molecule and $\overline{c^2}$ is the mean square speed. Setting this expression for pV equal to $Nk_B T$ gives

$$\frac{1}{2}m\overline{c^2} = \frac{3}{2}k_B T, \quad (1.3)$$

so the mean translational kinetic energy of a molecule depends on the temperature and on nothing else. The root mean square speed $c_{\text{rms}} = \sqrt{\overline{c^2}} = \sqrt{3k_B T/m}$ therefore falls as the mass of the molecule rises, and at room temperature it comes out at several hundred metres a second for ordinary gases.

The direct evidence that molecules move in this way is Brownian motion. A small grain suspended in water, watched through a microscope, jitters about on an irregular path, as Robert Brown found when he looked at the tiny particles released from pollen grains. The water molecules strike the grain from all sides, and the number arriving from one side in a short interval does not exactly match the number arriving from the other. The jittering motion is more vigorous for smaller particles and at higher temperatures, and the molecular picture requires both of those dependences, since a smaller grain is struck by fewer molecules in a given time and so suffers a larger fractional imbalance, and the molecules of a warmer liquid move faster.

1.2 Pressure and the Distribution of Molecular Speeds

The school derivation of the pressure of a gas assumes that every molecule has the same speed, and then replaces that speed by a root mean square at the end. It is easy to see why that assumption, although convenient, is unrealistic; the molecules of a gas collide

constantly, and a collision leaves the two molecules with speeds different from the ones they had before. In this section we do the calculation allowing the molecules to have whatever velocities they like, and we then work out what those velocities are.

A word about notation before we continue. In Section 1.1 the amount of gas in moles was written ν . From here to the end of this part, n denotes the number of molecules per unit volume, so that the ideal gas equation reads $p = nk_B T$.

Pressure from a Distribution of Velocities

Let the gas be held in a container, and let us look at a flat piece of the wall of area A , perpendicular to the x -axis, as indicated in Figure 1. We write $f(v_x) dv_x$ for the fraction of the molecules whose x -component of velocity lies between v_x and $v_x + dv_x$, so that the number of such molecules per unit volume is $n f(v_x) dv_x$. The function g is what we are eventually after, and the pressure can be written down before we know it.

Let us take the molecules of one such group, with $v_x > 0$. In a short time dt such a molecule reaches the wall if it starts within a perpendicular distance $v_x dt$ of it, since that is how far it travels in the x -direction. The molecules of the group that arrive at the patch in the time dt are therefore those in a slab of volume $A v_x dt$ next to the wall, drawn in Figure 1, and there are $n g(v_x) dv_x A v_x dt$ of them.

Each of these molecules bounces elastically off the wall, which reverses v_x and leaves v_y and v_z alone, so each delivers momentum $2mv_x$ to the wall. Adding up the contributions of all the groups with $v_x > 0$, the momentum delivered to the patch in time dt is

$$d\Pi_{\text{wall}} = \int_0^\infty 2mv_x \cdot n g(v_x) A v_x dt dv_x. \quad (1.4)$$

The force on the patch is $d\Pi_{\text{wall}}/dt$, and the pressure is that force divided by A :

$$p = 2mn \int_0^\infty v_x^2 g(v_x) dv_x. \quad (1.5)$$

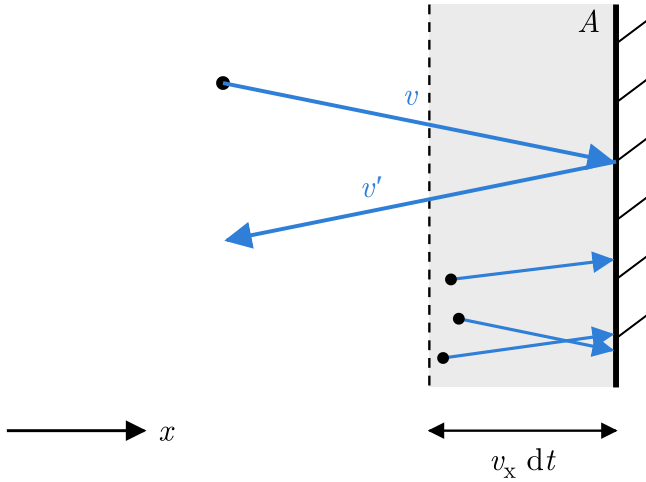


Figure 1. Molecules with a given x -component of velocity v_x reach the shaded patch of wall, of area A , within a time dt if they lie in the slab of thickness $v_x dt$ next to it. A molecule that bounces elastically off the wall has its x -component of velocity reversed and its other two components unchanged, so it delivers momentum $2mv_x$.

There are no distinguished directions in this system, and so g is an even function of v_x , and the integral of $v_x^2 g$ over the positive half of the range is half of the integral over the whole range, which is by definition the mean of v_x^2 . Hence

$$p = nm\overline{v_x^2}. \quad (1.6)$$

The same absence of a preferred direction means that the three components contribute equally to $\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}$, so that $\overline{v_x^2} = \frac{1}{3}\overline{v^2}$ and

$$p = \frac{1}{3}nm\overline{v^2}. \quad (1.7)$$

This is the school result of Section 1.1, now obtained from an average over a distribution of velocities. Comparing it with $p = nk_B T$ gives $\frac{1}{2}m\overline{v^2} = \frac{3}{2}k_B T$ as before, and $\frac{1}{2}m\overline{v_x^2} = \frac{1}{2}k_B T$ for each component separately.

The Isothermal Atmosphere

To find the distribution g itself we look at a different problem, one whose answer can be obtained by mechanics alone. Let us consider a column of gas in a uniform gravitational field, all of it at one temperature T , and let us ask how its density varies with height.

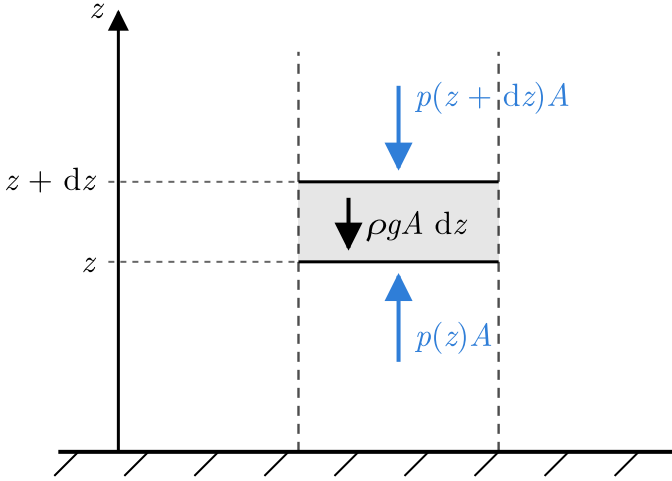


Figure 2. A slab of the atmosphere between heights z and $z + dz$, of cross-sectional area A . The gas below pushes it up with a force $p(z)A$, the gas above pushes it down with a force $p(z + dz)A$, and its weight is $\rho g A dz$.

Let us take a slab of the column between heights z and $z + dz$, of cross-sectional area A , as in Figure 2. The slab is in equilibrium, so the forces on it balance. The gas below pushes it up with a force $p(z)A$, the gas above pushes it down with a force $p(z + dz)A$, and its weight $\rho g A dz$ acts downwards. Balancing the three forces and cancelling the area,

$$p(z + dz) - p(z) = -\rho g dz, \quad \text{so} \quad \frac{dp}{dz} = -\rho g. \quad (1.8)$$

The density of the gas is $\rho = nm$ and its pressure is $p = nk_B T$, so at constant temperature $dp/dz = k_B T dn/dz$, and the balance of forces becomes a separable first-order equation for n ,

$$\frac{1}{n} \frac{dn}{dz} = -\frac{mg}{k_B T}. \quad (1.9)$$

Integrating both sides from the ground upwards gives $\ln [n(z)/n(0)] = -mgz/k_B T$, that is,

$$n(z) = n(0) e^{-mgz/k_B T}. \quad (1.10)$$

From the form of the exponential we can conclude that the density falls off faster for heavier molecules and more slowly at higher temperatures.

The Boltzmann Factor

The quantity mgz in the exponent is the potential energy of a molecule at height z . Since the molecules are alike, the fraction of them found at that height is proportional to $n(z)$, and therefore the chance of finding a particular molecule at a height where its energy is $E = mgz$ carries the factor $e^{-E/k_B T}$.

We now take this result, obtained for the one case of a molecule in a uniform gravitational field, as a general principle, which we assume for the rest of this part: for a system in equilibrium at temperature T , the probability of finding it in a particular state of energy E is proportional to

$$e^{-E/k_B T} = e^{-\beta E}, \quad \beta = \frac{1}{k_B T}. \quad (1.11)$$

This quantity is called the Boltzmann factor, and β is a convenient shorthand for $1/k_B T$ that we shall use throughout. Although we have extrapolated this form from a single example, it agrees with experiment wherever it has been tested, and we shall use it freely in what follows.

The Distribution of Velocities

A molecule of a gas at rest in a container has energy $\frac{1}{2}mv^2 = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2)$, and by the general principle just stated the prob-

ability of finding it with a velocity in a small range about (v_x, v_y, v_z) is proportional to

$$e^{-\beta m v^2/2} = e^{-\beta m v_x^2/2} e^{-\beta m v_y^2/2} e^{-\beta m v_z^2/2}. \quad (1.12)$$

In writing the right-hand side we have used the fact that the exponential of a sum is the product of the exponentials. The right-hand side is a product of three factors, one for each component, so the three components are independent of one another, and since the three factors are identical in form each component is distributed in the same way. The distribution of a single component is therefore

$$g(v_x) = C e^{-\beta m v_x^2/2}, \quad (1.13)$$

with C a constant to be determined by the requirement that the total probability is one. The integral we need for that is

$$I(a) = \int_{-\infty}^{\infty} e^{-a v^2} dv, \quad (1.14)$$

and it cannot be done by finding an antiderivative, since $e^{-a v^2}$ has none that can be written in terms of the elementary functions. The integral can nevertheless be evaluated, and the first step is to write the square of $I(a)$ as an integral over a plane whose two coordinates are x and y ,

$$I(a)^2 = \int_{-\infty}^{\infty} e^{-a x^2} dx \int_{-\infty}^{\infty} e^{-a y^2} dy = \int e^{-a(x^2+y^2)} dA, \quad (1.15)$$

where $dS = dx dy$ is an element of area of the plane, and x and y are here simply the two coordinates of that plane. The integrand depends only on the distance $r = \sqrt{x^2 + y^2}$ from the origin, so we slice the plane into thin rings of radius r and width dr , each of area $2\pi r dr$, and add up what the rings contribute:

$$I(a)^2 = \int_0^{\infty} e^{-a r^2} 2\pi r dr. \quad (1.16)$$

The substitution $u = r^2$, for which $du = 2r dr$, turns this into an elementary integral,

$$I(a)^2 = \pi \int_0^\infty e^{-au} du = \frac{\pi}{a}, \quad (1.17)$$

so that $I(a) = \sqrt{\pi/a}$. Taking $a = \beta m/2$ and using this result to fix C ,

$$g(v_x) = \sqrt{\frac{m}{2\pi k_B T}} e^{-mv_x^2/2k_B T}. \quad (1.18)$$

We can check this distribution against the pressure. The mean of v_x^2 is $\int v_x^2 g dv_x$, and integrating by parts with $u = v_x$ and $dw = v_x e^{-av_x^2} dv_x$ leaves $1/2a$ times the normalising integral, so that $\overline{v_x^2} = 1/2a = k_B T/m$. Putting this into $p = nm\overline{v_x^2}$ gives $p = nk_B T$, the ideal gas equation we started from.

The Distribution of Speeds

Most of the time we care about the speed of a molecule and not about the direction in which it is going. To get the distribution of speeds we think of the velocity of a molecule as a point in a three-dimensional space whose axes are v_x , v_y , and v_z . The probability density at a point of this velocity space is the product of the three one-component distributions, $(m/2\pi k_B T)^{3/2} e^{-\beta mv^2/2}$, which depends only on the distance v from the origin.

All the velocities with speed between v and $v + dv$ lie in a thin spherical shell of radius v and thickness dv , drawn in Figure 3, whose volume is $4\pi v^2 dv$. The density is the same everywhere on the shell, so the probability $F(v) dv$ that a molecule has a speed in that range is the density times the volume of the shell:

$$F(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}. \quad (1.19)$$

This distribution is called the Maxwell-Boltzmann distribution of molecular speeds. The factor v^2 comes from the volume of the shell

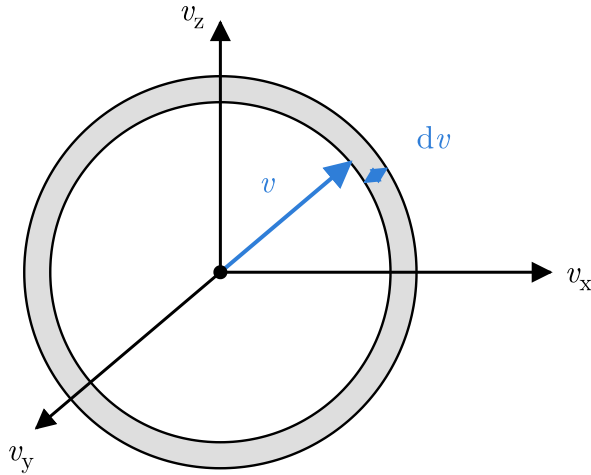


Figure 3. Velocity space, with a molecule's velocity marked as a point at distance v from the origin. All the velocities with speed between v and $v + dv$ lie in the shaded spherical shell, whose volume is $4\pi v^2 dv$.

and takes the distribution to zero at $v = 0$, while the exponential takes it to zero at large v , so the curve has a peak somewhere between. It is drawn for two temperatures in Figure 4.

Mean, Root Mean Square, and Most Probable Speeds

Of the averages that can be formed from the distribution of speeds, three are commonly quoted, and they differ from one another by numerical factors of order one. The mean speed is

$$\bar{v} = \int_0^\infty v F(v) dv = 4\pi \left(\frac{a}{\pi}\right)^{3/2} \int_0^\infty v^3 e^{-av^2} dv, \quad (1.20)$$

where we have again written $a = m/2k_B T$. We do this integral by parts, taking $u = v^2$ and $dw = ve^{-av^2} dv$, so that $du = 2v dv$ and $w = -e^{-av^2}/2a$:

$$\int_0^\infty v^3 e^{-av^2} dv = \left[-\frac{v^2 e^{-av^2}}{2a} \right]_0^\infty + \frac{1}{a} \int_0^\infty v e^{-av^2} dv. \quad (1.21)$$

The boundary term vanishes at both limits, since v^2 is zero at the lower limit and the exponential falls faster than any power at the upper one. The integral that remains is elementary, because ve^{-av^2} is the derivative of $-e^{-av^2}/2a$, and it equals $1/2a$. Hence

$$\int_0^\infty v^3 e^{-av^2} dv = \frac{1}{a} \cdot \frac{1}{2a} = \frac{1}{2a^2}. \quad (1.22)$$

Putting this into the expression for $\bar{v}$ and restoring $a = m/2k_B T$ gives

$$\bar{v} = \frac{2}{\sqrt{\pi a}} = \sqrt{\frac{8k_B T}{\pi m}}. \quad (1.23)$$

The mean square speed we already know, since $\overline{v^2} = 3\overline{v_x^2} = 3k_B T/m$, so the root mean square speed is $v_{\text{rms}} = \sqrt{3k_B T/m}$. The most probable speed is the position of the peak of F , and since the constants in front do not depend on v they do not move the peak, so we set the derivative of $v^2 e^{-av^2}$ to zero:

$$2v e^{-av^2} - 2av^3 e^{-av^2} = 0, \quad (1.24)$$

which gives $av^2 = 1$, that is, $v_p = \sqrt{2k_B T/m}$. The three speeds are in the ratio

$$v_p : \bar{v} : v_{\text{rms}} = \sqrt{2} : \sqrt{8/\pi} : \sqrt{3}, \quad (1.25)$$

so they rise in that order, and no two of them differ by more than about a fifth. All three are proportional to $\sqrt{T/m}$, so as the temperature rises the distribution stretches along the v -axis and flattens, in such a way that the area under it stays equal to one.

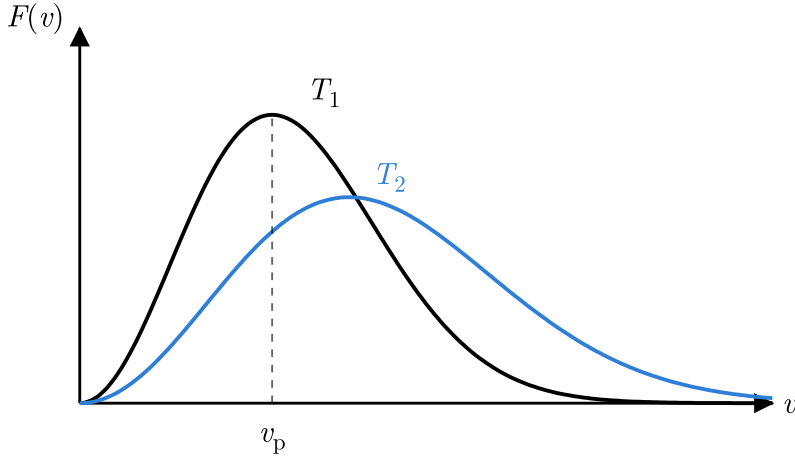


Figure 4. The Maxwell-Boltzmann distribution of molecular speeds at two temperatures, with $T_2 > T_1$. The dashed line marks the most probable speed at the lower temperature. The two curves enclose the same area, so the curve at the higher temperature is broader and lower.

Equipartition and Heat Capacities

The calculation of $\overline{v_x^2}$ gave $\frac{1}{2}m\overline{v_x^2} = \frac{1}{2}k_B T$, and the same holds for the y - and z -components, so each of the three terms in the kinetic energy of a molecule contributes $\frac{1}{2}k_B T$ to its mean energy. We take this equal contribution from each term in the energy as a second general principle, again assumed here without a general proof: for a system in equilibrium at temperature T , every independent term in its energy that is quadratic in a coordinate or in a velocity component, and whose variable ranges continuously over all values, contributes $\frac{1}{2}k_B T$ to the mean energy. Such a term is called a degree of freedom, and the principle itself is called the equipartition of energy.

Let us consider what equipartition gives for the heat capacity of a gas. The heat capacity at constant volume, C_V , is the heat needed to raise the temperature of the gas by one degree while the volume is held fixed, and no work is done on a gas at fixed volume, so the heat supplied all goes into the internal energy U and $C_V = dU/dT$ at fixed volume.

A monatomic gas, such as helium or argon, has molecules whose energy is the kinetic energy of the centre of mass and nothing else, which is three quadratic terms. Its internal energy is $U = \frac{3}{2}Nk_B T$ and its heat capacity is

$$C_V = \frac{3}{2}Nk_B, \quad (1.26)$$

which for one mole is $\frac{3}{2}R$. A diatomic molecule can also rotate, and the energy of rotation about each of the three perpendicular axes through its centre of mass, which we learned to calculate in the notes on mechanics and dynamical systems, is quadratic in the corresponding angular velocity, so equipartition as stated adds three rotational terms to the three translational ones. What is observed, however, is that rotation about the line joining the atoms does not contribute, and that only the two axes perpendicular to that line count. Adding those two rotational terms to the three translational ones gives $U = \frac{5}{2}Nk_B T$ and $C_V = \frac{5}{2}Nk_B$, or $\frac{5}{2}R$ per mole. The two atoms can also vibrate along the line joining them, which adds a kinetic and a potential term and brings the total to $\frac{7}{2}R$ per mole.

Measurement gives a rather more complicated story, sketched in Figure 5. Over a wide range of ordinary temperatures a diatomic gas does have a heat capacity close to $\frac{5}{2}R$ per mole, and at high temperatures the measured value climbs towards $\frac{7}{2}R$. At low temperatures, however, it falls below $\frac{5}{2}R$ and approaches $\frac{3}{2}R$, as though the two rotational degrees of freedom had stopped contributing altogether. Equipartition does not explain the fall at low temperatures, since a quadratic term either appears in the energy of a molecule or it does not, and the principle attaches no temperature to it. A classical model of the kind we have been using gives no account of this behaviour, nor of the rotation about the line joining the atoms that was left out above.

1.3 Effusion, Collisions, and the Mean Free Path

The molecules of a gas keep striking the walls of the container, and if a small hole is made in a wall they stream out through it. Between one wall and the next they keep running into one another. We work out first the rate at which molecules strike a unit area of wall, then

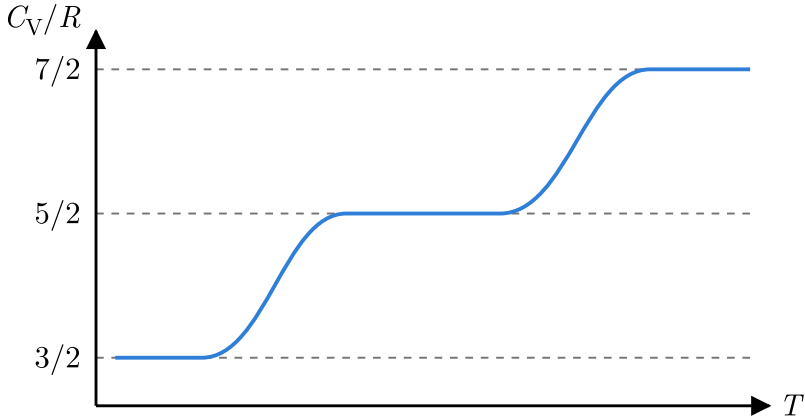


Figure 5. The measured heat capacity at constant volume of a diatomic gas, in units of R per mole, against temperature on a logarithmic scale. The value rises from $\frac{3}{2}$ at low temperatures through a wide plateau at $\frac{5}{2}$ and towards $\frac{7}{2}$ at high temperatures.

what happens when they are allowed through, and finally how far a molecule travels between collisions.

The Flux at a Surface

Let us define the flux Φ as the number of molecules striking unit area of the wall in unit time. The counting was done in Section 1.2; the molecules with x -velocity in the range dv_x about v_x that reach a patch of wall of area A in a time dt are those in a slab of volume $Av_x dt$, and there are $n g(v_x) dv_x Av_x dt$ of them. Dividing by A and by dt and adding up the groups with $v_x > 0$,

$$\Phi = n \int_0^{\infty} v_x g(v_x) dv_x. \quad (1.27)$$

With the distribution of a single component from Section 1.2, and writing $a = m/2k_B T$ as before, the integral is elementary, $\int_0^{\infty} v_x e^{-av_x^2} dv_x = 1/2a$, so that

$$\Phi = n \sqrt{\frac{a}{\pi}} \cdot \frac{1}{2a} = \frac{n}{2\sqrt{\pi a}} = n \sqrt{\frac{k_B T}{2\pi m}}. \quad (1.28)$$

Comparing this with the mean speed $\bar{v} = \sqrt{8k_B T/\pi m}$ obtained in Section 1.2, the square roots match up to a factor of four, and the flux can be written in the form in which it is usually quoted,

$$\Phi = \frac{1}{4} n \bar{v}. \quad (1.29)$$

The factor $n\bar{v}$ is what one would guess by multiplying a density by a speed, and the factor of a quarter comes from the fact that only half the molecules are moving towards the wall, and that those which are approach it obliquely rather than head on.

Effusion

Suppose now that a small hole of area A is opened in the wall, with a vacuum on the far side. If the hole is small compared with the mean free path, which is the average distance a molecule travels between collisions and which we work out at the end of this section, the molecules leave one at a time, without colliding in the hole and without setting up an organised flow towards it. The gas inside is then undisturbed, and the molecules that leave are those that would have struck that piece of wall had it still been there. Escape of this kind is called effusion, and if N is the number of molecules left in the vessel, they leave at the rate $\frac{1}{4} n \bar{v} A$, so that

$$\frac{dN}{dt} = -\frac{1}{4} n \bar{v} A. \quad (1.30)$$

Since $\bar{v} = \sqrt{8k_B T/\pi m}$, two gases at the same temperature and the same number density effuse at rates in inverse proportion to the square roots of their molecular masses. A light gas leaks out of a small hole faster than a heavy one, a result known as Graham's law.

The molecules in the emerging beam are not a fair sample of the gas inside. In the flux integral above, each group of molecules is weighted by its velocity component v_x towards the hole, because

faster molecules sweep out a longer slab in the time dt and so reach the hole from further away, and for the molecules of a given speed v that weight, averaged over the directions in which they travel, is proportional to v . The distribution of speeds in the beam therefore carries one extra factor of v compared with the distribution inside,

$$F_{\text{beam}}(v) \propto v^3 e^{-mv^2/2k_B T}, \quad (1.31)$$

and the beam is richer in fast molecules than the gas it came from. The two distributions are drawn in Figure 6.

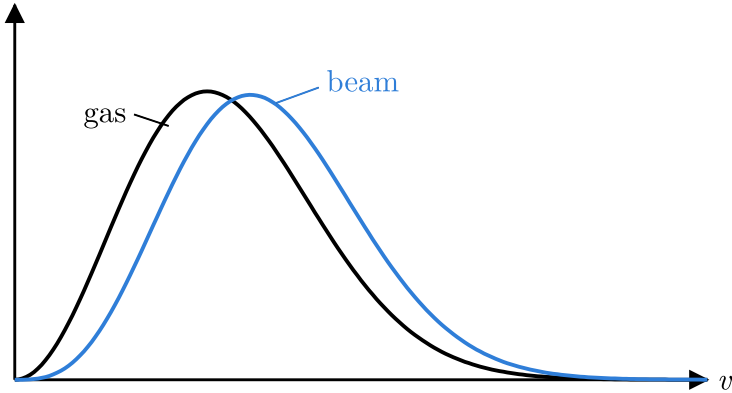


Figure 6. The distribution of speeds in the gas, proportional to $v^2 e^{-mv^2/2k_B T}$, and in a beam effusing from a small hole, proportional to $v^3 e^{-mv^2/2k_B T}$, both at the same temperature and each scaled to unit area. The extra factor of v shifts the peak of the beam distribution to a higher speed and thickens the tail.

Effusion can be used to separate isotopes. Let two isotopes of the same element be present as molecules of masses m_1 and m_2 , with $m_1 < m_2$. When their number densities are equal they effuse at rates in the ratio $\sqrt{m_2/m_1}$, so the gas that comes through the hole is slightly richer in the lighter isotope than the gas left behind. The masses of two isotopes of a heavy element differ by a very small fraction, so the square root of their ratio is barely above one and a single stage achieves very little. Feeding the enriched gas from each

stage to the next multiplies the enrichment factor by $\sqrt{m_2/m_1}$ each time, and a long cascade produces a useful separation. Uranium was separated in this way, as the gas uranium hexafluoride, in cascades of thousands of stages.

Collisions and the Cross-Section

We now turn from the walls to the collisions the molecules have with each other. Let us model a molecule as a hard sphere of diameter d . A pair of such molecules collide when the distance between their centres is less than d , so a moving molecule collides with any molecule whose centre lies within a distance d of the line along which its centre travels. The molecule therefore sweeps out a cylinder of radius d , drawn in Figure 7, and collides with every molecule whose centre lies inside it. The cross-sectional area of that cylinder,

$$\sigma = \pi d^2, \quad (1.32)$$

is called the collision cross-section. Real molecules are not hard spheres, and for them σ is best understood as a measured area that describes how effectively they deflect one another, which for most gases comes out at a few times the geometrical area of a molecule.

In a time t the molecule sweeps a cylinder of volume $\sigma \bar{v}t$, and if the centres of the other molecules are scattered at random with n per unit volume, the number inside the cylinder is $n\sigma \bar{v}t$. Dividing the length of the path by the number of collisions along it gives the mean free path as $\lambda = \bar{v}t/(n\sigma \bar{v}t) = 1/n\sigma$.

The Factor of Root Two

The calculation above treats the other molecules as though they were standing still, and they are not. What governs whether two molecules meet is the motion of one relative to the other, so the speed that belongs in the numerator, the length of the path swept per unit time, is the mean relative speed $\bar{v}_{\text{rel}}$, while the speed that belongs in the mean free path, which is a distance travelled through the container, is $\bar{v}$.

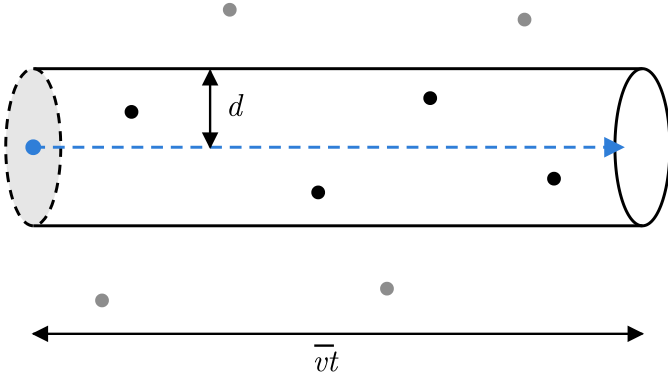


Figure 7. A molecule travelling along the dashed line sweeps out a cylinder of radius d and length $\bar{v}t$ in a time t . It collides with the molecules whose centres lie inside the cylinder, drawn as solid dots, and misses those outside, drawn faint.

The mean relative speed is larger than the mean speed by a factor of $\sqrt{2}$. To see where the factor comes from, let us take two molecules with velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ and write their total kinetic energy in terms of the velocity of their centre of mass, $\mathbf{V} = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$, and their relative velocity, $\mathbf{v}_{\text{rel}} = \mathbf{v}_1 - \mathbf{v}_2$:

$$\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}(2m)V^2 + \frac{1}{2}\left(\frac{1}{2}m\right)v_{\text{rel}}^2. \quad (1.33)$$

The Boltzmann factor for the pair is the exponential of minus β times this energy, which is a product of one factor depending on $\mathbf{V}$ and another depending on $\mathbf{v}_{\text{rel}}$. Since the change of variables from $\mathbf{v}_1$ and $\mathbf{v}_2$ to $\mathbf{V}$ and $\mathbf{v}_{\text{rel}}$ is linear, a small range of the old variables corresponds to a range of the new ones of a fixed proportional size, and the relative velocity is therefore distributed exactly like the velocity of a single molecule, with the mass m replaced by $\frac{1}{2}m$. Since the mean speed is proportional to $1/\sqrt{m}$, halving the mass multiplies it by $\sqrt{2}$, and $\bar{v}_{\text{rel}} = \sqrt{2}\bar{v}$.

Putting the relative speed into the collision rate and keeping $\bar{v}$ for the distance travelled, the number of collisions in a time t is $n\sigma\sqrt{2}\bar{v}t$ while the distance covered is $\bar{v}t$, so

$$\lambda = \frac{1}{\sqrt{2}n\sigma}, \quad \tau = \frac{\lambda}{\bar{v}} = \frac{1}{\sqrt{2}n\sigma\bar{v}}, \quad (1.34)$$

where τ is the mean time between collisions.

How the Mean Free Path Depends on Pressure

Using $n = p/k_B T$ to eliminate the number density,

$$\lambda = \frac{k_B T}{\sqrt{2}\sigma p}, \quad (1.35)$$

so at a fixed temperature the mean free path is inversely proportional to the pressure. At atmospheric pressure it is small compared with the size of any ordinary container, and a molecule makes an enormous number of collisions between one wall and the next. As the gas is pumped out the mean free path grows, and at low enough pressure it exceeds the size of the container, at which point the molecules travel from wall to wall without meeting one another. The condition for effusion stated earlier, that the hole be small compared with λ , is therefore easy to satisfy in an evacuated vessel and hard to satisfy at atmospheric pressure.

1.4 Transport in Gases

A gas that is left alone comes to equilibrium, with the same temperature, the same composition, and the same mean velocity everywhere. If one part of it is disturbed, the disturbance spreads, and it spreads because the molecules carry things about as they move. We now set up an argument based on the mean free path and apply it to three different quantities: to the sideways momentum of the molecules, which gives the viscosity of a gas; to their energy, which gives the thermal conductivity; and to the molecules themselves, which gives the diffusion coefficient.

The Mean-Free-Path Argument

Let χ be the mean value, per molecule, of some quantity carried by the molecules, and suppose that χ varies with position along the

x -direction and is uniform in the other two directions. Let us look at an imaginary plane at $x = x_0$, perpendicular to the x -axis, and count what crosses it.

From the left, molecules cross the plane at the rate $\frac{1}{4}n\bar{v}$ per unit area, which is the flux calculated in Section 1.3, and from the right at the same rate, since the gas is in mechanical equilibrium and there is no net flow of molecules across the plane. A molecule arriving at the plane brings with it the value of χ that it acquired where it last collided, on the assumption that a single collision gives a molecule the mean value of χ at the place where the collision happened. Let ℓ be the mean perpendicular distance back to that last collision. The net transport of χ across unit area of the plane per unit time is then the difference between what arrives from the left and what arrives from the right,

$$J_\chi = \frac{1}{4}n\bar{v} [\chi(x_0 - \ell) - \chi(x_0 + \ell)] . \quad (1.36)$$

The mean free path is very small compared with the distance over which χ changes appreciably, so we expand each term to first order in ℓ , and the two zeroth-order terms cancel:

$$J_\chi = -\frac{1}{2}n\bar{v}\ell \frac{d\chi}{dx} . \quad (1.37)$$

The distance ℓ is not the mean free path itself, because the molecules arrive from every direction. A molecule travelling at an angle θ to the normal covers a mean free path λ along its line of flight but advances only $\lambda \cos \theta$ in the x -direction. Each direction has to be weighted by the rate at which the molecules travelling that way cross the plane, which is proportional to $\cos \theta$, and the fraction of directions lying within $d\theta$ of θ is proportional to the area $2\pi \sin \theta d\theta$ of the corresponding band of a sphere of unit radius. Averaging $\cos \theta$ over the directions of arrival with these two weightings gives

$$\ell = \lambda \frac{\int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta}{\int_0^{\pi/2} \cos \theta \sin \theta d\theta} = \lambda \frac{1/3}{1/2} = \frac{2}{3}\lambda , \quad (1.38)$$

where both integrals are done by substituting $w = \cos \theta$. Then

$$J_\chi = -\frac{1}{3}n\bar{v}\lambda \frac{d\chi}{dx}. \quad (1.39)$$

The factor of $\frac{1}{3}$ should not be taken too seriously, for reasons given at the end of this section.

Viscosity

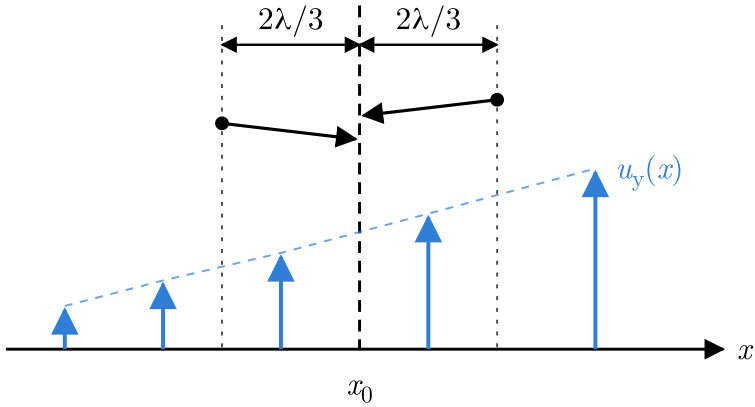


Figure 8. A gas drifting in the y -direction with a speed $u_y(x)$ that increases with x , shown by the vertical arrows. Molecules crossing the plane at x_0 last collided on average a distance $\frac{2}{3}\lambda$ away, so those arriving from the right bring more y -momentum than those arriving from the left.

Let the gas be in slow steady motion in the y -direction, with a drift velocity $u_y(x)$ that varies with x , as in Figure 8. The drift is slow compared with the thermal speed, so it barely disturbs the distribution of molecular velocities, and each molecule carries the y -momentum mu_y of the place it came from. Putting $\chi = mu_y$ into the general result,

$$J_{p_y} = -\frac{1}{3}nm\bar{v}\lambda \frac{du_y}{dx}. \quad (1.40)$$

The rate at which y -momentum crosses unit area of the plane is a force per unit area exerted by the gas on one side on the gas on the other, acting along y and therefore parallel to the plane. A stress of that kind is called a shear stress, and a fluid in which the shear stress is proportional to the velocity gradient, as it is in air and in water, is called Newtonian, the constant of proportionality being the viscosity η , defined by

$$J_{p_y} = -\eta \frac{du_y}{dx}. \quad (1.41)$$

Comparing the two expressions for J_{p_y} ,

$$\eta = \frac{1}{3}nm\bar{v}\lambda = \frac{1}{3}\rho\bar{v}\lambda. \quad (1.42)$$

Thermal Conductivity

Now let the gas be at rest but hotter in some places than in others, with a temperature $T(x)$. Let c be the heat capacity of the gas at constant volume per molecule, that is C_V/N , so that the mean energy carried by a molecule at temperature T is $\chi = cT$, up to an additive constant that does not enter the gradient. The general result with this choice of χ gives

$$J_E = -\frac{1}{3}nc\bar{v}\lambda \frac{dT}{dx}, \quad (1.43)$$

which is the rate at which energy crosses unit area. Comparing it with the definition of the thermal conductivity κ , namely $J_E = -\kappa dT/dx$, which expresses in symbols the law of conduction described in Section 1.1,

$$\kappa = \frac{1}{3}nc\bar{v}\lambda. \quad (1.44)$$

Diffusion

Finally, let the gas be uniform in pressure and temperature, but let some of the molecules be labelled in a way that does not change how they move, for instance by being a different isotope of the same

species, and let the number density $n_1(x)$ of the labelled molecules vary with position while the total density n stays uniform. The quantity carried is now the label itself, so the counting is done directly on the labelled molecules; they cross the plane from the left at the rate $\frac{1}{4}n_1(x_0 - \ell)\bar{v}$ and from the right at the rate $\frac{1}{4}n_1(x_0 + \ell)\bar{v}$, and the same expansion as before gives the net flux of labelled molecules,

$$J = -\frac{1}{3}\bar{v}\lambda \frac{dn_1}{dx}. \quad (1.45)$$

A flux proportional to the gradient of the density in this way is called Fick's law, $J = -D dn_1/dx$, and the constant D is called the diffusion coefficient. Comparing the two expressions for J ,

$$D = \frac{1}{3}\bar{v}\lambda. \quad (1.46)$$

How the Coefficients Depend on Pressure and Temperature

The three coefficients contain the combination $\bar{v}\lambda$, and the mean free path found in Section 1.3 is $\lambda = 1/\sqrt{2}n\sigma$, which is inversely proportional to the density. In the viscosity the density appears once in front and once in λ , and the two factors of the density cancel:

$$\eta = \frac{m\bar{v}}{3\sqrt{2}\sigma} = \frac{2}{3\sigma} \sqrt{\frac{mk_B T}{\pi}}. \quad (1.47)$$

The viscosity of a gas does not depend on its pressure, and it grows as $\sqrt{T}$. The independence of pressure is surprising, since one might expect a thinner gas to be less viscous. The reason the density cancels is that halving it halves the number of molecules available to carry momentum across the plane and at the same time doubles the distance each of them carries it from. The same cancellation happens in the thermal conductivity, $\kappa = c\bar{v}/3\sqrt{2}\sigma$, which is also independent of pressure and also grows as $\sqrt{T}$. The diffusion coefficient behaves differently, because only one factor of the density ap-

pears in it; $D = \bar{v}/3\sqrt{2}n\sigma$ is inversely proportional to the pressure at fixed temperature and proportional to $T^{3/2}$ at fixed pressure.

Comparing the expressions for the three coefficients, we find the following relations between them,

$$\kappa = \frac{c}{m} \eta, \quad D = \frac{\eta}{\rho}, \quad (1.48)$$

where c/m is the heat capacity per unit mass at constant volume. These relations are more useful than the individual coefficients, because the cross-section σ has cancelled out of them, and σ is the one quantity in the theory that has to be taken from experiment.

How Well the Argument Works

The prediction that the viscosity of a gas is independent of its pressure was made by Maxwell and tested by him experimentally in the 1860s, and it holds well over a wide range of pressures. Nothing in the behaviour of liquids would have suggested it, which is what makes the agreement such good evidence for the kinetic picture. The predicted growth as $\sqrt{T}$ is right in outline, though measured viscosities grow somewhat faster, because real molecules are soft rather than hard; a fast molecule penetrates further into the repulsive field of another before being turned aside, so the effective cross-section falls as the temperature rises.

The numerical factors are less good than the dependences. The relation $\kappa = c\eta/m$ gives the right order of magnitude, but measurement puts the ratio $\kappa m/c\eta$ at about two and a half for monatomic gases and somewhat less for diatomic ones, and the relation $D = \eta/\rho$ is out by a factor of about one and a half in the same direction.

The argument is rough in several places. The assumption that a molecule carries the mean value of χ from a fixed distance back and takes on the mean value of its new surroundings in a single collision is crude, since a collision changes a molecule's velocity by a random amount, and the molecules that travel much further than λ before colliding are not properly counted. The faster molecules both travel further between collisions and cross the plane more often, so speed and distance of travel are correlated in a way that using $\bar{v}$ and λ

separately ignores. In the calculation of κ the density was taken as uniform while the temperature varies, which cannot hold exactly at constant pressure. A more careful treatment follows how the distribution of velocities is itself distorted by the gradient; that distortion is not treated in this course, and allowing for it changes the numerical factors while leaving the dependence on pressure and temperature as we have found it.

The argument also has a range of validity, set by the mean free path. The calculations above assumed that λ is small compared with the size of the container, so that a molecule makes many collisions in crossing it. When the gas is pumped down far enough for λ to exceed the size of the container, momentum and energy are carried straight from one wall to the other, the mean free path no longer enters, and the viscosity and thermal conductivity become proportional to the pressure. That change of behaviour is what makes a vacuum flask work, since pumping the space between its two walls down far enough leaves a thermal conductivity that falls in proportion to the pressure that remains.

Problems

§1.1 Thermal Physics at School

1. Show that the density of an ideal gas of molar mass M_m at pressure p and temperature T is $\rho = pM_m/RT$.
2. A vessel of volume V holds two ideal gases, of N_1 and N_2 molecules, at temperature T , and they do not react. Assuming that each contributes to the pressure as though the other were absent, show that the total pressure is the sum of the two partial pressures, and that the fraction of the total pressure contributed by the first gas is $N_1/(N_1 + N_2)$.
3. Using the school results $pV = \frac{1}{3}Nm\overline{c^2}$ and $pV = Nk_B T$, show that two gases at the same temperature, whose molecules have masses m_1 and m_2 , have root mean square speeds in the ratio $c_{1,\text{rms}}/c_{2,\text{rms}} = \sqrt{m_2/m_1}$.
4. A cube of an isotropic solid has side ℓ and expands by $\Delta\ell = \alpha\ell\Delta T$ in each direction. Show that to first order in $\alpha\Delta T$ the

fractional change in its volume is $3\alpha \Delta T$.

§1.2 Pressure and the Distribution of Molecular Speeds

1. Show, by writing $\overline{v^2} = \int_0^\infty v^2 F(v) dv$ with the Maxwell-Boltzmann distribution and integrating by parts twice, that $\overline{v^2} = 3k_B T/m$, in agreement with the result obtained from the components.
2. Show that the mean of $|v_x|$ for a single velocity component is $\overline{|v_x|} = \sqrt{2k_B T/\pi m}$, and hence that it is equal to $\frac{1}{2}\bar{v}$.
3. For the isothermal atmosphere of this section, show that the mean height of a molecule above the ground is $\bar{z} = k_B T/mg$, and that the mean potential energy of a molecule is therefore $k_B T$. Why does equipartition not give $\frac{1}{2}k_B T$ here?
4. In a simple model of a crystal, each atom sits in a potential well and has energy $\frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) + \frac{1}{2}k(x^2 + y^2 + z^2)$, where x , y , and z are measured from the bottom of the well. Use equipartition to find the mean energy of an atom and show that the heat capacity of one mole of the crystal is $3R$.

§1.3 Effusion, Collisions, and the Mean Free Path

1. Show that the mean speed of the molecules in an effusing beam is $\frac{3}{8}\pi\bar{v}$, where $\bar{v}$ is the mean speed in the gas. You may use the integral $\int_0^\infty v^4 e^{-av^2} dv = \frac{3}{8}\sqrt{\pi/a^5}$, which was obtained in the solutions to Section 1.2.
2. A vessel of volume V contains gas at number density n_0 and has a small hole of area A in its wall, with a vacuum outside. The temperature is held fixed. Show that the density inside goes as $n(t) = n_0 e^{-t/t_0}$ and find t_0 in terms of V , A , and $\bar{v}$.
3. A gas contains two isotopic species of molecular masses m_1 and m_2 , with $m_1 < m_2$. Show that the ratio of the numbers of the two species in the effusing beam exceeds the ratio inside the vessel by the factor $\sqrt{m_2/m_1}$, and hence that s successive stages of effusion, each fed by the beam from the one before, give an enrichment factor $(m_2/m_1)^{s/2}$.

4. Show that the number of collisions a molecule makes per unit time is proportional to $p/\sqrt{T}$, and that the total number of collisions per unit time in unit volume of the gas is proportional to $p^2/T^{3/2}$.

§1.4 Transport in Gases

1. Show that the thermal conductivity of a gas is independent of its pressure, and that the diffusion coefficient is proportional to $T^{3/2}/p$.
2. For a monatomic gas, use equipartition to find the heat capacity c of one molecule at constant volume, and show that the mean-free-path argument then gives $\kappa = 3k_B\eta/2m$.
3. A gas fills the gap between two large parallel plates a distance L apart. One plate is held still and the other is dragged along in its plane at a steady speed u_0 . In the steady state no momentum accumulates anywhere in the gas, so J_{p_y} is the same at every value of x . Show that the drift velocity varies linearly across the gap and that the force per unit area needed to drag the plate is $\eta u_0/L$.
4. The gas between the plates of the previous problem is now pumped out until λ is much larger than L . Molecules then cross from one plate to the other without colliding on the way. Taking a molecule leaving a plate to carry away the drift velocity of that plate, show that the force per unit area is of order $\frac{1}{4}\bar{n}\bar{v}mu_0$, and explain why the gas now behaves as though its viscosity were proportional to the pressure.

The Laws of Thermodynamics

2.1 Thermodynamics at School

Chapter 1 looked at a gas from the inside, as a swarm of atoms and molecules, and worked out the pressure, the distribution of speeds, and the rate at which momentum and energy are carried about. This part looks at the same systems from the outside, using only quantities that can be measured with a thermometer, a pressure gauge, and a ruler. Much of what follows was already met at school, and this section collects those results in one place. All of it was covered thoroughly there, so the results below are stated and used rather than derived a second time.

Temperature and Thermal Equilibrium

When two bodies at different temperatures are placed in thermal contact, energy passes from the hotter to the colder, and it keeps passing until the two temperatures are equal. The bodies are then said to be in thermal equilibrium, and the energy transferred in this way is called heat. Temperature is what two bodies in thermal equilibrium have in common, and a thermometer works because it is a small body brought into equilibrium with a larger one.

If body A is in thermal equilibrium with body C , and body B is in thermal equilibrium with the same body C , then A and B are

found to be in thermal equilibrium with each other. Without this experimental fact a thermometer would tell us nothing about any body other than itself.

Temperatures in thermodynamics are measured on the absolute scale, in kelvin. The dilute-gas thermometer is used to define it as follows: a fixed amount of gas is held at constant volume, its pressure is measured, and the temperature is taken proportional to that pressure. In the limit of low density all gases agree with one another, since the molecules are then far enough apart for the forces between them to have no appreciable effect, and the equation of state is

$$pV = \nu RT = Nk_B T, \quad (2.1)$$

where ν is the number of moles, N the number of molecules, R the gas constant, and k_B Boltzmann's constant.

Heat Capacity, Latent Heat, and Calorimetry

Supplying heat Q to a body of mass m raises its temperature by an amount ΔT given by

$$Q = mc \Delta T, \quad (2.2)$$

where c is the specific heat capacity. A chemist is more likely to use the molar heat capacity, the heat needed to raise one mole through one kelvin. A gas has two such specific heats, one measured at constant volume and one at constant pressure, and the constant-pressure value is the larger of the two, because a gas heated at constant pressure expands and pushes the atmosphere back as it does so.

At a melting point or a boiling point, heat supplied changes the phase of the material without changing its temperature, and the heat needed is $Q = mL$ with L the specific latent heat of fusion or of vaporisation. Calorimetry rests on the formula for the heat capacity and the formula for the latent heat together with the conservation of energy: in a well-insulated container, the total heat given up by

the bodies that cool is equal to the total heat taken in by the bodies that warm.

The First Law

Energy can be given to a system as heat and as work, and the school statement of the first law is

$$Q = \Delta U + W, \quad (2.3)$$

where Q is the heat supplied to the system, W is the work done by the system on its surroundings, and ΔU is the change in the internal energy. For a gas expanding slowly against a piston, the work done by the gas as the volume increases by dV is $p dV$, so

$$W = \int_{V_1}^{V_2} p dV, \quad (2.4)$$

which is the area under the curve traced out on a graph of pressure against volume. For an expansion at constant pressure this area is simply $p(V_2 - V_1)$. For an isothermal expansion of an ideal gas we substitute $p = \nu RT/V$ and obtain $W = \nu RT \ln(V_2/V_1)$.

A change in which no heat enters or leaves the gas is called adiabatic. For an ideal gas undergoing a slow adiabatic change,

$$pV^\gamma = \text{constant}, \quad TV^{\gamma-1} = \text{constant}, \quad (2.5)$$

where γ is the ratio of the heat capacity at constant pressure to the heat capacity at constant volume, equal to $5/3$ for a monatomic gas and close to $7/5$ for the common diatomic gases.

Engines, Refrigerators, and Heat Pumps

A heat engine works in cycles, taking heat Q_h from a hot body, delivering work W , and rejecting heat Q_c to a cold body. Since the engine returns to its starting condition at the end of each cycle, its internal energy is unchanged over the cycle and $W = Q_h - Q_c$. Its efficiency is

$$\eta = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h}. \quad (2.6)$$

A refrigerator is the same device run backwards; work is supplied, heat Q_c is taken from a cold body, and heat $Q_h = Q_c + W$ is delivered to a warmer one. Its performance is measured by the coefficient of performance Q_c/W , since what is wanted from a refrigerator is the heat removed from inside it. A heat pump is the same device again, used to warm a building rather than to cool a cold store, and for it the coefficient of performance is Q_h/W .

No engine working between a hot body at T_h and a cold body at T_c can have an efficiency greater than the Carnot value

$$\eta_C = 1 - \frac{T_c}{T_h}, \quad (2.7)$$

and the corresponding limits on the coefficients of performance are $T_c/(T_h - T_c)$ for a refrigerator and $T_h/(T_h - T_c)$ for a heat pump.

The Second Law at School

The school version of the second law is the statement that the entropy of an isolated system never decreases. Entropy is introduced through the general principle that when heat Q enters a body held at a fixed absolute temperature T , the entropy of that body changes by

$$\Delta S = \frac{Q}{T}. \quad (2.8)$$

Heat flows unaided from a hot body to a cold one because doing so raises the entropy of the two together: the cold body gains Q/T_c while the hot one loses the smaller amount Q/T_h . The reverse flow would lower the total entropy, and it is never observed.

In the sections that follow we put this material on a firmer footing. Section 2.2 develops the mathematics of functions of two variables and restates the first law with it; Section 2.3 examines heat engines and the Carnot cycle; Section 2.4 works through the cycles used in

real engines; Section 2.5 states the second law carefully and shows that it forces entropy to exist as a function of state; and Section 2.6 calculates entropy changes in a range of processes.

2.2 Functions of State and the First Law

The equation of state tells us that the pressure, volume, and temperature of a fixed quantity of gas are not independent: fix any two and the third follows. Quantities such as the internal energy are therefore functions of two variables, and to work with them we need partial derivatives. We develop the mathematics as far as it is used here, and then apply it to the first law.

Functions of Two Variables and Partial Derivatives

Let $f(x, y)$ be a function of two variables. The partial derivative of f with respect to x is the ordinary derivative taken with y held fixed,

$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad (2.9)$$

and the partial derivative with respect to y is defined in the same way with x held fixed. The subscript on the bracket says which variable is being held fixed. In most of mathematics the subscript is dropped as obvious, and in thermodynamics it is often written explicitly, because the same system can be described by p and V , or by p and T , or by T and V , and the derivative of the internal energy with respect to temperature at constant volume is a different quantity from the derivative with respect to temperature at constant pressure.

Taking two partial derivatives in succession gives a second derivative, and for the functions that appear in thermodynamics the order in which the two derivatives are taken makes no difference:

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y}\right). \quad (2.10)$$

The conditions under which the two mixed second derivatives are equal are not treated in this course, since they belong to a course on analysis, and we shall take the equality as given for every function used here.

The Total Differential

If x changes by a small amount dx and y by a small amount dy , the change in f is the sum of the change caused by each, to first order in the small quantities:

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy. \quad (2.11)$$

This expression is called the total differential of f . Integrating it along any path in the xy -plane from a point P_1 to a point P_2 gives $f(P_2) - f(P_1)$, whatever the path, since the integrand is the change in f at each step.

Exact and Inexact Differentials

Suppose instead that we are handed an expression of the form

$$\delta g = M(x, y) dx + N(x, y) dy, \quad (2.12)$$

and we ask whether there is a function $g(x, y)$ of which this is the total differential. If there is such a function, then $A = (\partial g / \partial x)_y$ and $B = (\partial g / \partial y)_x$, and since the mixed second derivatives of g are equal,

$$\left(\frac{\partial A}{\partial y} \right)_x = \left(\frac{\partial B}{\partial x} \right)_y. \quad (2.13)$$

An expression that passes this test, which we met in the notes on mechanics as the condition for a force in a plane to be conservative, is called an exact differential, and it is written with a d . An expression that fails it is called inexact and is written with a δ , as above, to warn us that there is no function whose change it measures.

As an example, consider $\delta g = y \, dx$, so that $A = y$ and $B = 0$. Then $(\partial A / \partial y)_x = 1$ while $(\partial B / \partial x)_y = 0$, and the differential is inexact. We can see what this means by integrating from the origin to the point (a, b) along two different routes. Going first along the x -axis and then straight up, y is zero on the whole of the first leg and dx is zero on the whole of the second, so the integral is zero. Going first up the y -axis and then across, dx is zero on the first leg and $y = b$ throughout the second, so the integral is ab . The value depends on the route taken, and no function of position can have these as its changes.

Reciprocity and the Cyclic Relation

Of the relations between partial derivatives, two are used often enough in thermodynamics to be worth deriving here. Let x , y , and z be three quantities connected by one equation, so that any one of them is a function of the other two.

The first relation is called reciprocity. Holding z fixed leaves one independent variable, so x and y are related by an ordinary function of one variable, and the derivative of that function and the derivative of its inverse are reciprocals:

$$\left(\frac{\partial x}{\partial y}\right)_z = \frac{1}{\left(\frac{\partial y}{\partial x}\right)_z}. \quad (2.14)$$

The second relation is obtained from the total differential of z as a function of x and y ,

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy. \quad (2.15)$$

Setting $dz = 0$, which means moving along a line of constant z , and dividing by dy ,

$$\left(\frac{\partial x}{\partial y}\right)_z = -\frac{\left(\frac{\partial z}{\partial y}\right)_x}{\left(\frac{\partial z}{\partial x}\right)_y}. \quad (2.16)$$

Multiplying through by the denominator, writing $(\partial z/\partial y)_x = 1/(\partial y/\partial z)_x$ by reciprocity, and rearranging gives the cyclic relation,

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1. \quad (2.17)$$

The minus sign on the right-hand side is easily forgotten, and that is why the cyclic relation is set down separately here.

The State of a Simple System

A simple system, by which we mean a fixed quantity of a single substance with no electric or magnetic fields acting and no surface effects to worry about, has its equilibrium state fixed by two variables. Any pair will do, and the equation of state gives the rest. A quantity determined by the state, such as the pressure, the volume, the temperature, or the internal energy, is called a function of state, and a change in it between two equilibrium states depends only on those two states.

Heat and work are of a different character. They describe energy in transit, and neither of them is a property that a system possesses, so it makes no sense to ask how much heat a gas contains. When a gas is taken from one state to another, the heat supplied and the work done separately depend on the route, and only their sum is the same whichever route is taken.

The First Law Again

The first law, written for a small change, is

$$dU = \delta Q + \delta W, \quad (2.18)$$

where δQ is the heat supplied to the system and δW is the work done on the system. The differential dU is exact, while δQ and δW are inexact, and that is what the first law asserts: the sum of the heat and the work is the change in a function of state, although neither the heat alone nor the work alone is.

The sign convention here differs from the school form $Q = \Delta U + W$, in which W meant the work done by the system on its surroundings. Both conventions describe the same physics, and δW here is minus the school W . The convention used from now on treats heat and work alike, both counted positive when they carry energy into the system, which makes the first law easy to remember.

For a gas that expands slowly by dV against a piston, the surroundings are pushed back and the work done on the gas is negative:

$$\delta W = -p dV. \quad (2.19)$$

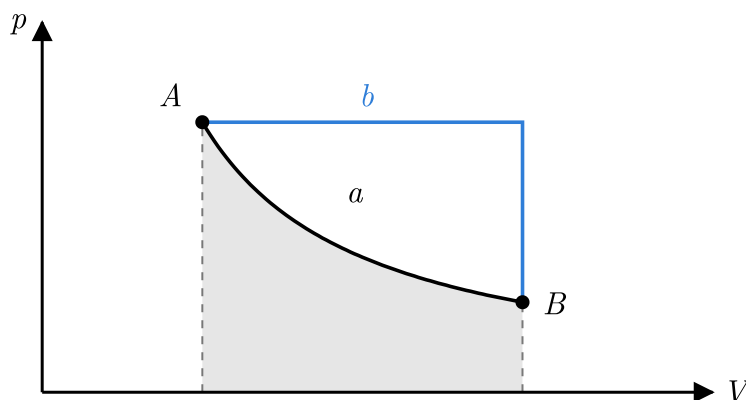


Figure 9. Paths a and b , each of them leading from the state A to the state B on the pressure–volume plane. Path a is an isothermal expansion and path b is an expansion at constant pressure followed by a fall in pressure at constant volume. The work done by the gas is the area under the path, shaded for path a , and it is larger for path b , while the change in internal energy is the same for both.

Figure 9 shows two paths between the same pair of states. The area under path b is larger than the area under path a , so the work done differs between the two paths, and since the change in internal energy is the same along both, the heat supplied must differ by the same amount as the work does.

Quasistatic and Reversible Changes

Writing $\delta W = -p dV$ assumes that the gas has a well-defined pressure p throughout the change, which requires the piston to move slowly enough for the gas to remain uniform. A change carried out slowly enough that the system passes through a continuous sequence of equilibrium states is called quasistatic, and only such a change can be drawn as a curve on the pressure–volume plane. A sudden expansion into a vacuum passes through states in which the gas has no single pressure, and it cannot be drawn as a curve.

A change is called reversible if it is quasistatic and, in addition, free of dissipation, so that running it backwards returns both the system and its surroundings to their original states. Friction between the piston and the cylinder walls, or heat flowing across a finite temperature difference, makes a change irreversible even when it is carried out arbitrarily slowly.

Heat Capacities

At constant volume no work is done, so $\delta Q = dU$ and the heat capacity at constant volume is

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V. \quad (2.20)$$

At constant pressure the gas expands as it warms, and $\delta Q = dU + p dV$ gives

$$C_p = \left(\frac{\partial U}{\partial T} \right)_p + p \left(\frac{\partial V}{\partial T} \right)_p. \quad (2.21)$$

For an ideal gas the internal energy depends on the temperature alone, a result that kinetic theory gave us in Section 1.2 and that Joule's experiments on expansion into a vacuum confirm. The first term is then the same whether the pressure or the volume is held fixed, and equals C_V . The second term follows from $V = \nu RT/p$, which gives $(\partial V/\partial T)_p = \nu R/p$. Hence

$$C_p - C_V = \nu R. \quad (2.22)$$

The difference is the work done in pushing the surroundings back for each unit rise in temperature, and the ratio of the two heat capacities is written $\gamma = C_p/C_V$.

Adiabatic Changes of an Ideal Gas

For a reversible adiabatic change, $\delta Q = 0$ and the first law reads $dU = -p dV$. With $dU = C_V dT$ and $p = \nu RT/V$ this becomes

$$C_V dT = -\frac{\nu RT}{V} dV, \quad (2.23)$$

which can be rearranged as $dT/T = -(\nu R/C_V) dV/V$. Since $\nu R/C_V = (C_p - C_V)/C_V = \gamma - 1$, integrating both sides gives

$$\ln T + (\gamma - 1) \ln V = \text{constant}, \quad TV^{\gamma-1} = \text{constant}, \quad (2.24)$$

and substituting $T = pV/\nu R$ turns this into pV^γ constant, the relation quoted in Section 2.1.

The slopes of the isotherm and of the adiabat are found by differentiating the relation that each of them obeys. Along an isotherm pV is constant, so $p dV + V dp = 0$ and

$$\left(\frac{\partial p}{\partial V} \right)_T = -\frac{p}{V}. \quad (2.25)$$

Along a reversible adiabat pV^γ is constant, so $\gamma p V^{\gamma-1} dV + V^\gamma dp = 0$, and dividing through by $V^{\gamma-1}$ gives an adiabatic slope of $-\gamma p/V$. Since γ is greater than one, the adiabat through a given state is steeper than the isotherm through it, as drawn in Figure 10. Compressing a gas adiabatically raises its pressure faster than compressing it isothermally, because the gas also grows hotter.

2.3 Heat Engines and the Carnot Cycle

An engine is a device that takes in heat and gives out work, and it has to do so over and over again, which means that its working

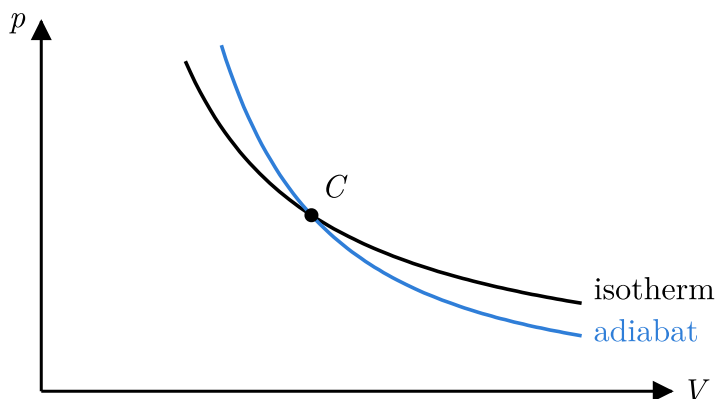


Figure 10. An isotherm and a reversible adiabat of an ideal gas drawn through the same state C , for $\gamma = 5/3$. The isothermal slope at C is $-p/V$ and the adiabatic slope is $-\gamma p/V$, so the adiabat is the steeper of the two curves.

substance must return to the same state at the end of each cycle. That single requirement, together with the first law and the experimental fact that no device can, as its sole result, take heat from a single reservoir and turn all of it into work, is enough to fix the highest efficiency any engine can reach.

Cyclic Devices

Since the internal energy is a function of state, it returns to its starting value at the end of a cycle, and the first law gives

$$\oint dU = 0, \quad W_{\text{by}} = Q_{\text{h}} - Q_{\text{c}}, \quad (2.26)$$

where the circle on the integral sign denotes an integral taken once round the cycle, Q_{h} is the heat taken in from the hot reservoir, Q_{c} the heat given out to the cold one, and W_{by} the work done by the engine on its surroundings. A reservoir here means a body so large that exchanging heat with it does not change its temperature, as

the sea does for the condensers of a power station built on the coast. The efficiency is defined as the work obtained divided by the heat taken from the hot reservoir,

$$\eta = \frac{W_{\text{by}}}{Q_{\text{h}}} = 1 - \frac{Q_{\text{c}}}{Q_{\text{h}}}, \quad (2.27)$$

and the coefficients of performance of a refrigerator and of a heat pump, Q_{c}/W and Q_{h}/W , where W is the work supplied to the device, follow from the same energy balance run in the opposite direction.

The Kelvin and Clausius Statements

Nothing in the first law forbids an engine with $Q_{\text{c}} = 0$, which could, for example, take heat from the sea and use it to power a ship, and nothing in the first law forbids a refrigerator that needs no work. Neither device has ever been built, and the second law begins as a statement that neither ever will be.

The Kelvin statement is that no process is possible whose sole result is the complete conversion of heat drawn from a single reservoir into work. The Clausius statement is that no process is possible whose sole result is the transfer of heat from a colder body to a hotter one. The words "sole result" are essential to both statements. A gas expanding isothermally does turn heat entirely into work, and there is no difficulty with that, because at the end of the expansion the gas occupies a larger volume, so the conversion of heat into work is not the sole result.

The Carnot Cycle

Carnot's cycle consists of two isothermal legs and two adiabatic legs, and it is reversible throughout. Let us work it out for an ideal gas, with the states labelled as in Figure 11.

Along AB the gas is held in contact with the hot reservoir at T_{h} and expands from V_A to V_B . Its temperature does not change, so its internal energy does not change either, and the heat taken in equals the work done by the gas, which was worked out for an isothermal expansion in Section 2.1:

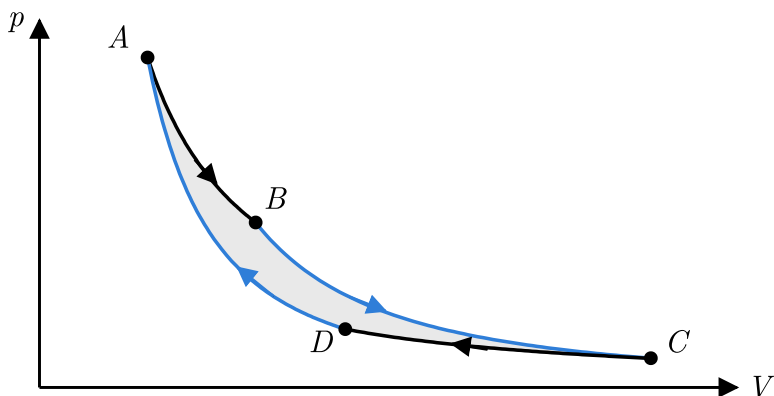


Figure 11. The Carnot cycle of an ideal gas on the pressure–volume plane, drawn for $\gamma = 5/3$ and $T_h = 2T_c$. The legs AB and CD are isotherms at T_h and T_c , and the legs BC and DA , drawn in blue, are adiabats. The enclosed area is the work done by the gas in one cycle.

$$Q_h = \int_{V_A}^{V_B} \frac{\nu R T_h}{V} dV = \nu R T_h \ln \frac{V_B}{V_A}. \quad (2.28)$$

Along BC the gas is insulated and expands further, cooling from T_h to T_c . Along CD it is held against the cold reservoir at T_c and compressed from V_C to V_D , giving out

$$Q_c = \nu R T_c \ln \frac{V_C}{V_D}. \quad (2.29)$$

Along DA it is insulated again and compressed back to its starting state, warming from T_c to T_h . The two adiabatic legs are governed by $TV^{\gamma-1}$ constant, which was derived in Section 2.2, so

$$T_h V_B^{\gamma-1} = T_c V_C^{\gamma-1}, \quad T_h V_A^{\gamma-1} = T_c V_D^{\gamma-1}. \quad (2.30)$$

Dividing the first of these equations by the second gives $(V_B/V_A)^{\gamma-1} = (V_C/V_D)^{\gamma-1}$, and since both volume ratios are posi-

tive, taking the $(\gamma - 1)$ th root of each side gives $V_B/V_A = V_C/V_D$, so the two logarithms in Q_h and Q_c are the same number. Hence

$$\frac{Q_c}{Q_h} = \frac{T_c}{T_h}, \quad \eta_C = 1 - \frac{T_c}{T_h}. \quad (2.31)$$

The efficiency depends on the two reservoir temperatures, and not on the size of the engine or on how far the gas was expanded.

Carnot's Theorem

Carnot's theorem states that no engine working between two given reservoirs can be more efficient than a reversible engine working between the same two reservoirs, and that all reversible engines between them have the same efficiency.

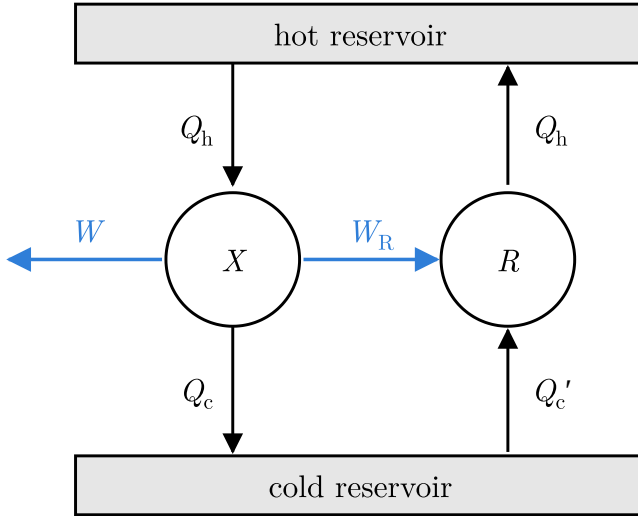


Figure 12. An engine X drives a reversible engine R backwards between the same two reservoirs. The heats exchanged with the hot reservoir are arranged to be equal, so that over one cycle of the pair the hot reservoir is unchanged. The work W_R passes from X to R , the heat pump R draws Q_c' from the cold reservoir, and the surplus work W leaves the assembly.

To prove the first part, let X be any engine and R a reversible one, both working between a reservoir at T_h and a reservoir at T_c , and

suppose that $\eta_X > \eta_R$. Run R backwards as a heat pump, driven by X , as in Figure 12, and adjust the sizes of the two engines so that the heat Q_h that R delivers to the hot reservoir is exactly the heat that X draws from it. The work produced by X in one cycle is $\eta_X Q_h$ and the work needed to drive R is $\eta_R Q_h$, so there is a surplus $W = (\eta_X - \eta_R)Q_h$, which is positive by assumption.

Now we look at what the pair of engines has done in one cycle. The hot reservoir has given out Q_h and taken back Q_h , so it is unchanged. The engine X has delivered $Q_h - \eta_X Q_h$ to the cold reservoir and the heat pump R has taken $Q_h - \eta_R Q_h$ from it, so the cold reservoir has lost the net amount $(\eta_X - \eta_R)Q_h$, which is exactly the surplus work. The sole result is therefore the complete conversion into work of heat drawn from the cold reservoir alone, and the Kelvin statement forbids it. Hence $\eta_X \leq \eta_R$.

The second part follows by applying the first part twice. If R_1 and R_2 are both reversible, then taking R_1 as the arbitrary engine gives $\eta_{R_1} \leq \eta_{R_2}$, and taking R_2 as the arbitrary engine gives $\eta_{R_2} \leq \eta_{R_1}$, so the two efficiencies are equal. The efficiency of a reversible engine is therefore fixed by the temperatures of the two reservoirs alone, and is independent of the substance the engine works with, so the calculation of the previous subsection gives the efficiency of every reversible engine.

2.4 Engine Cycles

Real engines do not follow the Carnot cycle. The isothermal legs would have to be run very slowly for the gas to stay at the reservoir temperature, and an engine that runs slowly delivers little power however efficient it is. The cycles used in practice replace the isotherms by legs that can be carried out quickly, and we work through the two best known of them here. Each is treated in the same way, by following a fixed quantity of ideal gas once round the cycle and adding up the heat taken in and given out along the separate legs.

The Otto Cycle

The Otto cycle is the idealisation of the petrol engine. The working substance is treated as a fixed quantity of ideal gas with constant heat capacities, and the cycle has four legs, shown in Figure 13. From state 1 to state 2 the gas is compressed adiabatically from volume V_1 to volume V_2 . From 2 to 3 heat Q_{in} is supplied at constant volume, which stands for the burning of the fuel. From 3 to 4 the gas expands adiabatically back to V_1 , and this leg delivers the work. From 4 back to 1 heat Q_{out} is rejected at constant volume, which represents the exhaust being swept out and fresh air drawn in.

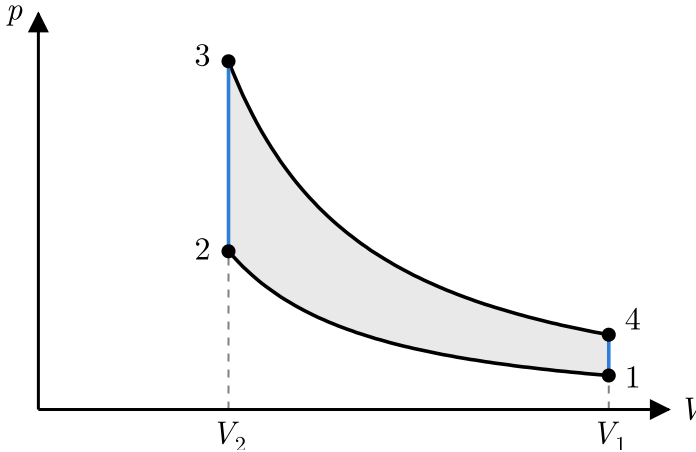


Figure 13. The Otto cycle on the pressure–volume plane, drawn for a compression ratio of three and $\gamma = 7/5$. The curved legs $1 \rightarrow 2$ and $3 \rightarrow 4$ are adiabats, and the vertical legs $2 \rightarrow 3$ and $4 \rightarrow 1$, drawn in blue, are the constant-volume heating and cooling. The shaded area is the work done per cycle.

Since no work is done on the two constant-volume legs, the heat supplied and rejected are

$$Q_{\text{in}} = C_V(T_3 - T_2), \quad Q_{\text{out}} = C_V(T_4 - T_1), \quad (2.32)$$

and the efficiency is

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}. \quad (2.33)$$

Write $r = V_1/V_2$ for the compression ratio. Applying $TV^{\gamma-1}$ constant to the leg $1 \rightarrow 2$ gives $T_2 = T_1 r^{\gamma-1}$, and applying it to the leg $3 \rightarrow 4$ gives $T_3 = T_4 r^{\gamma-1}$. Substituting these two results,

$$\eta = 1 - \frac{T_4 - T_1}{r^{\gamma-1}(T_4 - T_1)} = 1 - r^{1-\gamma}. \quad (2.34)$$

The efficiency depends on the compression ratio and on γ , and not on how much heat is supplied. It rises as r rises, which is one reason engine designers favour a high compression ratio. The limit on r comes from the heating of the gas during the compression; the temperature after compression is $T_1 r^{\gamma-1}$, and if that temperature is high enough the fuel and air mixture ignites unaided before the spark arrives. The resulting pressure wave hits the piston at the wrong moment and damages the engine, and this phenomenon is known as knocking. The compression ratio that can be used is fixed by the temperature at which the particular fuel ignites.

The Diesel Cycle

In a diesel engine the air alone is compressed, so there is no mixture to ignite early and the compression ratio can be much higher. The fuel is injected into the hot air at the end of the compression stroke and burns as it is injected, which keeps the pressure roughly constant while the piston begins to move outwards. The Diesel cycle therefore replaces the constant-volume heating of the Otto cycle by a constant-pressure heating, as drawn in Figure 14.

The heat supplied along the constant-pressure leg is $Q_{\text{in}} = C_p(T_3 - T_2)$, and the heat rejected is $Q_{\text{out}} = C_V(T_4 - T_1)$ as before. Hence

$$\eta = 1 - \frac{T_4 - T_1}{\gamma(T_3 - T_2)}. \quad (2.35)$$

Write $r = V_1/V_2$ for the compression ratio and $r_c = V_3/V_2$ for the cut-off ratio, which measures how far into the expansion stroke the

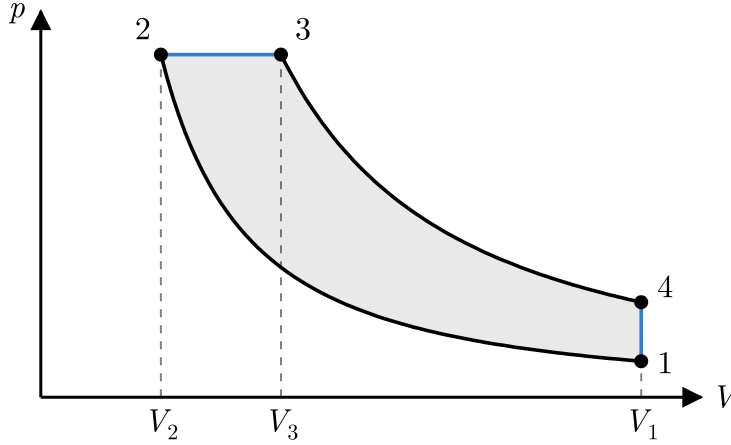


Figure 14. The Diesel cycle, drawn for a compression ratio of five, a cut-off ratio of two, and $\gamma = 7/5$. Heat is supplied along the constant-pressure leg $2 \rightarrow 3$ and rejected along the constant-volume leg $4 \rightarrow 1$, both drawn in blue.

injection continues. The adiabatic leg $1 \rightarrow 2$ gives $T_2 = T_1 r^{\gamma-1}$. The constant-pressure leg $2 \rightarrow 3$ gives $T_3 = T_2 r_c$. The adiabatic leg $3 \rightarrow 4$ runs from $V_3 = r_c V_2$ to $V_4 = V_1 = r V_2$, so $T_4 = T_3 (r_c/r)^{\gamma-1}$. Substituting for T_3 and then for T_2 in this last expression,

$$T_4 = T_2 r_c \left(\frac{r_c}{r} \right)^{\gamma-1} = T_1 r^{\gamma-1} r_c \frac{r_c^{\gamma-1}}{r^{\gamma-1}} = T_1 r_c^\gamma, \quad (2.36)$$

the powers of r cancelling between the numerator and the denominator. Therefore $T_4 - T_1 = T_1 (r_c^\gamma - 1)$ and $T_3 - T_2 = T_1 r^{\gamma-1} (r_c - 1)$, so

$$\eta = 1 - \frac{1}{\gamma} \frac{r_c^\gamma - 1}{r^{\gamma-1} (r_c - 1)}. \quad (2.37)$$

Letting r_c tend to one recovers the Otto result, as it should, since the heating then happens at constant volume: writing $r_c = 1 + \epsilon$, the last factor becomes $[(1 + \epsilon)^\gamma - 1]/\epsilon$, which tends to γ as $\epsilon \rightarrow 0$ and cancels the γ in front of $r^{\gamma-1}$. For any r_c greater than one the

last factor makes the diesel efficiency lower than the Otto efficiency at the same compression ratio. Diesel engines are nonetheless the more efficient in practice, because the compression ratios they can use are two or three times larger.

Indicator Diagrams and Indicated Power

An indicator is an instrument that records the pressure in a cylinder against the position of the piston as the engine runs, and the closed curve it draws is called an indicator diagram. The area enclosed by the curve is the net work done by the gas on the piston in one cycle, since the work done on any leg is the area under that leg, and the outward and return legs are traversed in opposite senses, so that their areas enter with opposite signs and only the enclosed area is left.

Figure 15 shows the diagram drawn by a four-stroke petrol engine, with the pressure rising by a factor of twenty between the start of the compression stroke and the peak that follows combustion. The large loop is the work the burning charge does on the piston. The exhaust and the induction strokes lie within a few per cent of atmospheric pressure on either side of it, so on the scale of the main diagram they fall on top of one another, and the inset draws them with the pressure scale magnified. The loop they enclose is traversed in the opposite sense from the large one, so the work it represents is done on the gas by the piston, and its area is therefore subtracted from the area of the large loop.

Let W be the net area of the indicator diagram, measured in energy units. An engine with N cylinders running at f revolutions per second then develops an indicated power of NWf if it completes a cycle every revolution, and $NWf/2$ if it is a four-stroke engine, which completes a cycle every second revolution. The indicated power is what the gas delivers to the pistons. The brake power, measured at the output shaft, is smaller, and the difference is the power absorbed by friction in the bearings, the piston rings, and the valve gear.

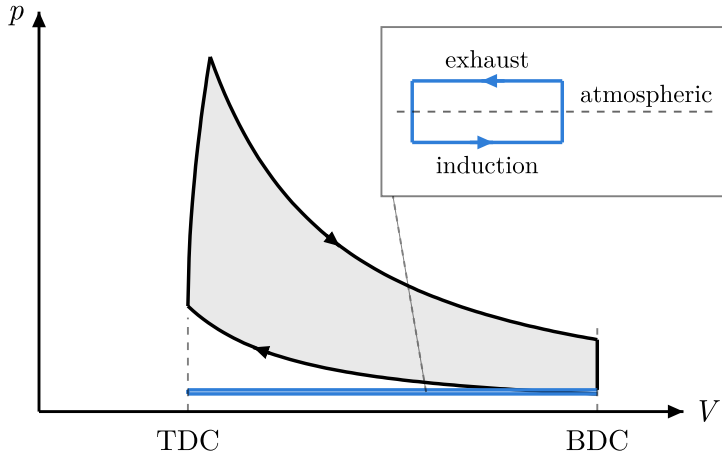


Figure 15. The indicator diagram of a four-stroke petrol engine. The piston stands at the top of its travel, called top dead centre, at the volume marked TDC, and at the bottom of its travel, called bottom dead centre, at the volume marked BDC. The charge is compressed along the lower curve, the pressure rises steeply while it burns, and the gas then expands along the upper curve, so that the large loop is traversed clockwise and the shaded area is the work the gas does on the piston. The exhaust and the induction strokes run so close to atmospheric pressure that they cannot be told apart at this scale, and the inset draws them again with the pressure scale magnified. The small loop they enclose is traversed anticlockwise.

Why Real Engines Fall Short

The efficiencies derived above are upper limits for the idealised cycles, and a real engine reaches perhaps half to two-thirds of them. The most important reasons for the shortfall are these:

1. The working substance is air and combustion products rather than a gas with a constant γ , and at the high temperatures inside a cylinder the heat capacities rise, which lowers γ and lowers the efficiency along with it.
2. Heat leaks through the cylinder walls to the cooling water during the whole cycle, so the expansion is not adiabatic.
3. Combustion takes a finite time and is never quite complete.
4. The valves open and close over a finite angle, so the constant-volume legs are approximations.
5. Drawing fresh air in and pushing exhaust out takes work from the engine.
6. Every leg is carried out at a finite rate, with pressure differences across the gas, so none of them is reversible.

2.5 The Second Law

Section 2.3 used the Kelvin statement to prove Carnot's theorem and left the Clausius statement standing beside it without saying how the two statements are related. What follows shows that they are the same statement in different words, then uses Carnot's theorem to define a temperature scale that refers to no particular substance, and finally derives the inequality from which entropy follows.

Equivalence of the Kelvin and Clausius Statements

The way to show that the Kelvin statement and the Clausius statement are equivalent is to show that a device violating either one can be used to build a device violating the other.

Suppose first that the Kelvin statement fails, so that we have a device which draws heat Q from a hot reservoir and turns all of it into work, with no other effect. Use that work to drive an ordinary refrigerator working between a cold reservoir and the same hot one, taking Q_c from the cold reservoir and delivering $Q_c + Q$ to the hot one. Adding the two devices together, the hot reservoir gives up Q and receives $Q_c + Q$, so it gains Q_c , and the cold reservoir loses Q_c . No work has been supplied from outside and nothing else has changed, so heat has passed from the colder body to the hotter one as the sole result, and the Clausius statement fails too.

Suppose instead that the Clausius statement fails, so that we have a device which carries heat Q_c from the cold reservoir to the hot one with no work supplied. Run an ordinary engine between the same reservoirs, sized so that the heat it rejects to the cold reservoir is exactly Q_c , while it takes Q_h from the hot reservoir and does work $Q_h - Q_c$. Adding the two devices together, the cold reservoir is unchanged, the hot reservoir has lost $Q_h - Q_c$, and work $Q_h - Q_c$ has been produced. Heat from a single reservoir has been converted entirely into work as the sole result, and the Kelvin statement fails too. The two statements are therefore equivalent, and either may be used as the second law.

The Thermodynamic Temperature Scale

Carnot's theorem tells us that the ratio Q_c/Q_h for a reversible engine depends only on the two reservoirs and not on the working substance. Let the reservoirs be labelled by readings θ_h and θ_c on any convenient empirical thermometer, such as a mercury-in-glass one, so that

$$\frac{Q_c}{Q_h} = f(\theta_h, \theta_c) \quad (2.38)$$

for some function f that is the same for every reversible engine.

Now we take three reservoirs, at θ_1 , θ_2 , and θ_3 , and run a reversible engine from θ_1 to θ_2 followed by a second reversible engine from θ_2 to θ_3 that takes in exactly the heat the first rejects. The pair together is a reversible engine from θ_1 to θ_3 , so

$$f(\theta_1, \theta_3) = f(\theta_1, \theta_2) f(\theta_2, \theta_3). \quad (2.39)$$

The reservoir at θ_2 has disappeared from the left-hand side, so the dependence on θ_2 on the right-hand side must cancel between the two factors. To see what kind of function of two variables can behave in that way, let us fix some reference reading θ_* and define the single-variable function $\phi(\theta) = f(\theta_*, \theta)$. Setting $\theta_1 = \theta_*$ in the relation above gives $f(\theta_*, \theta_3) = f(\theta_*, \theta_2) f(\theta_2, \theta_3)$, and dividing by $f(\theta_*, \theta_2)$,

$$f(\theta_2, \theta_3) = \frac{f(\theta_*, \theta_3)}{f(\theta_*, \theta_2)} = \frac{\phi(\theta_3)}{\phi(\theta_2)}, \quad (2.40)$$

so that $f(\theta_a, \theta_b) = \phi(\theta_b)/\phi(\theta_a)$ for any pair of readings. We define the thermodynamic temperature to be $T = \phi(\theta)$, up to the overall constant that is fixed by choosing a reference point, and then

$$\frac{Q_c}{Q_h} = \frac{T_c}{T_h}. \quad (2.41)$$

This definition refers to no substance. The calculation of Section 2.3 used an ideal gas, whose temperature was defined by $pV = \nu RT$, and gave exactly the same ratio, so the thermodynamic scale and the ideal-gas scale are proportional to each other. Assigning the same value, 273.16 K, to the triple point of water on both scales makes the constant of proportionality equal to one, and the two scales then agree everywhere.

The Clausius Inequality

Let a system be taken round a cycle, not necessarily reversible, during which it receives heat in small amounts δQ . We want a statement about the sum of $\delta Q/T$ around the cycle, where T is the temperature of the system at the moment it receives each amount of heat.

The construction is drawn in Figure 16. Rather than let the system take heat from wherever it pleases, we supply every amount of heat through a small reversible engine C that draws on one large

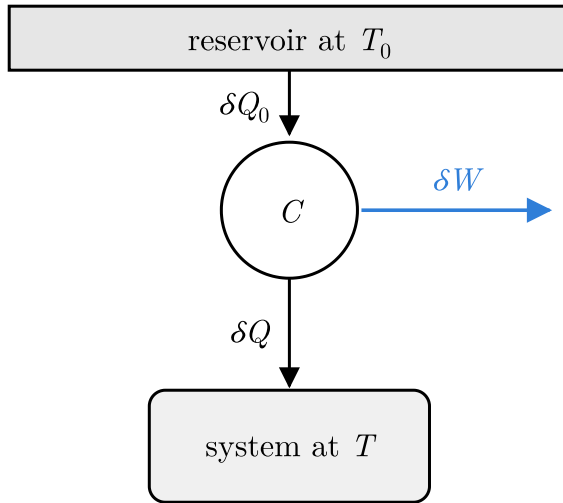


Figure 16. The construction used to derive the Clausius inequality. A small reversible engine C draws heat δQ_0 from a single large reservoir at T_0 , delivers δQ to the system at the temperature T at which the system is receiving it, and puts out the work $\delta W = \delta Q_0 - \delta Q$.

reservoir held at a fixed temperature T_0 . When the system is at temperature T and is to receive δQ , the engine C works between T_0 and T , and since it is reversible,

$$\frac{\delta Q_0}{T_0} = \frac{\delta Q}{T}, \quad \delta Q_0 = T_0 \frac{\delta Q}{T}. \quad (2.42)$$

This relation holds with signs included, so it also covers the stages at which the system gives heat out and C runs as a heat pump.

Over one complete cycle of the system, the system returns to its initial state and so does each auxiliary engine, since C is itself cyclic. The total heat withdrawn from the reservoir at T_0 is

$$Q_0 = T_0 \oint \frac{\delta Q}{T}, \quad (2.43)$$

and since nothing in the assembly has changed its state, the first law says that all of this heat has come out as work. If Q_0 were positive, the sole result of the whole operation would be the complete conversion into work of heat drawn from the single reservoir at T_0 , which the Kelvin statement forbids. Hence $Q_0 \leq 0$, and since T_0 is positive,

$$\oint \frac{\delta Q}{T} \leq 0. \quad (2.44)$$

This result is called the Clausius inequality. It holds for any cycle, reversible or not.

Entropy as a Function of State

Suppose now that the cycle is reversible. Then it can be run backwards, and running it backwards reverses the sign of every δQ while leaving every T alone. The inequality applied to the reversed cycle reads $-\oint \delta Q_{\text{rev}}/T \leq 0$, and the two inequalities together force

$$\oint \frac{\delta Q_{\text{rev}}}{T} = 0. \quad (2.45)$$

Consider any two equilibrium states A and B , and any two reversible paths from A to B . Going out along the first path and back along the second makes a reversible cycle, so the integral of $\delta Q_{\text{rev}}/T$ along the first path equals the integral along the second. The integral therefore depends only on the end points, and we may define a function of state S , called the entropy, by

$$S(B) - S(A) = \int_A^B \frac{\delta Q_{\text{rev}}}{T}, \quad dS = \frac{\delta Q_{\text{rev}}}{T}. \quad (2.46)$$

The subscript on δQ_{rev} matters. The integral is taken along any reversible path between the two states, and once the difference has been computed along one such path it is the entropy difference between those states however the system is actually taken from one to the other.

2.6 Entropy

Entropy was defined in Section 2.5 by an integral along a reversible path. Let us use that definition to work out the entropy change in a series of processes, beginning with the simplest, and then draw out what the results have in common.

Heating a Body

Let a body of constant heat capacity C_p be warmed at constant pressure from T_1 to T_2 . A reversible path between the two states is available: bring the body into contact with a succession of reservoirs, each one very slightly hotter than the body at that moment, so that no heat ever crosses a finite temperature difference. Along that path $\delta Q_{\text{rev}} = C_p dT$, and

$$\Delta S = \int_{T_1}^{T_2} \frac{C_p dT}{T} = C_p \ln \frac{T_2}{T_1}. \quad (2.47)$$

The entropy change of the body is the same whether it was in fact warmed in this careful way or dropped straight into a hot bath, because entropy is a function of state. What differs between

the two ways of warming the body is the entropy change of the surroundings.

Expansion of an Ideal Gas

For an ideal gas the first law along a reversible path reads $dU = T dS - p dV$ with $dU = C_V dT$, so

$$dS = \frac{C_V dT}{T} + \frac{\nu R dV}{V}, \quad (2.48)$$

and integrating between two states, with C_V taken constant,

$$\Delta S = C_V \ln \frac{T_2}{T_1} + \nu R \ln \frac{V_2}{V_1}. \quad (2.49)$$

Let us consider first a reversible isothermal expansion. A gas at temperature T is held in contact with a reservoir at the same temperature, and pushes a piston out slowly from V_1 to V_2 . The temperature does not change, so $\Delta S_{\text{gas}} = \nu R \ln(V_2/V_1)$. The gas has absorbed heat $Q = \nu RT \ln(V_2/V_1)$ from the reservoir, whose entropy has therefore fallen by $Q/T = \nu R \ln(V_2/V_1)$. The entropy gained by the gas and the entropy lost by the reservoir cancel. In any reversible process the entropy of the system and its surroundings taken together stays the same in just this way.

Let us take next the same gas expanding into a vacuum. The gas is held in one half of a rigid insulated box, the partition is removed, and the gas fills the whole box, doubling its volume from V_1 to $V_2 = 2V_1$. No heat enters and no work is done, since there is nothing for the gas to push against, so the internal energy is unchanged, and for an ideal gas that means the temperature is unchanged as well. The entropy change is

$$\Delta S_{\text{gas}} = \nu R \ln \frac{V_2}{V_1} = \nu R \ln 2, \quad (2.50)$$

while the surroundings are untouched, so the entropy of the whole has risen. This process is called a free expansion, and it is irreversible: the gas will not gather itself back into one half of the box.

Bodies in Thermal Contact

Let a quantity of hot water of mass m , constant specific heat c , and temperature T_1 be mixed with an equal mass of cold water at T_2 in an insulated vessel. Energy conservation gives the final temperature $T_f = (T_1 + T_2)/2$, and each portion changes its entropy by the formula derived above for heating a body, so

$$\Delta S = mc \ln \frac{T_f}{T_1} + mc \ln \frac{T_f}{T_2} = mc \ln \frac{T_f^2}{T_1 T_2}. \quad (2.51)$$

The entropy change is positive whenever the two temperatures differ, because

$$4T_f^2 - 4T_1 T_2 = (T_1 + T_2)^2 - 4T_1 T_2 = (T_1 - T_2)^2 > 0, \quad (2.52)$$

so $T_f^2 > T_1 T_2$, the argument of the logarithm exceeds one, and ΔS is positive.

A simpler case is heat Q passing from a body at T_h to a body at T_c , with both so large that neither temperature shifts. The hot body loses Q/T_h and the cold body gains Q/T_c , so

$$\Delta S = Q \left(\frac{1}{T_c} - \frac{1}{T_h} \right) > 0. \quad (2.53)$$

Mixing Two Gases

Let a rigid insulated box be divided by a partition, with ν_1 moles of one ideal gas in a volume V_1 on one side and ν_2 moles of a different ideal gas in a volume V_2 on the other, both at the same temperature. When the partition is removed, each gas spreads through the whole volume $V = V_1 + V_2$ without being impeded by the other, since the molecules of an ideal gas exert no forces on one another. Each gas therefore undergoes a free expansion, and

$$\Delta S = \nu_1 R \ln \frac{V}{V_1} + \nu_2 R \ln \frac{V}{V_2}. \quad (2.54)$$

For equal amounts of the two gases in equal volumes this is $2\nu R \ln 2$. If the gas on the two sides is the same gas at the same pressure and temperature, removing the partition changes nothing that can be measured, and the entropy change is zero. The formula above must therefore be used only when the two gases are genuinely different.

The Direction of Change and the Work Available

In every irreversible process above, the entropy of the system together with its surroundings increased, and in the reversible process it stayed the same. The Clausius inequality gives this result in general. Let a system be taken from A to B by any process and then returned from B to A along a reversible path. The two legs together make a cycle, so the Clausius inequality gives $\int_A^B \delta Q/T + \int_B^A \delta Q_{\text{rev}}/T \leq 0$, and the second integral is $S(A) - S(B)$ by the definition of entropy. Hence $S(B) - S(A) \geq \int_A^B \delta Q/T$, with equality when the process is reversible. Applied to a system and its surroundings taken together, so that no heat crosses the outer boundary, the right-hand side is zero and the total entropy cannot fall.

Entropy also measures how much of an energy transfer can still be turned into work. Let us take again the heat Q passing directly from the body at T_h to the body at T_c . Had that heat been passed through a reversible engine instead, work $Q(1 - T_c/T_h)$ would have come out of it. Comparing that amount of work with the entropy generated,

$$T_c \Delta S = Q \left(1 - \frac{T_c}{T_h} \right), \quad (2.55)$$

so the work that can no longer be obtained is the entropy generated multiplied by the temperature of the coldest body available. Every irreversible process generates entropy, and every irreversible process leaves less work obtainable than a reversible one would have done.

The Carnot Cycle on the Temperature–Entropy Plane

Since entropy is a function of state, a reversible cycle can be drawn on a diagram of temperature against entropy as readily as on the

pressure–volume plane, and the Carnot cycle is drawn that way here.

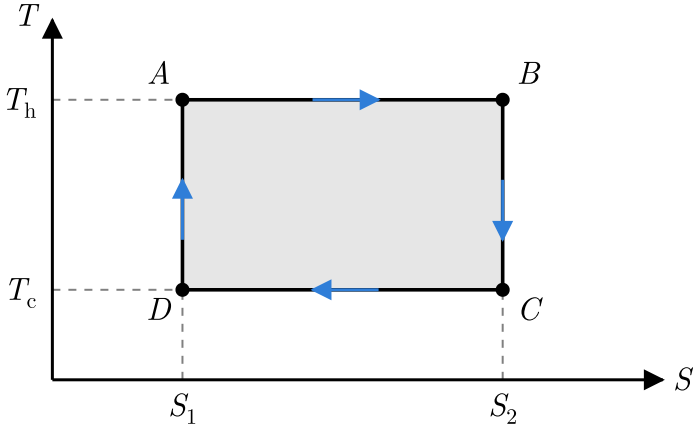


Figure 17. The Carnot cycle on the temperature–entropy plane. The isothermal legs AB and CD are horizontal and the adiabatic legs BC and DA are vertical, so the cycle is a rectangle whose area, $(T_h - T_c)(S_2 - S_1)$, is the work done in one cycle.

An isothermal leg is a line of constant T , and a reversible adiabatic leg has $\delta Q_{\text{rev}} = 0$ and therefore constant S , so the cycle is the rectangle of Figure 17. Along the upper leg the working substance absorbs $Q_h = T_h(S_2 - S_1)$, and along the lower leg it gives out $Q_c = T_c(S_2 - S_1)$. The work done per cycle is the difference, which is the area of the rectangle, and the efficiency

$$\eta = 1 - \frac{T_c(S_2 - S_1)}{T_h(S_2 - S_1)} = 1 - \frac{T_c}{T_h} \quad (2.56)$$

can be read straight off the diagram. The result of Section 2.3, which was obtained there by integrating along each leg of the cycle for an ideal gas, follows here from the lengths of the sides of a rectangle, and it holds for any working substance.

The Combined First and Second Laws

For a reversible change, $\delta Q_{\text{rev}} = T \, dS$ and $\delta W_{\text{rev}} = -p \, dV$, so the first law $dU = \delta Q + \delta W$ becomes

$$dU = T \, dS - p \, dV. \quad (2.57)$$

Every quantity appearing in this relation is a function of state, and dU , dS , and dV are the differences between the values at two neighbouring equilibrium states. The relation therefore holds between any two neighbouring equilibrium states, however the system is actually taken from one to the other. If the change is carried out irreversibly, the heat absorbed is less than $T \, dS$ and the work done on the system is greater than $-p \, dV$ by the same amount, and their sum is unaltered. We have already used the fact that the relation holds however the change is carried out, in computing the entropy change of a free expansion, where the actual process absorbed no heat.

Problems

§2.1 Thermodynamics at School

1. A gas expands at constant pressure p from volume V_1 to volume V_2 while absorbing heat Q . Find the change in its internal energy.
2. A mass m_1 of water at temperature T_1 is poured into a well-insulated container holding a mass m_2 of ice at its melting point T_0 . The specific heat capacity of water is c and the specific latent heat of fusion of ice is L . Find the condition on m_1 , m_2 , $T_1 - T_0$, c , and L for all the ice to melt, and find the final temperature when it does.
3. An ideal gas at pressure p_1 and temperature T_1 occupying volume V_1 is compressed slowly and adiabatically to volume V_2 . Find the final pressure and the final temperature.
4. Show that the coefficient of performance of a reversible heat pump exceeds that of a reversible refrigerator working between

the same two bodies by exactly one, and say in words why.

§2.2 Functions of State and the First Law

1. Show that $\delta W = -p dV$ is an inexact differential for an ideal gas, by writing it in terms of dT and dV and applying the test for exactness.
2. Verify the cyclic relation explicitly for an ideal gas by computing $(\partial p/\partial V)_T$, $(\partial V/\partial T)_p$, and $(\partial T/\partial p)_V$ from $pV = \nu RT$ and multiplying them together.
3. A gas obeys the equation of state $p(V - \nu b) = \nu RT$, where b is a constant. Find the work done on the gas in a reversible isothermal compression from V_1 to V_2 , and compare it with the ideal-gas result.
4. Show that the work done on an ideal gas in a reversible adiabatic change from (p_1, V_1) to (p_2, V_2) is $(p_2 V_2 - p_1 V_1)/(\gamma - 1)$.

§2.3 Heat Engines and the Carnot Cycle

1. A reversible engine uses a gas obeying $p(V - \nu b) = \nu RT$, whose internal energy depends only on the temperature. Show that on an adiabat $T(V - \nu b)^{\gamma-1}$ is constant, and hence that a Carnot cycle worked with this gas again has efficiency $1 - T_c/T_h$.
2. A Carnot engine works between T_h and an intermediate temperature T_m , and all the heat it rejects is taken in by a second Carnot engine working between T_m and T_c . Show that the efficiency of the pair, defined as the total work divided by the heat drawn from the reservoir at T_h , is $1 - T_c/T_h$, whatever the value of T_m .
3. Show that an engine working between two reservoirs must satisfy $Q_c/Q_h \geq T_c/T_h$, and say what happens when the equality holds.
4. A body of constant heat capacity C_p is cooled from T_1 to T_2 by a reversible refrigerator that rejects heat to a reservoir at T_h . Find the total work required.

§2.4 Engine Cycles

1. Show that the efficiency of the Otto cycle can be written $\eta = 1 - T_1/T_2$, where T_1 and T_2 are the temperatures at the start and the end of the compression stroke, and show that it is also equal to $1 - T_4/T_3$.
2. Show that for $r_c > 1$ the quantity $(r_c^\gamma - 1)/[\gamma(r_c - 1)]$ is greater than one, and hence that at equal compression ratios the Diesel cycle is less efficient than the Otto cycle.
3. The Joule cycle consists of two adiabats and two constant-pressure legs, with heat supplied at the higher pressure p_2 and rejected at the lower pressure p_1 . Show that its efficiency is $1 - (p_1/p_2)^{(\gamma-1)/\gamma}$.
4. A four-stroke engine has N cylinders, each with an indicator diagram of area W , and runs at f revolutions per second. Write down the indicated power, and explain why the factor of two appears.

§2.5 The Second Law

1. Show that two reversible adiabats of a system cannot cross. Assume that they do cross, close the region between them with an isotherm to make a cycle, and apply the Kelvin statement.
2. The function ϕ was built in the text from a reference reading θ_* . Show that building it instead from a different reference reading θ'_* replaces ϕ by a constant multiple of itself, so that every ratio of thermodynamic temperatures is unaffected by the choice, and say how the remaining overall factor is fixed in practice.
3. A system is taken round a cycle during which it exchanges heat only with two reservoirs, absorbing Q_1 from the one at T_1 and Q_2 from the one at T_2 , with either quantity allowed to be negative. Show that $Q_1/T_1 + Q_2/T_2 \leq 0$, and that the equality holds when the cycle is reversible.
4. An ideal gas is taken reversibly round a cycle made of an isothermal expansion at T_1 from V_1 to V_2 , a cooling at constant volume

V_2 to T_2 , an isothermal compression at T_2 back to V_1 , and a heating at constant volume V_1 back to T_1 . Evaluate $\oint \delta Q/T$ directly and check that it is zero.

§2.6 Entropy

1. A body of constant heat capacity C_p at temperature T_1 is dropped into a large reservoir at T_2 . Find the entropy change of the body, of the reservoir, and of the two together, and show that the total is positive for any $T_1 \neq T_2$. You may use the inequality $\ln x \geq 1 - 1/x$.
2. Show that for an ideal gas $\Delta S = C_p \ln(T_2/T_1) - \nu R \ln(p_2/p_1)$, and check that it agrees with the expression in terms of T and V given in the text.
3. Let there be two identical bodies, each of constant heat capacity C , at temperatures T_1 and T_2 . Find the common final temperature if they are simply placed in contact, and find it again if instead a reversible engine is run between them until they reach a common temperature. Find the work obtained in the second case.
4. A quantity of ideal gas at temperature T and pressure p occupies a volume V , and an equal volume of a different ideal gas is at the same temperature and pressure alongside it. Find the entropy change when the partition between them is removed, and state what the answer becomes if the two gases are the same.

Thermodynamic Potentials

3.1 The Thermodynamic Potentials

The combined statement of the first and second laws, obtained in Section 2.6, is

$$dU = T dS - p dV. \quad (3.1)$$

When this equation is interpreted as a relation between differentials, it tells us which variables the internal energy is most naturally written in terms of. If we know U as a function of S and V , then comparing the equation above with the total differential of $U(S, V)$ from Section 2.2,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV, \quad (3.2)$$

gives the temperature and the pressure by differentiation:

$$T = \left(\frac{\partial U}{\partial S} \right)_V, \quad p = - \left(\frac{\partial U}{\partial V} \right)_S. \quad (3.3)$$

A single function $U(S, V)$ therefore contains the equation of state and the thermal properties of the substance together. We call S and

V the natural variables of U , and a function that carries this much information about a system is called a thermodynamic potential.

Why We Want Other Variables

The difficulty with $U(S, V)$ is experimental. Nobody has an instrument that measures the entropy of a gas, and few experiments are done in a rigid sealed container. What we control in a laboratory is the temperature, by putting the system in a bath, and the pressure, by leaving the system open to the atmosphere, or by loading a piston, or by pumping gas in or out of a closed vessel. We would like potentials whose natural variables are T and p , and we would like to get them without losing any of the information that $U(S, V)$ holds. In the sections that follow we construct three such functions, then give each a physical interpretation, and finally use the relations between them to compute quantities that cannot be measured directly.

The Legendre Transform

The mathematical operation that exchanges a variable for the slope with respect to that variable is called the Legendre transform. We met it in the notes on mechanics and dynamical systems, where it carried us from the Lagrangian to the Hamiltonian, although the sign convention used there is the opposite of the one that is customary in thermodynamics and that we adopt below. It is easiest to see for a function of one variable. Let $f(x)$ be a function whose slope

$$s = \frac{df}{dx} \tag{3.4}$$

increases steadily with x , so that each value of the slope belongs to exactly one point of the curve. We may then use s in place of x as the label of a point. Define

$$g(s) = f(x) - sx, \tag{3.5}$$

with x understood as the value at which the slope is s . Now we take the differential of both sides, using the product rule on the

term sx ,

$$dg = df - d(sx) = df - s dx - x ds. \quad (3.6)$$

The differential of f is $df = (df/dx) dx$, which is $s dx$ by the definition of s , so the first two terms on the right cancel and

$$dg = -x ds. \quad (3.7)$$

The differential of g contains ds and no dx , so g is a function of the slope alone, and comparing the equation above with $dg = (dg/ds) ds$,

$$x = -\frac{dg}{ds}. \quad (3.8)$$

Nothing has been lost in the exchange. Starting from $g(s)$ we recover x by the equation above, and the definition of g rearranges to $f = g + sx$, which returns the original function. The geometry behind the transform is shown in Figure 18; the tangent to the curve at the point x has slope s , so a point of that tangent at horizontal position X stands at the height $f + s(X - x)$, and putting $X = 0$ shows that the tangent meets the vertical axis at the height $f - sx$, which is $g(s)$. If we are given the height of the intercept for every slope, we have been given the whole family of tangent lines, and the curve is the envelope of that family.

As an example, consider the function $f(x) = \frac{1}{2}kx^2$, for which $s = kx$ and so $x = s/k$. Then

$$g(s) = \frac{1}{2}k \left(\frac{s}{k}\right)^2 - s \frac{s}{k} = -\frac{s^2}{2k}, \quad (3.9)$$

and $-dg/ds = s/k$, which is indeed x .

Enthalpy, Helmholtz, and Gibbs

We now apply the transform to $U(S, V)$, first in V , then in S , and finally in both. The recipe of the previous subsection is to subtract

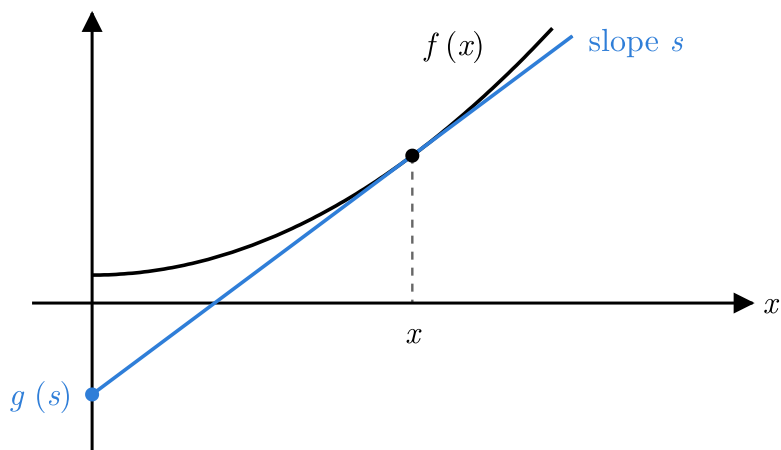


Figure 18. The Legendre transform of a function $f(x)$. The tangent drawn at the point x has slope $s = df/dx$ and cuts the vertical axis at $f - sx$, which is the transformed function $g(s)$. For the tangent drawn here $g(s)$ is negative, so the intercept lies below the origin. Giving the intercept for every slope specifies the curve $f(x)$ just as completely as giving f at every x .

from the function the variable we are removing multiplied by the slope with respect to that variable. The slope of U with respect to V is $(\partial U/\partial V)_S = -p$, so exchanging the volume for the pressure gives

$$U - \left(\frac{\partial U}{\partial V} \right)_S V = U - (-p)V = U + pV, \quad (3.10)$$

and the slope with respect to S is $(\partial U/\partial S)_V = T$, so exchanging the entropy for the temperature gives $U - TS$. Carrying out the two exchanges together both adds pV and subtracts TS . We call these three transformed functions H , F , and G :

$$H = U + pV, \quad F = U - TS, \quad G = U - TS + pV. \quad (3.11)$$

These are called the enthalpy, the Helmholtz function, and the Gibbs function. To find the differential of each we differentiate the defining expression and then use $dU = T dS - p dV$ to remove dU . For the enthalpy,

$$\begin{aligned} dH &= dU + p dV + V dp \\ &= (T dS - p dV) + p dV + V dp = T dS + V dp, \end{aligned} \quad (3.12)$$

where the two terms in dV cancel, as the transform was arranged to make them do. For the Helmholtz function,

$$\begin{aligned} dF &= dU - T dS - S dT \\ &= (T dS - p dV) - T dS - S dT = -S dT - p dV, \end{aligned} \quad (3.13)$$

and for the Gibbs function the terms in dS and the terms in dV cancel together,

$$dG = dU - T dS - S dT + p dV + V dp = -S dT + V dp. \quad (3.14)$$

The natural variables are (S, p) for the enthalpy, (T, V) for the Helmholtz function, and (T, p) for the Gibbs function, and the four potentials are laid out with their transforms in Figure 19. Since the natural variables of the Gibbs function are the temperature and the pressure, which are the quantities we can control in the laboratory, it is the potential a chemist might reach for first.

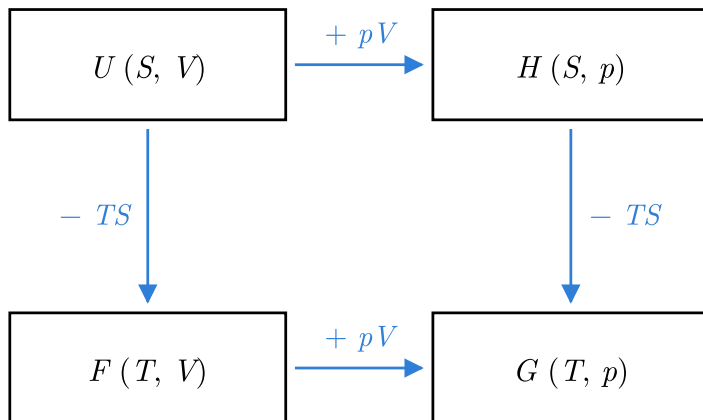


Figure 19. The four thermodynamic potentials with their natural variables in brackets. Moving to the right adds pV and exchanges the volume for the pressure; moving downwards subtracts TS and exchanges the entropy for the temperature.

The Enthalpy and the Heat at Constant Pressure

Each of the three new functions has a direct physical interpretation, and we take them in turn. We write δQ for a small quantity of heat and δW for a small quantity of work done on the system, keeping the symbol d for the differential of a function of state, as in Section 2.2.

Suppose a system is held at a fixed pressure p and does no work other than pushing its boundary, so that $\delta W = -p dV$. The first law gives $dU = \delta Q - p dV$, and since p is constant,

$$dH = dU + p dV = \delta Q. \quad (3.15)$$

The change in enthalpy is the heat taken in during a change at constant pressure. This is why the heat evolved by a reaction in an open vessel is quoted as an enthalpy change, and why the heat capacity at constant pressure is

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p, \quad (3.16)$$

which is the counterpart of $C_V = (\partial U / \partial T)_V$ from Section 2.2.

Work at Constant Temperature

Now let a system be in contact with a reservoir at temperature T . The Clausius inequality of Section 2.5 says that the heat it takes in during a small change satisfies $\delta Q \leq T \, dS$, with equality when the change is reversible. Combining this inequality with the first law,

$$\delta W = dU - \delta Q \geq dU - T \, dS. \quad (3.17)$$

The work done by the system is $-\delta W$, so

$$-\delta W \leq T \, dS - dU = -d(U - TS) = -dF, \quad (3.18)$$

where the temperature was taken through the differential because it is held constant by the reservoir. Integrating over a change from one state to another at the same temperature, the work done by the system satisfies

$$W_{\text{by}} \leq F_i - F_f = -\Delta F. \quad (3.19)$$

The decrease in the Helmholtz function is the largest amount of work the system can do at constant temperature, as when a gas held in a water bath expands against a piston, and a reversible change delivers exactly that amount. The internal energy of the system may fall by more than the work it does, and the remainder of the fall is heat given to the reservoir.

Equilibrium at Constant Temperature and Pressure

Suppose instead that the system sits in a reservoir at temperature T and is surrounded by an atmosphere at pressure p , which does work $-p dV$ on it as it changes volume. Any further work done on the system, such as the work of an electric current driving a chemical cell, we call $\delta W'$. Putting $\delta W = -p dV + \delta W'$ into the inequality $\delta W \geq dU - T dS$ found in the previous subsection,

$$\delta W' \geq dU - T dS + p dV = d(U - TS + pV) = dG, \quad (3.20)$$

again using the constancy of T and p . If the system does no work beyond pushing the atmosphere aside, then $\delta W' = 0$ and

$$dG \leq 0. \quad (3.21)$$

A system held at constant temperature and pressure can only move in a direction that lowers its Gibbs function, and it comes to rest when G has reached its smallest available value. The same argument with the volume held fixed instead of the pressure gives $dF \leq 0$, so the Helmholtz function plays the same part for a system in a rigid container. The statement that G falls at constant temperature and pressure, and the statement that F falls at constant temperature and volume, are how the second law is most often used in practice, since a chemist working at the bench controls T and p and never the entropy.

3.2 Maxwell Relations and Their Uses

Each of the four potentials of Section 3.1 has a differential of the form $d\Phi = X dx + Y dy$, where x and y are its natural variables. Comparing this with the total differential of $\Phi(x, y)$ from Section 2.2 identifies the coefficients as

$$X = \left(\frac{\partial \Phi}{\partial x} \right)_y, \quad Y = \left(\frac{\partial \Phi}{\partial y} \right)_x, \quad (3.22)$$

and differentiating X with respect to y and Y with respect to x gives the second derivative of Φ taken in the two possible orders,

$$\left(\frac{\partial X}{\partial y}\right)_x = \frac{\partial^2 \Phi}{\partial y \partial x}, \quad \left(\frac{\partial Y}{\partial x}\right)_y = \frac{\partial^2 \Phi}{\partial x \partial y}. \quad (3.23)$$

Section 2.2 showed that the order in which two partial derivatives are taken makes no difference to the result, so the two right-hand sides are equal, and therefore

$$\left(\frac{\partial X}{\partial y}\right)_x = \left(\frac{\partial Y}{\partial x}\right)_y. \quad (3.24)$$

Each of the four potentials is a function of state, so each is a well-defined function of the two natural variables belonging to it, and the equation above therefore holds for U , for H , for F , and for G alike.

The Maxwell Relations

For a system whose only work is done by a change of volume there are four relations of this kind, and we write them down in the order of the potentials. To see how one of them is obtained, we take $dU = T dS - p dV$, so that $x = S$ and $y = V$ while $X = T$ and $Y = -p$. Substituting these four identifications into the condition on the mixed second derivatives gives

$$\left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{\partial(-p)}{\partial S}\right)_V = -\left(\frac{\partial p}{\partial S}\right)_V, \quad (3.25)$$

and reading off the coefficients of $dH = T dS + V dp$ in the same way, with $x = S$, $y = p$, $X = T$, and $Y = V$,

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p. \quad (3.26)$$

The remaining two come from $dF = -S dT - p dV$, where $X = -S$ and $Y = -p$ so that the two minus signs cancel, and from $dG =$

$-S dT + V dp$, where $X = -S$ and $Y = V$ so that one minus sign is left over:

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V, \quad \left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p. \quad (3.27)$$

These are called the Maxwell relations. The relations that come from F and from G are the ones used most, because the left-hand side of each contains the entropy, which no instrument measures, while the right-hand side contains only p , V , and T , which an ordinary laboratory can measure. The relations therefore let us express quantities that cannot be measured directly in terms of quantities that can.

Expansivity and Compressibility

Among these measurable quantities there are a couple that appear so often that they are given names. The isobaric expansivity is the fractional change of volume per unit rise in temperature at fixed pressure,

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_p, \quad (3.28)$$

and the isothermal compressibility is the fractional decrease in volume per unit rise in pressure at fixed temperature,

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T. \quad (3.29)$$

The minus sign in κ_T makes it positive, since every substance in stable equilibrium shrinks when squeezed. For an ideal gas, differentiating $V = Nk_B T/p$ at fixed pressure gives $(\partial V/\partial T)_p = Nk_B/p = V/T$ and so $\alpha = 1/T$, while differentiating it at fixed temperature gives $(\partial V/\partial p)_T = -Nk_B T/p^2 = -V/p$ and so $\kappa_T = 1/p$.

We shall also need the rate of change of pressure with temperature at fixed volume, which is awkward to measure because a solid or a liquid held at constant volume develops enormous pressures. We

can get it from α and κ_T instead. Writing the total differential of $V(T, p)$,

$$dV = \left(\frac{\partial V}{\partial T} \right)_p dT + \left(\frac{\partial V}{\partial p} \right)_T dp, \quad (3.30)$$

and setting $dV = 0$ to hold the volume fixed, we can solve for the ratio dp/dT along that path:

$$\left(\frac{\partial p}{\partial T} \right)_V = - \frac{(\partial V / \partial T)_p}{(\partial V / \partial p)_T} = - \frac{V\alpha}{-V\kappa_T} = \frac{\alpha}{\kappa_T}. \quad (3.31)$$

The Energy Equation

Let us consider how the internal energy of a substance changes when we expand it at fixed temperature. Dividing the combined law $dU = T dS - p dV$ through by dV along a path on which the temperature is held fixed,

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial S}{\partial V} \right)_T - p, \quad (3.32)$$

and the Maxwell relation from the Helmholtz function turns the entropy derivative into a pressure derivative:

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial p}{\partial T} \right)_V - p. \quad (3.33)$$

This is called the energy equation, and the right-hand side is known as soon as the equation of state is known. Let us apply it to an ideal gas. With $p = Nk_B T/V$ we have $(\partial p / \partial T)_V = Nk_B/V = p/T$, so

$$\left(\frac{\partial U}{\partial V} \right)_T = T \frac{p}{T} - p = 0. \quad (3.34)$$

The internal energy of an ideal gas depends on its temperature alone, whatever its volume. In Section 2.2 that independence of the volume was taken from experiment, following Joule's observation

that a gas expanding freely into a vacuum changes its temperature very little. Here the same statement comes out of the equation of state and the second law together.

The Difference of the Heat Capacities

We know from Section 2.2 that C_p exceeds C_V , because a substance heated at constant pressure also expands and does work on its surroundings. The general size of the difference follows from the Maxwell relations. Writing both heat capacities in terms of the entropy, using $\delta Q = T dS$ for a reversible change,

$$C_V = T \left(\frac{\partial S}{\partial T} \right)_V, \quad C_p = T \left(\frac{\partial S}{\partial T} \right)_p. \quad (3.35)$$

Let us take S as a function of T and V , so that its total differential is

$$dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV, \quad (3.36)$$

and let the substance follow a path of constant pressure, along which the volume changes with the temperature at the rate $(\partial V / \partial T)_p$. Dividing the total differential by dT along that path,

$$\left(\frac{\partial S}{\partial T} \right)_p = \left(\frac{\partial S}{\partial T} \right)_V + \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_p, \quad (3.37)$$

so that, multiplying through by T and subtracting,

$$C_p - C_V = T \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_p = T \left(\frac{\partial p}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_p, \quad (3.38)$$

where the Maxwell relation from the Helmholtz function was used again. Writing the derivative of p with respect to T as α / κ_T and the derivative of V with respect to T as $V\alpha$, both from the subsection on the expansivity and the compressibility,

$$C_p - C_V = T \frac{\alpha}{\kappa_T} V \alpha = \frac{TV\alpha^2}{\kappa_T}. \quad (3.39)$$

Everything on the right is measurable, and since κ_T is positive and α appears squared, the difference is never negative. It vanishes where the expansivity vanishes, as it does for water near four degrees above the ice point, and there C_p and C_V are equal. For an ideal gas, putting $\alpha = 1/T$ and $\kappa_T = 1/p$ into the formula gives $C_p - C_V = TV(1/T^2)p = pV/T = Nk_B$, which is the result obtained for an ideal gas in the problems of Section 3.1.

Entropy from Measurable Quantities

The entropy itself can now be found from measurements. Let us take S as a function of T and p , whose total differential is

$$dS = \left(\frac{\partial S}{\partial T} \right)_p dT + \left(\frac{\partial S}{\partial p} \right)_T dp. \quad (3.40)$$

The first coefficient is C_p/T , from the expression for C_p written down above. The second is $-(\partial V/\partial T)_p$ by the Maxwell relation from the Gibbs function, and that derivative is $V\alpha$ by the definition of the expansivity, so the second coefficient is $-V\alpha$ and

$$dS = \frac{C_p}{T} dT - V\alpha dp. \quad (3.41)$$

Since S is a function of state, we may integrate this expression along any convenient path between two states. The path drawn in Figure 20 runs first along the isobar $p = p_0$, where only the term in dT contributes, and then along the isotherm at the final temperature, where only the term in dp contributes:

$$S(T, p) = S(T_0, p_0) + \int_{T_0}^T \frac{C_p(T', p_0)}{T'} dT' - \int_{p_0}^p V(T, p') \alpha(T, p') dp'. \quad (3.42)$$

A calorimeter supplies C_p along the first leg and a measurement of thermal expansion supplies $V\alpha$ along the second leg, so the entropy of a substance relative to a reference state is an experimental quantity.

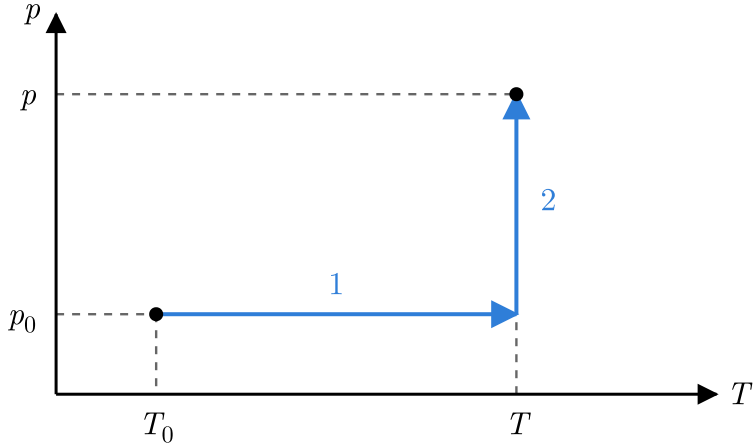


Figure 20. A path in the (T, p) plane used to find the entropy of a substance from measurements. Along leg 1 the pressure is held at p_0 and the heat capacity C_p is integrated; along leg 2 the temperature is held at T and the product $V\alpha$ is integrated. Since S is a function of state, any other path between the same two points gives the same answer.

The Adiabatic Stretching of a Rod

The method developed in the previous sections, a Legendre transform followed by a Maxwell relation, applies equally well to systems in which the work is done by something other than a change of volume. As an example, consider a thin rod of length L held under a tension f . Stretching it by dL does work $f dL$ on the rod, so the combined law reads $dU = T dS + f dL$, with f in the place of $-p$ and L in the place of V . The Legendre transform that exchanges the entropy for the temperature and the length for the tension is $\Phi = U - TS - fL$, whose differential is

$$d\Phi = dU - T dS - S dT - f dL - L df = -S dT - L df, \quad (3.43)$$

so its natural variables are the temperature and the tension. The coefficient of dT is $-S$ and the coefficient of df is $-L$, so the condition on the mixed second derivatives reads $(\partial(-S)/\partial f)_T = (\partial(-L)/\partial T)_f$, and cancelling the minus signs,

$$\left(\frac{\partial S}{\partial f}\right)_T = \left(\frac{\partial L}{\partial T}\right)_f. \quad (3.44)$$

Now suppose we stretch the rod fast enough that no heat has time to enter or leave it, and gently enough for the change to be reversible, so that its entropy stays fixed. To find how its temperature responds we write the total differential of $S(T, f)$,

$$dS = \left(\frac{\partial S}{\partial T}\right)_f dT + \left(\frac{\partial S}{\partial f}\right)_T df, \quad (3.45)$$

and set it to zero, since the entropy does not change. Rearranging what is left and then using the Maxwell relation for the rod obtained above,

$$\left(\frac{\partial T}{\partial f}\right)_S = -\frac{(\partial S/\partial f)_T}{(\partial S/\partial T)_f} = -\frac{T}{C_f} \left(\frac{\partial L}{\partial T}\right)_f, \quad (3.46)$$

where $C_f = T(\partial S/\partial T)_f$ is the heat capacity of the rod at fixed tension. A metal rod lengthens when it is heated at fixed tension, so $(\partial L/\partial T)_f$ is positive and the temperature falls as the tension is raised. A rubber band behaves the other way round, contracting when warmed at fixed tension, and it therefore warms when stretched, which is easily confirmed.

3.3 The Chemical Potential

Every relation so far has been written for a system with a fixed number of particles. Many systems of interest do not keep their number of particles fixed; a liquid in contact with its vapour passes molecules across the surface, a gas leaks through a porous plug into a second vessel, and a reacting mixture destroys molecules of one kind and makes molecules of another. To describe such systems we have to let N vary and ask what accompanies a change in it.

Adding a Particle

Let us take a system whose state is fixed by its entropy, its volume, and the number of particles it contains, so that $U = U(S, V, N)$. The total differential of this function has three terms,

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,N} dS + \left(\frac{\partial U}{\partial V}\right)_{S,N} dV + \left(\frac{\partial U}{\partial N}\right)_{S,V} dN. \quad (3.47)$$

The first two coefficients are T and $-p$ as before, since nothing in the argument of Section 3.1 assumed that N could change. We give the third one the symbol μ and call it the chemical potential,

$$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}, \quad (3.48)$$

so that the combined law becomes

$$dU = T dS - p dV + \mu dN. \quad (3.49)$$

In words, μ is the energy that must be supplied to add one particle to the system while holding its entropy and volume fixed. Adding the term μdN to the combined law adds it unchanged to the differential of each of the other three potentials, since the Legendre transforms of Section 3.1 touch only the pairs (T, S) and (p, V) . In particular

$$dG = -S dT + V dp + \mu dN, \quad \mu = \left(\frac{\partial G}{\partial N}\right)_{T,p}. \quad (3.50)$$

The Gibbs Function per Particle

For a system made of a single substance, the chemical potential has a simple interpretation. Let us hold the temperature and the pressure fixed and imagine doubling the amount of material, by placing two identical systems side by side and removing the partition between them. The temperature and the pressure are unchanged, because both are intensive, and the Gibbs function of the pair is

twice that of one system, because U , S , and V each double. The same argument applies for any factor in place of two, so at fixed T and p the Gibbs function is proportional to N :

$$G(T, p, N) = N \phi(T, p), \quad (3.51)$$

where ϕ depends on the temperature and the pressure alone. Differentiating with respect to N at fixed T and p ,

$$\mu = \left(\frac{\partial G}{\partial N} \right)_{T, p} = \phi(T, p), \quad (3.52)$$

so the chemical potential of a pure substance is its Gibbs function per particle, $\mu = G/N$. Nothing of the kind holds for the other potentials; $(\partial U / \partial N)_{S, V}$ is not U/N , because holding S and V fixed while adding particles is not the same as scaling the system up.

Equilibrium between Systems in Contact

Let two systems, labelled 1 and 2, be joined so that they may exchange energy, volume, and particles, and let the pair be isolated, as in Figure 21. The totals $U_1 + U_2$, $V_1 + V_2$, and $N_1 + N_2$ are then fixed, so that $dU_2 = -dU_1$, $dV_2 = -dV_1$, and $dN_2 = -dN_1$. Solving the combined law for dS gives

$$dS = \frac{1}{T} dU + \frac{p}{T} dV - \frac{\mu}{T} dN. \quad (3.53)$$

Adding the entropy changes of the two systems, and using the three relations between their changes to write everything in terms of the changes in system 1,

$$dS_{\text{tot}} = \left(\frac{1}{T_1} - \frac{1}{T_2} \right) dU_1 + \left(\frac{p_1}{T_1} - \frac{p_2}{T_2} \right) dV_1 - \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2} \right) dN_1. \quad (3.54)$$

The second law of Section 2.5 says that an isolated system reaches equilibrium when its entropy is as large as it can be, and at a maximum the entropy is stationary against every allowed change. The

three quantities dU_1 , dV_1 , and dN_1 can be varied independently, so each bracket must vanish separately, by the argument that carried us from Hamilton's principle to Lagrange's equations in the mechanics notes, and therefore

$$T_1 = T_2, \quad p_1 = p_2, \quad \mu_1 = \mu_2. \quad (3.55)$$

Equality of temperature is thermal equilibrium and equality of pressure is mechanical equilibrium, both of them familiar from Section 2.5. Equality of chemical potential is the condition for equilibrium with respect to the exchange of particles, as it is between a liquid and the vapour above it in a sealed flask, and it is the reason μ was introduced.

Which Way the Particles Move

Away from equilibrium the entropy is still rising, and the sign of each term in the expression for dS_{tot} tells us which way the corresponding exchange goes. Suppose the two systems have already come to the same temperature T and the same pressure, and only particles can still move. Then

$$dS_{\text{tot}} = -\frac{\mu_1 - \mu_2}{T} dN_1 \geq 0. \quad (3.56)$$

If $\mu_1 > \mu_2$, the entropy rises only if dN_1 is negative. Particles leave the system with the higher chemical potential and arrive in the system with the lower one, and they go on doing so until the two chemical potentials are equal. The name is chosen by analogy with the gravitational potential, in the sense that matter runs downhill in μ just as a ball runs downhill in height.

The Chemical Potential of an Ideal Gas

For a pure substance we may divide the differential of G at fixed N by the number of particles, since $\mu = G/N$, and obtain

$$d\mu = -s dT + v dp, \quad (3.57)$$

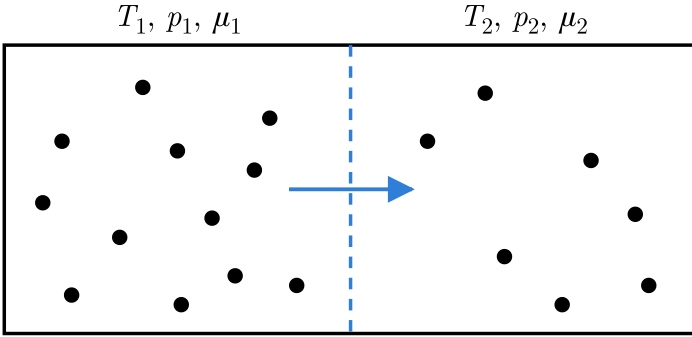


Figure 21. A pair of systems separated by a wall through which particles can pass. When the temperatures and pressures are equal, particles cross from the system with the larger chemical potential to the system with the smaller one, and the flow stops when $\mu_1 = \mu_2$.

where $s = S/N$ and $v = V/N$ are the entropy and the volume per particle. Reading off the coefficients of dp and of dT gives the following derivatives:

$$\left(\frac{\partial\mu}{\partial p}\right)_T = \frac{V}{N}, \quad \left(\frac{\partial\mu}{\partial T}\right)_p = -\frac{S}{N}. \quad (3.58)$$

For an ideal gas $V/N = k_B T/p$, so at a fixed temperature $d\mu = (k_B T/p) dp$. Integrating both sides from some reference pressure p_0 up to the pressure p , and writing $\mu_0(T)$ for the value of the chemical potential at p_0 ,

$$\mu(T, p) - \mu_0(T) = k_B T \int_{p_0}^p \frac{dp'}{p'} = k_B T [\ln p']_{p_0}^p, \quad (3.59)$$

so that

$$\mu(T, p) = \mu_0(T) + k_B T \ln \frac{p}{p_0}. \quad (3.60)$$

The quantity $\mu_0(T)$ is a function of temperature that thermodynamics alone leaves undetermined and that has to be measured.

The shape of the curve is drawn in Figure 22. Compressing a gas raises its chemical potential, so a gas pushed to a high pressure in one vessel drives molecules into any vessel held at a lower pressure, which is what we expect. The rise is logarithmic and therefore slow, and as the pressure falls towards zero the chemical potential falls without limit, which is why every substance gives off molecules into a space in which its vapour is very dilute. The derivative of μ with respect to temperature shows that μ falls as the temperature rises at fixed pressure, since the entropy per particle is positive.

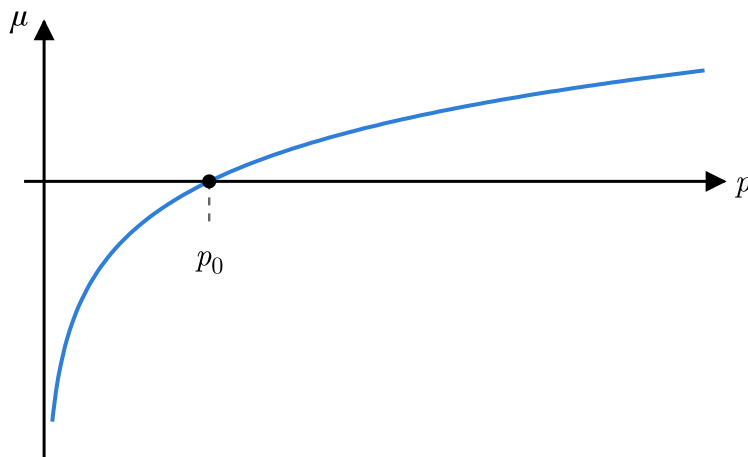


Figure 22. The chemical potential of an ideal gas against pressure at fixed temperature, drawn with the reference value μ_0 taken as the zero of the vertical scale. The curve is $k_B T \ln(p/p_0)$, rising slowly at high pressure and falling steeply towards $-\infty$ as $p \rightarrow 0$.

3.4 Chemical Equilibrium

A mixture of gases that can react is a system in which the numbers of particles of each kind change while the total amount of material stays the same. Section 3.3 gave the condition for equilibrium when particles move from one place to another, and the same reasoning applied to particles changing from one kind into another gives the condition for a reaction to be at equilibrium.

Writing a Reaction

A reaction is written as a relation among the chemical species X_i taking part,

$$\sum_i \nu_i X_i = 0, \quad (3.61)$$

where the integers ν_i are called the stoichiometric coefficients, counted positive for a species produced by the reaction and negative for a species consumed by it. The dissociation of a diatomic molecule into its two atoms, usually written $A_2 \rightleftharpoons 2A$, becomes $2A - A_2 = 0$ in this notation, so that $\nu_A = 2$ and $\nu_{A_2} = -1$.

The convention has the advantage that the changes in the particle numbers are all proportional to one another. If the reaction advances a little, the numbers change by

$$dN_i = \nu_i d\xi, \quad (3.62)$$

where the single variable ξ measures how far the reaction has gone and is called the extent of the reaction. The coefficients record that two atoms of A appear for every molecule of A_2 destroyed.

The Condition for Equilibrium

Let the mixture be held at constant temperature and pressure. Section 3.1 showed that it then moves in whatever direction lowers its Gibbs function, and comes to rest at the minimum. The differential of G for a mixture is

$$dG = -S dT + V dp + \sum_i \mu_i dN_i, \quad (3.63)$$

with one chemical potential for each species, defined as $\mu_i = (\partial G / \partial N_i)_{T,p,N_{j \neq i}}$. At fixed T and p the first two terms vanish, and substituting $dN_i = \nu_i d\xi$,

$$\left(\frac{\partial G}{\partial \xi} \right)_{T,p} = \sum_i \nu_i \mu_i. \quad (3.64)$$

The Gibbs function is stationary when this sum vanishes, so the condition for chemical equilibrium is

$$\sum_i \nu_i \mu_i = 0. \quad (3.65)$$

Away from equilibrium the sign of the sum tells us which way the reaction runs. If $\sum_i \nu_i \mu_i$ is negative then G falls as ξ increases, and the reaction proceeds from left to right; if it is positive the reaction runs backwards. The shape of $G(\xi)$ is drawn in Figure 23.

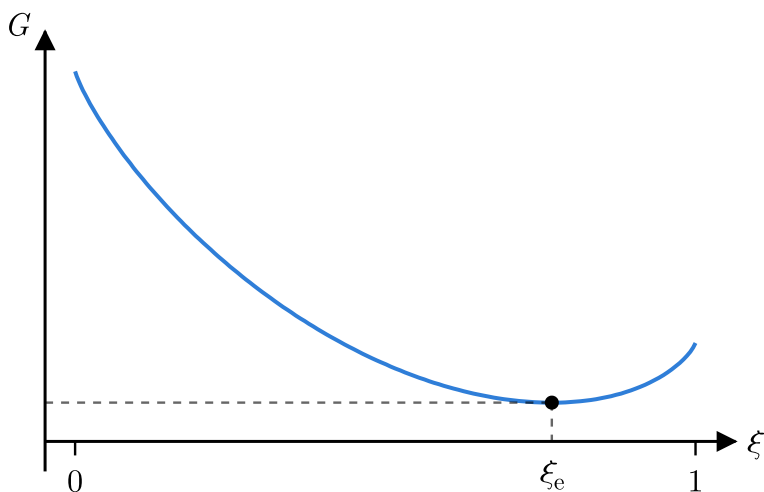


Figure 23. The Gibbs function of a reacting mixture against the extent of the reaction ξ , at fixed temperature and pressure, with the extent scaled so that $\xi = 0$ for pure reactants and $\xi = 1$ for complete conversion. The slope is $\sum_i \nu_i \mu_i$, and the mixture comes to rest at the minimum ξ_e , where the slope vanishes. The slope is $-\infty$ at $\xi = 0$ and $+\infty$ at $\xi = 1$, because the chemical potential of a species present in a vanishing amount tends to $-\infty$.

The Law of Mass Action

To turn the equilibrium condition into a relation between the amounts of the species present, we need their chemical potentials, and for a mixture of ideal gases they are the ones found in Section

3.3. The molecules of an ideal gas do not interact, so each species behaves as though the others were absent, exerting the partial pressure $p_i = (N_i/N)p$ it would exert alone in the same volume at the same temperature, where N is the total number of molecules. Its chemical potential is therefore

$$\mu_i = \mu_{0i}(T) + k_B T \ln \frac{p_i}{p_0}. \quad (3.66)$$

Substituting this into $\sum_i \nu_i \mu_i = 0$ and separating the terms,

$$k_B T \sum_i \nu_i \ln \frac{p_i}{p_0} = - \sum_i \nu_i \mu_{0i}(T). \quad (3.67)$$

A coefficient in front of a logarithm can be taken inside it as a power, and a sum of logarithms is the logarithm of a product, so the left-hand side may be rewritten as

$$k_B T \sum_i \ln \left(\frac{p_i}{p_0} \right)^{\nu_i} = k_B T \ln \prod_i \left(\frac{p_i}{p_0} \right)^{\nu_i}. \quad (3.68)$$

Dividing through by $k_B T$ and taking the exponential of both sides,

$$\prod_i \left(\frac{p_i}{p_0} \right)^{\nu_i} = K(T), \quad K(T) = \exp \left(- \frac{\sum_i \nu_i \mu_{0i}(T)}{k_B T} \right). \quad (3.69)$$

This is called the law of mass action, and $K(T)$ the equilibrium constant. The partial pressures at equilibrium may take any values, provided the particular combination on the left has the value K , and that value depends on the temperature and on nothing else. The pressure and the composition of the mixture do not enter K , although they certainly enter the individual partial pressures.

How the Equilibrium Constant Depends on Temperature

Write $\Delta\phi = \sum_i \nu_i \mu_{0i}$ for the change in Gibbs function per reaction event with every species at the reference pressure, so that

$\ln K = -\Delta\phi/k_B T$. To differentiate $\ln K$ with respect to temperature we need the temperature dependence of μ_{0i}/T for each species. Dividing $G = H - TS$, which follows from the definitions of G and H in Section 3.1, by the number of particles gives $\mu = h - Ts$, where h and s are the enthalpy and the entropy per particle, and Section 3.3 gave $(\partial\mu/\partial T)_p = -s$. Differentiating the quotient μ/T by the quotient rule, then substituting $-s$ for $(\partial\mu/\partial T)_p$ and $h - Ts$ for μ ,

$$\frac{\partial}{\partial T} \left(\frac{\mu}{T} \right)_p = \frac{1}{T} \left(\frac{\partial\mu}{\partial T} \right)_p - \frac{\mu}{T^2} = -\frac{s}{T} - \frac{h - Ts}{T^2} = -\frac{h}{T^2}. \quad (3.70)$$

Applying this result to each species in the sum,

$$\frac{d \ln K}{dT} = -\frac{1}{k_B} \sum_i \nu_i \frac{d}{dT} \left(\frac{\mu_{0i}}{T} \right) = \frac{\sum_i \nu_i h_{0i}}{k_B T^2} = \frac{\Delta h}{k_B T^2}. \quad (3.71)$$

This is called the van 't Hoff relation, and $\Delta h = \sum_i \nu_i h_{0i}$ the enthalpy of reaction, which by the interpretation of the enthalpy given in Section 3.1 is the heat absorbed when the reaction advances by one event at constant pressure. A reaction that absorbs heat has $\Delta h > 0$, and its equilibrium constant grows as the temperature rises. A reaction that gives out heat has $\Delta h < 0$, and its equilibrium constant falls as the temperature rises.

The Dissociation of a Diatomic Molecule

Let us work an example. Consider a vessel containing N_0 molecules of the diatomic gas A_2 , held at temperature T and total pressure p , and let a fraction α of them dissociate into atoms. The numbers present at equilibrium are $N_0(1 - \alpha)$ molecules and $2N_0\alpha$ atoms, so the total is $N_0(1 + \alpha)$ and the partial pressures are

$$p_{A_2} = \frac{1 - \alpha}{1 + \alpha} p, \quad p_A = \frac{2\alpha}{1 + \alpha} p. \quad (3.72)$$

With $\nu_A = 2$ and $\nu_{A_2} = -1$, the law of mass action reads

$$\frac{(p_A/p_0)^2}{p_{A_2}/p_0} = K(T), \quad (3.73)$$

and substituting the two partial pressures, with the numerator divided by $(1 + \alpha)^2$ and the denominator by $(1 + \alpha)$,

$$\frac{4\alpha^2 p^2 / (1 + \alpha)^2}{p_0 p (1 - \alpha) / (1 + \alpha)} = \frac{4\alpha^2}{(1 + \alpha)(1 - \alpha)} \frac{p}{p_0} = \frac{4\alpha^2}{1 - \alpha^2} \frac{p}{p_0} = K(T). \quad (3.74)$$

To find the degree of dissociation we invert both sides and rearrange. Inverting gives $(1 - \alpha^2)p_0/4\alpha^2 p = 1/K$, and multiplying out and collecting the terms in α^{-2} ,

$$\frac{1}{\alpha^2} - 1 = \frac{4p}{p_0 K(T)}, \quad \text{so} \quad \alpha = \left(1 + \frac{4p}{p_0 K(T)}\right)^{-1/2}, \quad (3.75)$$

which is drawn against pressure in Figure 24. At a fixed temperature the dissociation is almost complete at low enough pressure, however small the equilibrium constant, and it is driven back towards whole molecules as the pressure is raised.

Le Chatelier's Principle

The behaviour of the dissociating gas is an instance of a general statement about how an equilibrium responds to a change in its conditions, known as Le Chatelier's principle: a system in equilibrium responds to a change imposed on it by moving in the direction that opposes the change. The response to a change of pressure and the response to a change of temperature can both be read off from the results above. Raising the pressure drove the dissociation back towards A_2 , which is the side of the reaction with fewer molecules, and a mixture with fewer molecules in the same volume exerts a smaller pressure, so the shift opposes the rise in pressure that caused it. In general, since $\prod_i (p_i/p_0)^{\nu_i}$ contains the total pressure raised to the power $\sum_i \nu_i$, raising p at fixed temperature shifts the equilibrium towards whichever side has the smaller number of molecules.

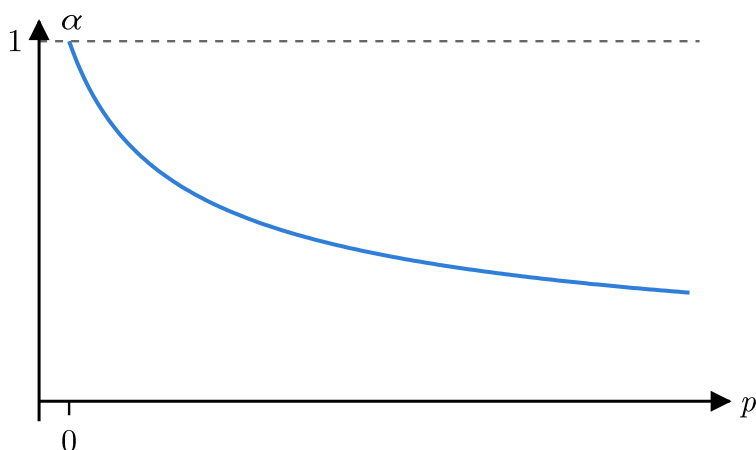


Figure 24. The degree of dissociation α of a diatomic gas against the total pressure at fixed temperature, from $\alpha = (1 + 4p/p_0K)^{-1/2}$. The degree of dissociation approaches unity as the pressure goes to zero, and at high pressure it goes as $p^{-1/2}$.

Raising the temperature has the effect given by the van 't Hoff relation. Breaking a chemical bond absorbs heat, so dissociation has $\Delta h > 0$ and K increases with temperature, and a hotter gas is more dissociated. The reaction moves in the direction that takes heat in, which tends to hold the temperature down. The principle is a consequence of the stability of the equilibrium, since the mixture sits at a minimum of the Gibbs function, where the curvature is positive, and the sign of that curvature fixes the direction in which the equilibrium moves when the conditions are changed. The general proof of the principle from the conditions of thermodynamic stability is not given here, and the statement about pressure and the statement about temperature were each obtained from the equilibrium condition and the sign of a derivative.

Problems

§3.1 The Thermodynamic Potentials

1. Find the Legendre transform $g(s)$ of $f(x) = e^x$, and verify that $-dg/ds$ returns x .

2. Show that $C_V = (\partial U/\partial T)_V$ and $C_p = (\partial H/\partial T)_p$ both follow from $C = \delta Q/dT$ under the appropriate constraint. Hence show that for an ideal gas $C_p - C_V = Nk_B$.
3. Starting from $F = U - TS$ and $S = -(\partial F/\partial T)_V$, show that $U = F - T(\partial F/\partial T)_V$, and obtain the corresponding expression for H in terms of G .
4. A system is held at constant temperature T and constant volume. Show that the total entropy of the system and the reservoir together increases by $-\Delta F/T$, and hence that F falling is the same statement as the total entropy rising.

§3.2 Maxwell Relations and Their Uses

1. Show that the temperature change of any substance under a reversible adiabatic change of pressure is given by $(\partial T/\partial p)_S = TV\alpha/C_p$. Apply this to an ideal gas to show that $T \propto p^{(\gamma-1)/\gamma}$ along such a path, where $\gamma = C_p/C_V$.
2. A gas obeys the equation of state $p = \frac{Nk_B T}{V - Nb} - \frac{aN^2}{V^2}$, where a and b are positive constants. Find $(\partial U/\partial V)_T$, and say what happens to the temperature of such a gas when it expands freely into a vacuum.
3. Show that $(\partial C_V/\partial V)_T = T(\partial^2 p/\partial T^2)_V$, and deduce that the heat capacity at constant volume of an ideal gas does not depend on the volume.
4. A block of solid is compressed reversibly and isothermally from p_0 to p . Show that the heat it gives out is $T \int_{p_0}^p V\alpha dp'$, and say which way the heat flows for a substance that expands on heating.

§3.3 The Chemical Potential

1. Show that $\mu = (\partial F/\partial N)_{T,V}$ and that $\mu = -T(\partial S/\partial N)_{U,V}$.
2. Let two vessels of the same ideal gas be held at the same temperature and joined by a narrow tube, the first at pressure p_1

and the second at pressure p_2 , with $p_1 > p_2$. Using the chemical potential, show that gas flows from the first vessel to the second, and that the flow stops when the pressures are equal.

3. A column of ideal gas stands in a uniform gravitational field at a single temperature T . The energy of a molecule at height z includes its potential energy mgz , so the chemical potential at that height is $\mu(T, p(z)) + mgz$, and this total must be the same at every height. Deduce that $p(z) = p(0) e^{-mgz/k_B T}$.
4. Starting from $G = \mu N$ and the differential of G , show that $S dT - V dp + N d\mu = 0$ for a pure substance.

§3.4 Chemical Equilibrium

1. A reaction has $\sum_i \nu_i = 0$, as in $A + B \rightleftharpoons C + D$. Show that the composition of the equilibrium mixture of ideal gases does not depend on the total pressure.
2. Assume the enthalpy of reaction Δh does not vary with temperature over the range of interest. Integrate the van 't Hoff relation to find $\ln[K(T_2)/K(T_1)]$, and check that K increases with temperature for a reaction that absorbs heat.
3. Show that as the extent of reaction ξ tends to zero, with only reactants present, the slope $\sum_i \nu_i \mu_i$ tends to $-\infty$. What does this say about whether a reaction that is thermodynamically allowed ever fails to start?
4. For the dissociation worked out in the text, show that at fixed pressure $\frac{d \ln \alpha}{dT} = \frac{1 - \alpha^2}{2} \frac{\Delta h}{k_B T^2}$, and comment on the sign for a dissociation.

Statistical Foundations and the Equilibrium Ensembles

4.1 Probability for Many Particles

Everything in Chapters 2 and 3 was built on quantities that can be measured with a thermometer, a pressure gauge, and a calorimeter. Nothing in those parts required us to know that matter is made of atoms and molecules. Chapter 1 did use those constituents, and it used two general principles that were stated without proof: that the speeds of the molecules of a gas follow the Maxwell–Boltzmann distribution, and that the probability of finding a molecule with energy E goes as $e^{-E/k_B T}$. In the sections that follow we start from the mechanics of many particles and the counting of possibilities, and arrive at both of those principles, at the entropy of Section 2.6, and at a general method for calculating thermodynamic quantities from the energy levels of a system. This first section sets out the probability theory that the counting will need.

Independent Objects with Two States

Let us consider a system of N objects, each of which can be in one of two states, as each molecule of a gas in a vessel is either in the left half of the vessel or in the right half. Each object is in

the first state with probability p and in the second with probability $q = 1 - p$, and the objects are independent of one another, so that the state of one has no influence on the state of another. Let n be the number of objects in the first state.

A particular arrangement in which n named objects are in the first state and the remaining $N - n$ are in the second has probability $p^n q^{N-n}$, since the probabilities of independent events multiply. The number of possible arrangements with n objects in the first state is the number of ways of choosing which n of the N they are, which is the binomial coefficient. So

$$P(n) = \binom{N}{n} p^n q^{N-n}, \quad \binom{N}{n} = \frac{N!}{n! (N-n)!}. \quad (4.1)$$

This is the binomial distribution, a result you should be familiar with from school. The binomial theorem shows that the probabilities add to one, since $\sum_n P(n) = (p + q)^N = 1$.

The Mean and the Variance

A convenient way to do the sums we need is to treat p and q as independent variables for the duration of the calculation, and to set $q = 1 - p$ only at the end. Since $\partial(p^n)/\partial p = np^{n-1}$, applying the operation $p \partial/\partial p$ to a term of the sum brings down a factor n . Applying it to the binomial theorem in the form $\sum_n \binom{N}{n} p^n q^{N-n} = (p + q)^N$ gives

$$\bar{n} = \sum_n n \binom{N}{n} p^n q^{N-n} = p \frac{\partial}{\partial p} (p + q)^N = Np(p + q)^{N-1} = Np, \quad (4.2)$$

where the last step uses $p + q = 1$. Applying $p \partial/\partial p$ to that result a second time, a calculation left as Problem 1 at the end of this section, gives $\overline{n^2} = Np + N(N-1)p^2$, and the variance is

$$\sigma_n^2 = \overline{n^2} - \bar{n}^2 = Np + N(N-1)p^2 - N^2p^2 = Np - Np^2 = Npq, \quad (4.3)$$

since the terms in $N^2 p^2$ cancel. The mean grows in proportion to N and the standard deviation $\sigma_n = \sqrt{Npq}$ grows only as $\sqrt{N}$.

The Relative Width

What matters for a physical system is the width of the distribution compared with the mean, since that is the fractional variation we would see in a measurement. From the mean and the variance found above,

$$\frac{\sigma_n}{\bar{n}} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{p}} \frac{1}{\sqrt{N}}. \quad (4.4)$$

The relative width falls as $1/\sqrt{N}$. For a handful of objects the fraction in the first state varies a great deal from one occasion to the next, and for a system with N of the order of Avogadro's number that fraction is fixed to a relative precision far below anything measurable. This is the reason the macroscopic properties of a large body have definite values. The pressure of a gas in a vessel, for example, does not visibly fluctuate, and the density of the air in a room is the same from one moment to the next, because the pressure and the density are both averages over enormous numbers of molecules, and the fractional spread of such an average is of order $1/\sqrt{N}$. Figure 25 shows how quickly the distribution narrows.

The Random Walk in One Dimension

The same distribution appears in a problem which at first sight has nothing to do with objects in two states. Suppose a walker starts at the origin and takes N steps along a line, each of length ℓ , each independently to the right with probability $\frac{1}{2}$ and to the left with probability $\frac{1}{2}$. If n of the steps are to the right then $N - n$ are to the left, and the final displacement is

$$x = n\ell - (N - n)\ell = (2n - N)\ell. \quad (4.5)$$

The mean of n is $N/2$, so $\bar{x} = 0$, which is what we should expect from the symmetry of the two directions. The variance of x is $4\ell^2$

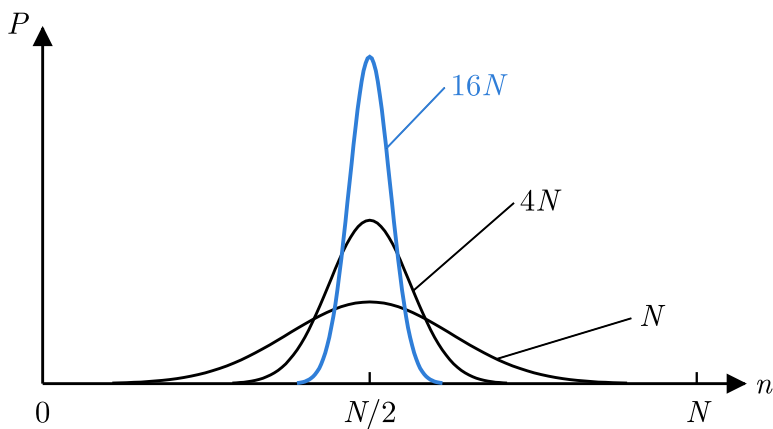


Figure 25. The binomial distribution with $p = q = \frac{1}{2}$, drawn for three systems whose sizes are in the ratio 1 : 4 : 16. The horizontal axis is scaled to the size of each system, so all three peak at $n = N/2$. The vertical axis is scaled in the same way, so that it shows the probability per unit of n/N , which is $NP(n)$, and the area under each curve is the same. The width of the peak relative to N falls as $1/\sqrt{N}$, and the height drawn to this scale rises as $\sqrt{N}$, so that the total probability remains one; the peak value $P(N/2)$ itself falls as $1/\sqrt{N}$.

times the variance of n , and with $p = q = \frac{1}{2}$ the variance of n is $N/4$, so

$$\overline{x^2} = N\ell^2. \quad (4.6)$$

The walker drifts nowhere on average, and the typical distance from the origin after N steps is $\sqrt{N}\ell$. Suppose now that one step is taken in every time interval τ , so that $N = t/\tau$. Then

$$\overline{x^2} = \frac{\ell^2}{\tau} t, \quad (4.7)$$

which grows in proportion to the time. A walker who travelled steadily in one direction would move a distance proportional to the time itself, so the random walk carries the walker away from the origin a good deal more slowly. The combination ℓ^2/τ has the dimensions of a length squared divided by a time, which are the dimensions of the diffusion coefficient D of Fick's law in Section 1.4. For a molecule in a gas, ℓ is the mean free path λ and τ is the mean time between collisions, and λ and τ are related by $\lambda = \bar{v}\tau$ as Section 1.3 found, so that the combination is $\lambda^2/\tau = \bar{v}\lambda$. The diffusion coefficient found in Section 1.4 by following the flux of molecules across a plane is $D = \frac{1}{3}\bar{v}\lambda$, which is the same product of the mean speed and the mean free path, to within a numerical factor that the crude picture of a walk in one dimension cannot be expected to get right.

Stirling's Approximation

The binomial coefficient contains factorials of large numbers, so we now work out a manageable expression for $\ln N!$. Since the logarithm of a product is the sum of the logarithms,

$$\ln N! = \sum_{n=1}^N \ln n. \quad (4.8)$$

The terms of this sum are values of the smooth function $\ln x$ at the integers, so the sum is close to the area under that function. Figure 26 illustrates the comparison. Each term $\ln n$ is the area of

a rectangle of unit width and height $\ln n$, drawn over the interval from $n - 1$ to n . Since $\ln x$ increases, the rectangle lies above the curve over that interval, and the sum of the rectangles is larger than the area under the curve from 1 to N . Drawing each rectangle over the interval from n to $n + 1$ instead puts it below the curve, and the sum is then smaller than the area under the curve from 1 to $N + 1$. The sum lies between two areas that differ by about $\ln N$, so

$$\ln N! = \int_1^N \ln x \, dx + O(\ln N). \quad (4.9)$$

The integral is done by parts, with $u = \ln x$ and $dv = dx$, so that $du = dx/x$ and $v = x$:

$$\begin{aligned} \int_1^N \ln x \, dx &= \left[x \ln x \right]_1^N - \int_1^N x \frac{dx}{x} \\ &= N \ln N - \left[x \right]_1^N = N \ln N - N + 1. \end{aligned} \quad (4.10)$$

When N is large the terms 1 and $O(\ln N)$ are negligible beside $N \ln N$, and we shall use Stirling's approximation in the form

$$\ln N! \approx N \ln N - N. \quad (4.11)$$

The Gaussian Near the Peak

We can now find the shape of the binomial distribution near its maximum. Consider the symmetric case $p = q = \frac{1}{2}$, for which $P(n) = \binom{N}{n} 2^{-N}$, and write $n = \frac{1}{2}N + m$, so that m measures the departure from the peak. Taking logarithms and using Stirling's approximation for each of the three factorials,

$$\begin{aligned} \ln P &= N \ln N - N - [n \ln n - n] \\ &\quad - [(N - n) \ln(N - n) - (N - n)] - N \ln 2. \end{aligned} \quad (4.12)$$

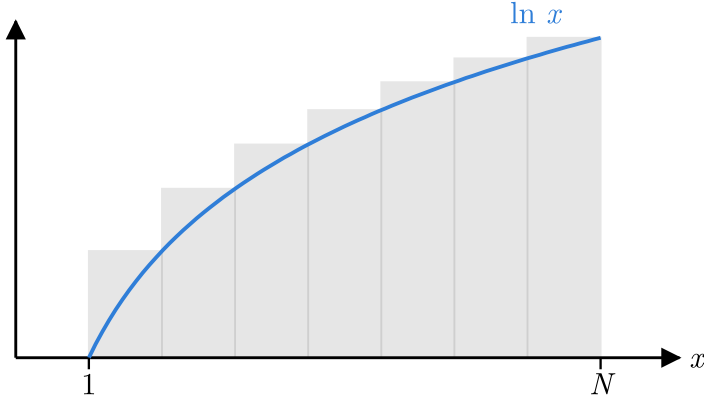


Figure 26. The sum $\ln N! = \sum_{n=1}^N \ln n$ drawn as rectangles of unit width, compared with the area under the curve $\ln x$. Each rectangle of height $\ln n$ is drawn over the interval from $n-1$ to n and so lies above the curve, which is why the sum exceeds the integral by an amount of order $\ln N$.

That substitution gives $\ln n = \ln(\frac{1}{2}N) + \ln(1 + 2m/N)$, and for $|m| \ll N$ the second logarithm is expanded by the series $\ln(1 + u) = u - \frac{1}{2}u^2 + \dots$ with $u = 2m/N$, which gives $2m/N - 2m^2/N^2$. Multiplying by n ,

$$n \ln n = (\frac{1}{2}N + m) \ln(\frac{1}{2}N) + (\frac{1}{2}N + m) \left(\frac{2m}{N} - \frac{2m^2}{N^2} \right), \quad (4.13)$$

and the second product, multiplied out and truncated after the terms in m^2 , is $m - m^2/N + 2m^2/N = m + m^2/N$, so that

$$n \ln n \approx (\frac{1}{2}N + m) \ln(\frac{1}{2}N) + m + \frac{m^2}{N}. \quad (4.14)$$

Since $N - n = \frac{1}{2}N - m$, the same expansion with m replaced by $-m$ gives

$$(N - n) \ln(N - n) \approx (\frac{1}{2}N - m) \ln(\frac{1}{2}N) - m + \frac{m^2}{N}. \quad (4.15)$$

Adding the expressions for $n \ln n$ and $(N - n) \ln(N - n)$, the terms linear in m cancel and the sum is $N \ln(\frac{1}{2}N) + 2m^2/N$. Putting that sum into the expression for $\ln P$, and noting that the terms $+n$ and $+(N - n)$ there add to N and so cancel the $-N$,

$$\ln P \approx N \ln N - N \ln(\tfrac{1}{2}N) - N \ln 2 - \frac{2m^2}{N} = -\frac{2m^2}{N}, \quad (4.16)$$

where the first three terms cancel because $N \ln(\frac{1}{2}N) = N \ln N - N \ln 2$. So, provided N is large, near its peak the binomial distribution is a Gaussian,

$$P(n) \approx A e^{-2m^2/N}, \quad m = n - \tfrac{1}{2}N, \quad (4.17)$$

where A is a constant. The calculation above does not give A correctly, because Stirling's approximation in the form we are using drops terms of order $\ln N$, which do not depend on m , and which are not small, so we fix A instead by requiring the probabilities to add to one. Neighbouring values of n differ by one, so each term of the sum over n stands for an interval of unit width in m and the sum can be replaced by an integral. Using the Gaussian integral $\int_{-\infty}^{\infty} e^{-am^2} dm = \sqrt{\pi/a}$ worked out in Section 1.2, with $a = 2/N$,

$$\sum_n P(n) \approx \int_{-\infty}^{\infty} A e^{-2m^2/N} dm = A \sqrt{\frac{\pi N}{2}} = 1, \quad (4.18)$$

so that $A = \sqrt{2/\pi N}$ and

$$P(n) \approx \sqrt{\frac{2}{\pi N}} e^{-2m^2/N}. \quad (4.19)$$

Comparing the exponent with the standard form $e^{-m^2/2\sigma^2}$ gives $\sigma = \sqrt{N}/2$, in agreement with $\sigma_n = \sqrt{Npq}$ found above, which for the case under consideration, $p = q = \frac{1}{2}$, is $\sqrt{N}/2$.

The Central Limit Theorem

The Gaussian shape obtained above is not special to two-state objects. Let $X = X_1 + X_2 + \cdots + X_N$ be a sum of N independent random quantities, where X_i has mean μ_i and variance s_i^2 , and assume that no single term dominates the sum. The central limit theorem states that as N grows the distribution of X approaches a Gaussian with mean $\sum_i \mu_i$ and variance $\sum_i s_i^2$, whatever the distributions of the individual X_i may be, provided each has a finite mean and a finite variance. The proof of the theorem needs mathematics beyond this course and we shall not attempt it, though the calculation above, in which a sum of N independent two-state variables came out Gaussian, is one instance of it.

The content of the theorem is that the detailed shape of the microscopic distribution is forgotten. To know how a macroscopic sum is distributed we need the mean and the variance of one contribution. For N identical contributions the mean of X grows as N and its standard deviation as $\sqrt{N}$, so the relative width again falls as $1/\sqrt{N}$. Almost every macroscopic quantity we have dealt with, from the energy of a gas to the number of molecules in a small region of it, is a sum of this kind.

4.2 Microstates, Multiplicity, and Entropy

The thermodynamics of Chapters 2 and 3 introduced the entropy through the heat absorbed in a reversible process, and left it as a function of state whose changes we could calculate and whose exact value we could not. In this section we count the ways in which a system can be arranged, and find that the counting reproduces the entropy, the temperature, the pressure, and the chemical potential, all from one assumption about probability.

Microstates and Macrostates

A microstate is a complete specification of the state of a system: the state of every one of its constituents. A macrostate is a specification of the few quantities we can measure, such as the total energy, the volume, and the number of particles. Many microstates belong to the same macrostate, since a small change in the state of one

constituent will usually leave the measured quantities as they were.

Let us take a small system to make the distinction concrete, say four objects, each of which can be in one of two states, of energies 0 and ε . A microstate says which of the two states each of the four objects is in, and there are 2^4 such specifications. A macrostate is specified only by the total energy. The macrostate of energy ε contains four microstates, since any one of the four objects may be the excited one. The macrostate of energy 2ε contains six, since there are six ways of choosing which two objects are excited. The macrostate of zero energy contains one. The general pattern is the binomial coefficient of Section 4.1: for N objects with n of them excited, and energy $U = n\varepsilon$, the number of microstates is $\binom{N}{n}$.

The Fundamental Assumption

The fundamental assumption of statistical mechanics is that an isolated system in equilibrium is equally likely to be found in any of the microstates accessible to it. Isolated means that the energy, the volume, and the number of particles are fixed, and accessible means consistent with those fixed values.

There is no proof of this assumption from mechanics alone, and it is adopted because we have no reason to prefer one microstate to another and because the results it gives agree with experiment. Its consequence is that the probability of a macrostate is proportional to the number of microstates it contains. The macrostate we find the system in is therefore the one that can be realised in the largest number of ways, provided the numbers of microstates in the different macrostates are as unequal as the counting of Section 4.1 suggests.

Multiplicity

Let $\Omega(U, V, N)$ be the number of microstates of an isolated system with energy U , volume V , and N particles. This is called the multiplicity. For the system of N two-state objects above, $\Omega = \binom{N}{n}$ with $n = U/\varepsilon$.

Now let us put two systems in contact through a rigid wall that passes energy but no particles, as in Figure 27, and isolate the pair

from everything else, so that $U = U_1 + U_2$ is fixed. A microstate of the combined system is a microstate of the first together with a microstate of the second, so for a given division of the energy the number of microstates is

$$\Omega_{\text{tot}}(U_1) = \Omega_1(U_1) \Omega_2(U - U_1). \quad (4.20)$$

Every microstate of the combined system is equally likely, so the probability of a particular division of the energy is proportional to this product.

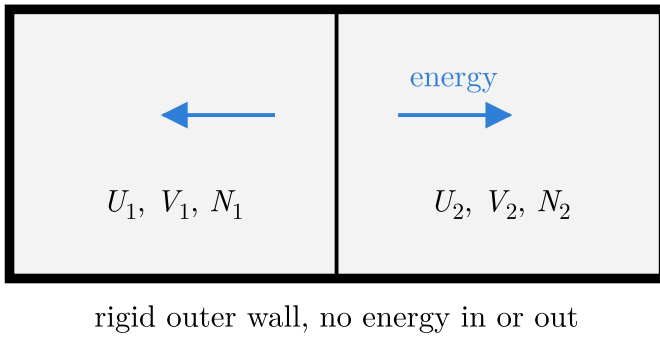


Figure 27. A pair of systems in contact through a rigid wall that passes energy but no particles. The pair is isolated, so $U_1 + U_2$ is fixed, and the division of the energy between the two systems, U_1 and $U - U_1$, is the only thing free to vary.

The Sharpness of the Peak

Let us find how sharp that probability distribution is, taking two systems whose multiplicities grow as a power of the energy, $\Omega_1 \propto U_1^{\alpha N_1}$ and $\Omega_2 \propto U_2^{\alpha N_2}$, with α a constant of order unity. Then

$$\ln \Omega_{\text{tot}} = \alpha N_1 \ln U_1 + \alpha N_2 \ln(U - U_1) + \text{constant}. \quad (4.21)$$

Differentiating with respect to U_1 and setting the derivative to zero gives $\alpha N_1/U_1 = \alpha N_2/U_2$, so the most probable division shares the

energy in proportion to the sizes, $U_1^* = UN_1/(N_1 + N_2)$. Differentiating a second time gives

$$\frac{\partial^2 \ln \Omega_{\text{tot}}}{\partial U_1^2} = -\frac{\alpha N_1}{U_1^2} - \frac{\alpha N_2}{U_2^2}, \quad (4.22)$$

which is of order $-N/U^2$ when the two systems are of comparable size N and the energy U is of order N times the energy per particle. Expanding $\ln \Omega_{\text{tot}}$ about its maximum to second order, in the same way that $\ln P$ was expanded in Section 4.1,

$$\ln \Omega_{\text{tot}}(U_1) \approx \ln \Omega_{\text{tot}}(U_1^*) - \frac{(U_1 - U_1^*)^2}{2 \Delta U_1^2}, \quad \frac{1}{\Delta U_1^2} = \left| \frac{\partial^2 \ln \Omega_{\text{tot}}}{\partial U_1^2} \right|, \quad (4.23)$$

so that Ω_{tot} is a Gaussian in U_1 whose width, from the size of the second derivative above, is

$$\Delta U_1 \sim \frac{U}{\sqrt{N}}, \quad (4.24)$$

and the fractional width of the peak is of order $1/\sqrt{N}$ (Figure 28). For two macroscopic bodies in thermal contact the division of energy between them is therefore fixed so sharply that no experiment could detect a departure from it. The peak is so narrow that on any scale we can draw it is a spike, and we may, for all practical purposes, treat the most probable division as the only one that occurs.

Entropy

The condition for the peak, written for a general pair of systems, is that

$$\frac{\partial \ln \Omega_1}{\partial U_1} = \frac{\partial \ln \Omega_2}{\partial U_2}. \quad (4.25)$$

When two systems are in thermal contact they reach equilibrium when this derivative takes the same value in each of them, and in

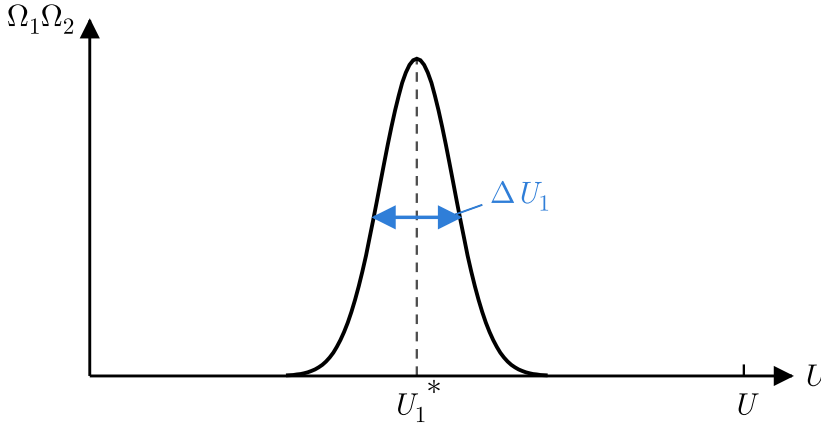


Figure 28. The number of microstates of the combined system, $\Omega_1(U_1)\Omega_2(U - U_1)$, as a function of the energy held by the first system. The peak at U_1^* has a width ΔU_1 of order $U/\sqrt{N}$, and for a macroscopic system it is far narrower than the curve drawn here.

thermodynamics the quantity that is the same for two bodies in thermal contact is the temperature. Something built from $\ln \Omega$ is therefore the link to Chapter 2. We define the entropy of an isolated system as

$$S = k_B \ln \Omega. \quad (4.26)$$

The logarithm is the right function to use because entropy is extensive, as Section 2.6 found from the heat absorbed, while multiplicities of a composite system always multiply. Let us put two independent systems side by side; each microstate of one can be combined with each microstate of the other, so $\Omega = \Omega_1\Omega_2$, and the only function that turns a product into a sum is the logarithm. The constant k_B in front is a choice of units, and it is Boltzmann's constant that makes the definition agree with the thermodynamic entropy.

The Second Law, Interpreted Statistically

Suppose an isolated system is prepared with a constraint, such as a partition holding all the gas in one half of a vessel, and the constraint is then removed. Every microstate that was accessible before is still accessible, and others become accessible as well, so Ω can only increase, and S with it. The system then spends almost all of its time in the macrostate with the largest multiplicity, because that macrostate contains almost all of the microstates. The increase of Ω when a constraint is removed, and the overwhelming weight of the largest macrostate, together make up the second law of Section 2.5, which is now interpreted as a statement about counting.

The statistical interpretation also adds something to the thermodynamic statement. Fundamentally speaking, nothing forbids the entropy of an isolated system from decreasing. A decrease by any measurable amount is merely overwhelmingly unlikely, since the departures from the peak are of relative size $1/\sqrt{N}$. For a system of a few particles the fluctuations are perfectly visible, and for a system of 10^{23} they are not.

Temperature, Pressure, and Chemical Potential

The combined statement of the first and second laws, from Sections 2.6 and 3.3, is $dU = T dS - p dV + \mu dN$, which rearranges to

$$dS = \frac{1}{T} dU + \frac{p}{T} dV - \frac{\mu}{T} dN, \quad (4.27)$$

so that

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V,N}, \quad \frac{p}{T} = \left(\frac{\partial S}{\partial V} \right)_{U,N}, \quad -\frac{\mu}{T} = \left(\frac{\partial S}{\partial N} \right)_{U,V}. \quad (4.28)$$

With $S = k_B \ln \Omega$ these three become definitions of T , p , and μ in terms of counting. The relation for the temperature turns the equilibrium condition found above into the statement that two systems in thermal contact end up at the same temperature, which is what a thermometer measures.

The relations for the pressure and the chemical potential follow in the same way from the same peak condition. If the wall between the two systems is free to move, the total volume $V_1 + V_2$ is fixed and the total entropy is maximised with respect to V_1 as well as U_1 , giving $p_1/T_1 = p_2/T_2$; with the temperatures already equal, the pressures are equal. If the wall also lets particles through, maximising with respect to N_1 gives $\mu_1/T_1 = \mu_2/T_2$, and hence equal chemical potentials, which is the condition for material equilibrium found in Section 3.3.

4.3 Counting States of an Ideal Gas

The definition $S = k_B \ln \Omega$ is of no use until we can count the microstates of a real system. Here we carry the counting out for a monatomic ideal gas, and obtain an expression for its entropy that contains the ideal gas law, the internal energy of Section 1.2, and a constant that no thermodynamic measurement could give us.

Position and Momentum

The state of one classical atom is given by its position and its momentum, which is six numbers, the same specification of a mechanical state that the Hamiltonian formulation used in the notes on mechanics and dynamical systems. The position and the momentum both vary continuously, so the number of states in any region is infinite unless we agree on how finely to divide the six-dimensional space of position and momentum into cells. Some such division is needed before the counting of Section 4.2 means anything in this context.

Quantum mechanics fixes the size of the cell. The quantisation of the momentum of a confined particle is not treated in this course, and we quote the result that for a particle confined to a length L the allowed momenta are evenly spaced, with a spacing h/L between neighbouring values, where h is Planck's constant. The number of states with position anywhere in the length L and with momentum in a range dp_x is therefore $L dp_x/h$, which is the area of that region of the x - p_x plane divided by h (Figure 29). Doing the same for all three directions, the number of single-particle states with position anywhere in a volume V and momentum in a small region of mo-

momentum space of volume d^3p is $V d^3p/h^3$. Each state occupies a cell of volume h^3 in the six-dimensional space.

If we had chosen a different cell volume, every multiplicity would be multiplied by a constant power of the ratio and every entropy would change by a constant times N . Differences of entropy at fixed N would be unaffected, and so would T and p , but the absolute entropy and the chemical potential would change. The value that agrees with measurement is h^3 .

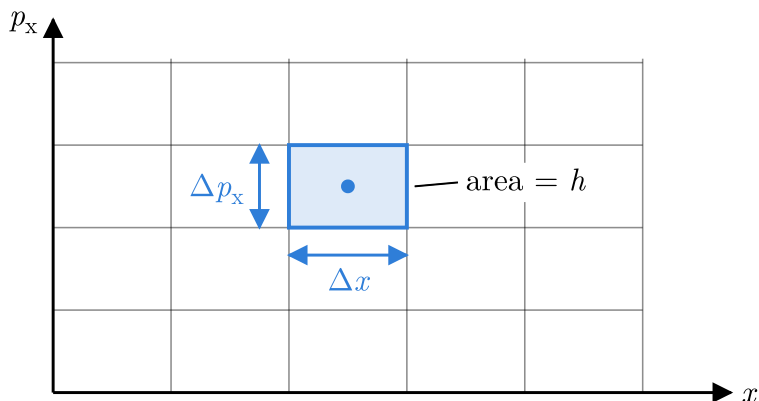


Figure 29. The plane of position and momentum for motion in one dimension, divided into cells of area $\Delta x \Delta p_x = h$. Each state of the particle corresponds to one cell. In three dimensions the space has six axes and each state occupies a cell of volume h^3 .

The Multiplicity of a Monatomic Gas

Let us count the microstates of N atoms of a monatomic gas, of mass m each, in a vessel of volume V , with total energy U . The atoms of an ideal gas do not interact, so the energy is entirely kinetic and

$$P_1^2 + P_2^2 + \cdots + P_N^2 = 2mU, \quad (4.29)$$

where P_i is the magnitude of the momentum of the i th atom. Each

atom may be anywhere in the vessel, so the positions contribute a factor V^N . The momenta contribute the volume of a thin shell at the surface defined by the equation above, divided by h^{3N} .

That equation says that the point whose $3N$ coordinates are the momentum components lies on the surface of a sphere of radius $R = \sqrt{2m\bar{U}}$ in $3N$ dimensions. The volume enclosed by a sphere of radius R in d dimensions is

$$V_d(R) = \frac{\pi^{d/2} R^d}{(d/2)!}, \quad (4.30)$$

and we quote this result rather than derive it, because the derivation requires an integration over d variables at once and the mathematics of this course does not extend that far. When d is odd, the factorial of a half-integer that appears in the denominator means the value of the gamma function, the standard smooth extension of the factorial to non-integer arguments, for which $x! = \Gamma(x + 1)$, and Stirling's approximation applies to it as it does to the factorial of an integer.

What the counting requires is the volume of a thin shell at the surface of the sphere rather than the volume of the whole sphere, and in many dimensions the volume of the shell is very nearly the volume of the whole sphere. A shell of thickness δ has volume

$$V_d(R) - V_d(R - \delta) = V_d(R) \left[1 - \left(1 - \frac{\delta}{R} \right)^d \right], \quad (4.31)$$

and since $x^d = e^{d \ln x}$ for any positive x , which is what raising a number to a power means, the quantity $(1 - \delta/R)^d$ is $e^{d \ln(1 - \delta/R)}$ with $\ln(1 - \delta/R)$ negative, so it falls exponentially as d grows and the bracket is indistinguishable from one once $d = 3N$ is of the order of Avogadro's number, for any fixed relative thickness δ/R we care to choose. Almost the whole volume of a sphere in many dimensions lies just beneath its surface, so we may use the volume of the whole sphere for the shell. With $d = 3N$,

$$\Omega = \frac{V^N}{h^{3N}} \frac{\pi^{3N/2} (2m\bar{U})^{3N/2}}{(3N/2)!}. \quad (4.32)$$

Taking the logarithm turns the products into sums,

$$\ln \Omega = N \ln V - 3N \ln h + \frac{3N}{2} \ln (2\pi m U) - \ln \left(\frac{3N}{2} \right)!, \quad (4.33)$$

and Stirling's approximation of Section 4.1 replaces the last term by $\frac{3N}{2} \ln \left(\frac{3N}{2} \right) - \frac{3N}{2}$. Writing $3N \ln h$ as $\frac{3N}{2} \ln h^2$ and gathering the three terms that carry a factor $\frac{3N}{2}$ into a single logarithm,

$$\frac{S}{k_B} = \ln \Omega = N \ln V + \frac{3N}{2} \ln \left(\frac{4\pi m U}{3N h^2} \right) + \frac{3N}{2}. \quad (4.34)$$

The Gibbs Paradox

There is something wrong with the entropy found above. Entropy is extensive: doubling the size of a system at fixed density and energy density should double it. Let us double N , V , and U together in the expression above. Every term doubles except $N \ln V$, which becomes $2N \ln 2V = 2N \ln V + 2N \ln 2$. The entropy comes out too large by $2N k_B \ln 2$.

The same discrepancy appears in a more physical form. Consider a vessel divided by a partition into two halves, each holding the same gas at the same density and temperature, and remove the partition. Nothing has changed; the gas on the left is the same gas as the gas on the right, at the same density and temperature, and putting the partition back restores the original state exactly. Yet the expression above gives an entropy increase of $N k_B \ln 2$, since each half has doubled the volume available to it. How can removing a partition between two identical samples increase the entropy?

The answer is that we have counted too many microstates. Our count treated the atoms as though they carried labels, so that swapping the positions and momenta of atom one and atom two gave a different microstate. Atoms of the same element carry no labels. In other words, atoms are indistinguishable, and an arrangement obtained by permuting the atoms among the same set of cells is the same microstate. Provided the atoms are spread thinly enough that they rarely share a cell, each microstate has been counted $N!$ times, and the correct multiplicity is smaller by that factor:

$$\Omega = \frac{1}{N!} \frac{V^N}{h^{3N}} \frac{\pi^{3N/2} (2mU)^{3N/2}}{(3N/2)!}. \quad (4.35)$$

The Sackur–Tetrode Entropy

Subtracting $\ln N! \approx N \ln N - N$ from the expression for S/k_B found above turns the first term $N \ln V$ into $N \ln V - N \ln N = N \ln(V/N)$, and the remaining $+N$ raises the last term from $\frac{3}{2}N$ to $\frac{5}{2}N$. So

$$S = Nk_B \left[\ln\left(\frac{V}{N}\right) + \frac{3}{2} \ln\left(\frac{4\pi mU}{3Nh^2}\right) + \frac{5}{2} \right], \quad (4.36)$$

which is called the Sackur–Tetrode equation. It is extensive, since doubling N , V , and U leaves the bracket unchanged and doubles the factor Nk_B in front, and it removes the difficulty with the partition, because V/N and U/N are the same before and after the partition is taken away.

Let us check the result against thermodynamics. Differentiating with respect to U at fixed V and N ,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V,N} = \frac{3Nk_B}{2U}, \quad (4.37)$$

so $U = \frac{3}{2}Nk_B T$, which is the internal energy of a monatomic gas given in Section 1.2. Differentiating with respect to V at fixed U and N ,

$$\frac{p}{T} = \left(\frac{\partial S}{\partial V} \right)_{U,N} = \frac{Nk_B}{V}, \quad (4.38)$$

which is the ideal gas law. The internal energy of a monatomic gas and the equation of state, both of which we assumed at the start of this course, come out of a calculation that counts cells in the space of position and momentum.

The Entropy of Mixing

Suppose now that two different gases are mixed, gas A occupying a volume V_A and gas B a volume V_B at the same temperature and pressure, separated by a partition which is then removed. Each gas expands into the total volume $V = V_A + V_B$, and since the two kinds of atom are distinguishable from each other, the count for each is unaffected by the presence of the other. Only the volume in the Sackur–Tetrode expression changes, so

$$\Delta S = N_A k_B \ln \frac{V}{V_A} + N_B k_B \ln \frac{V}{V_B} > 0. \quad (4.39)$$

Mixing two different gases increases the entropy, and the increase does not depend on how different they are. Mixing a gas with itself does not increase the entropy, because the factor $1/N!$ makes the entropy depend on V/N , which is unchanged. There is no single expression covering both the mixing of two different gases and the mixing of a gas with itself: the entropy change jumps from zero to its full value as soon as the two gases differ at all.

Entropy as Missing Information

There is a way of writing the entropy that does not assume all microstates are equally likely. Let p_i be the probability of microstate i . Then

$$S = -k_B \sum_i p_i \ln p_i. \quad (4.40)$$

When Ω microstates are equally likely and the rest are inaccessible, every non-zero p_i equals $1/\Omega$, and the sum has Ω terms each equal to $-(1/\Omega) \ln(1/\Omega)$, so $S = k_B \ln \Omega$ and the definition in terms of the probabilities agrees with the definition in terms of the multiplicity. The general expression is zero when one microstate has probability one, since then the system is known exactly, and it is largest when all the probabilities are equal, which is when we know least about which microstate the system occupies. When read this way, the entropy measures how much about the microscopic state of the system is unknown to us.

4.4 The Boltzmann Factor and the Partition Function

Counting the microstates of an isolated system is possible for an ideal gas and awkward for almost anything else, because the constraint of fixed total energy ties the parts of the system together. Laboratory systems are rarely isolated in any case; they sit in a room at a fixed temperature and exchange energy with their surroundings. We turn instead to the probability distribution for a system held at a fixed temperature, which turns out to be much easier to use.

A System in Contact with a Reservoir

Let a small system be in thermal contact with a large one, which we call the reservoir, and let the pair together be isolated with total energy E_{tot} . Let E_i be the energy of the small system when it is in its microstate i , and let $\Omega_R(E)$ be the multiplicity of the reservoir at energy E . Large means that the reservoir is so much bigger than the small system that its temperature does not change appreciably when the small system takes energy from it.

When the small system is in the definite microstate i , the reservoir holds the remaining energy $E_{\text{tot}} - E_i$, and the number of microstates of the combined system is $1 \times \Omega_R(E_{\text{tot}} - E_i)$. By the fundamental assumption of Section 4.2 every microstate of the combined isolated system is equally likely, so

$$p_i \propto \Omega_R(E_{\text{tot}} - E_i). \quad (4.41)$$

The multiplicity of the reservoir varies enormously with its energy, so we expand its logarithm rather than the multiplicity itself:

$$\ln \Omega_R(E_{\text{tot}} - E_i) = \ln \Omega_R(E_{\text{tot}}) - E_i \left(\frac{\partial \ln \Omega_R}{\partial E} \right)_{E_{\text{tot}}} + \cdots. \quad (4.42)$$

The derivative is $(1/k_B)(\partial S_R/\partial E)$, which is $1/k_B T$ by the definition of temperature in Section 4.2, where T is the temperature of the reservoir. The next term in the expansion involves $\partial(1/k_B T)/\partial E =$

$-1/(k_B T^2 C_R)$, where C_R is the heat capacity of the reservoir, and that term is smaller than the term we have kept by a factor of order $E_i/(TC_R)$, which vanishes as the reservoir is made large. Dropping it and taking the exponential,

$$p_i \propto e^{-E_i/k_B T}. \quad (4.43)$$

This is the Boltzmann factor, which we assumed as a general principle in Section 1.2 when writing down the distribution of molecular speeds. It is now derived as a consequence of counting the microstates of the reservoir. The energy E_i appears with a minus sign because every unit of energy the small system holds is a unit the reservoir does not have, and a reservoir with less energy has fewer microstates.

The Partition Function

Writing $\beta = 1/k_B T$ and fixing the constant of proportionality by requiring the probabilities to add to one,

$$p_i = \frac{e^{-\beta E_i}}{Z}, \quad Z = \sum_i e^{-\beta E_i}. \quad (4.44)$$

The quantity Z is called the partition function, or, more fully, the canonical partition function. The collection of microstates of a system held at a fixed temperature, each weighted by its probability p_i , is called the canonical ensemble, and the adjective is attached to Z whenever the ensemble it belongs to has to be named. The sum runs over microstates and not over energy levels, so a level of energy E shared by w different states contributes $w e^{-\beta E}$ to the sum.

The mean energy of the system follows by differentiation. Since $\partial e^{-\beta E_i}/\partial \beta = -E_i e^{-\beta E_i}$,

$$\bar{E} = \sum_i E_i p_i = \frac{1}{Z} \sum_i E_i e^{-\beta E_i} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}. \quad (4.45)$$

Examples of the Method

We will work through two examples, chosen to show how the sums appearing in a partition function are done.

Let us take first a system with just two states, of energies 0 and ε . The partition function has just two terms,

$$Z = 1 + e^{-\beta\varepsilon}, \quad (4.46)$$

and the mean energy is

$$\bar{E} = -\frac{\partial}{\partial\beta} \ln(1 + e^{-\beta\varepsilon}) = \frac{\varepsilon e^{-\beta\varepsilon}}{1 + e^{-\beta\varepsilon}} = \frac{\varepsilon}{e^{\beta\varepsilon} + 1}. \quad (4.47)$$

At low temperature, $\beta\varepsilon$ is large and $\bar{E} \approx \varepsilon e^{-\beta\varepsilon}$, which is exponentially small: the system is almost always in its lower state. At high temperature, $\beta\varepsilon$ is small, $e^{\beta\varepsilon} \approx 1$, and $\bar{E} \rightarrow \frac{1}{2}\varepsilon$, which says that the two states are then almost equally likely. The probability of the upper state is $1/(e^{\beta\varepsilon} + 1)$, which agrees with the fraction of excited objects obtained by counting microstates in Problem 2 of Section 4.2.

Let us take next a harmonic oscillator, whose classical motion was studied in the notes on mechanics and dynamical systems, and whose energy levels we quote from quantum mechanics as $E_n = (n + \frac{1}{2})\hbar\omega$ with $n = 0, 1, 2, \dots$, where ω is the angular frequency of the oscillator and $\hbar = h/2\pi$, with h the Planck constant of Section 4.3. The partition function is a geometric series with common ratio $e^{-\beta\hbar\omega}$,

$$Z = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}, \quad (4.48)$$

so that $\ln Z = -\frac{1}{2}\beta\hbar\omega - \ln(1 - e^{-\beta\hbar\omega})$ and, differentiating that logarithm with respect to β ,

$$\bar{E} = -\frac{\partial \ln Z}{\partial\beta} = \frac{\hbar\omega}{2} + \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1}, \quad (4.49)$$

where the last step follows on multiplying the numerator and the denominator of the second term by $e^{\beta\hbar\omega}$.

When $k_B T$ is much larger than $\hbar\omega$ the denominator is approximately $\beta\hbar\omega$ and $\bar{E} \approx k_B T$.

Grouping States by Energy

When a system has very many closely spaced levels it is convenient to group them. Let $g(E) dE$ be the number of microstates with energy between E and $E + dE$, a quantity called the density of states. The probability that the energy lies in that range is then

$$P(E) dE = \frac{g(E)e^{-\beta E}}{Z} dE. \quad (4.50)$$

The density of states rises with energy, because there are more ways of arranging a system at high energy than at low energy, while the Boltzmann factor falls. Their product is small at both ends and has a peak in between, as Figure 30 shows. The most probable energy is the energy at which the rise of g is exactly offset by the fall of the exponential, and it is a different quantity from the most probable microstate, which is always the one of lowest energy, since the Boltzmann factor is largest there.

The Distribution of Molecular Speeds

Let us apply the Boltzmann factor to a single atom of a dilute gas, treating that one atom as the small system and all the other atoms as the reservoir. The states of the atom were counted in Section 4.3, where the number with momentum of magnitude in a range $d(mv)$ about mv came to the volume of a thin shell in momentum space divided by h^3 and multiplied by V . Writing that shell in terms of the speed, the number of states with speed between v and $v + dv$ is $4\pi V m^3 v^2 dv / h^3$, and the energy of the atom is $E = \frac{1}{2}mv^2$, so the probability that its speed lies in that range is proportional to $v^2 e^{-mv^2/2k_B T} dv$.

Absorbing all the constant factors into one constant C to be determined, the distribution of speeds is

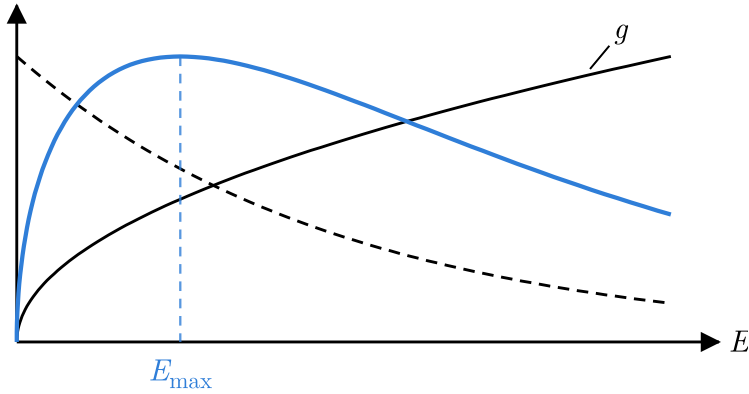


Figure 30. The density of states $g(E)$, which rises with energy, the Boltzmann factor $e^{-\beta E}$, which falls (dashed), and their product (blue), which is the probability of finding the system with energy near E . The three curves are each drawn to a convenient vertical scale. The product peaks at the most probable energy $E_{\max}$.

$$f(v) = C v^2 e^{-mv^2/2k_B T}. \quad (4.51)$$

The constant follows from the requirement that $\int_0^\infty f(v) dv = 1$. The Gaussian integral $\int_0^\infty e^{-av^2} dv = \frac{1}{2}\sqrt{\pi/a}$ is half the integral worked out in Section 1.2, since the integrand is even. Differentiating both sides of it under the integral sign with respect to a brings down a factor $-v^2$ in the integrand, so that

$$\int_0^\infty v^2 e^{-av^2} dv = -\frac{\partial}{\partial a} \left(\frac{1}{2} \sqrt{\frac{\pi}{a}} \right) = \frac{1}{4} \sqrt{\frac{\pi}{a^3}}. \quad (4.52)$$

With $a = m/2k_B T$, the normalisation condition reads $\frac{1}{4}C\sqrt{\pi/a^3} = 1$, which gives $C = 4a^{3/2}/\sqrt{\pi}$ and

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}, \quad (4.53)$$

which is the Maxwell–Boltzmann distribution written down in Section 1.2. The factor v^2 comes from the counting of states in a shell and the exponential from the Boltzmann factor, which is why the distribution vanishes at $v = 0$ and falls away at large v .

4.5 Free Energy and Fluctuations

The partition function was introduced in Section 4.4 as a normalising sum, and the mean energy came out of it by one differentiation. We shall find that Z carries all of the thermodynamics of Chapter 3, through a single relation to the Helmholtz free energy, and shall examine how much the energy of a system held at fixed temperature actually varies.

The Free Energy from the Partition Function

We return to a system in contact with a reservoir at temperature T , with the probabilities $p_i = e^{-\beta E_i}/Z$ of Section 4.4. Its entropy is given by the expression $S = -k_B \sum_i p_i \ln p_i$ of Section 4.3. Taking the logarithm of the probability,

$$\ln p_i = -\beta E_i - \ln Z, \quad (4.54)$$

and substituting into the entropy,

$$\begin{aligned} S &= -k_B \sum_i p_i (-\beta E_i - \ln Z) = k_B \beta \sum_i p_i E_i + k_B \ln Z \sum_i p_i \\ &= \frac{\bar{E}}{T} + k_B \ln Z, \quad (4.55) \end{aligned}$$

where the probabilities add to one in the second sum. Multiplying through by $-T$ and rearranging,

$$-k_B T \ln Z = \bar{E} - TS. \quad (4.56)$$

The right-hand side is the Helmholtz free energy $F = U - TS$ of Section 3.1, with $\bar{E}$ playing the part of the internal energy. So

$$F = -k_B T \ln Z. \quad (4.57)$$

We can check that the F defined this way obeys the thermodynamic relation $S = -(\partial F / \partial T)_{V,N}$ of Section 3.1. Differentiating $F = -k_B T \ln Z$ with respect to T , and using $\partial / \partial T = -(1/k_B T^2) \partial / \partial \beta$ together with $\partial \ln Z / \partial \beta = -\bar{E}$,

$$\left(\frac{\partial F}{\partial T} \right)_{V,N} = -k_B \ln Z - k_B T \frac{\bar{E}}{k_B T^2} = -k_B \ln Z - \frac{\bar{E}}{T} = -S, \quad (4.58)$$

which is the relation required.

The Recipe

The remaining thermodynamic quantities are obtained from F by the derivatives of Section 3.1. The volume and the particle number enter the partition function through the energy levels, since $E_i = E_i(V, N)$, and

$$S = - \left(\frac{\partial F}{\partial T} \right)_{V,N}, \quad p = - \left(\frac{\partial F}{\partial V} \right)_{T,N}, \quad \mu = \left(\frac{\partial F}{\partial N} \right)_{T,V}, \quad (4.59)$$

with $U = F + TS$ recovering the internal energy.

What the Free Energy Balances

Section 3.1 showed that a system held at fixed temperature and volume comes to equilibrium at the state of least Helmholtz free energy. The statistical expression $F = \bar{E} - TS$ shows what is being made small. The mean energy $\bar{E}$ is made small by putting the system into its lowest-lying states, and the entropy term $-TS$ is made small, since it enters with a minus sign, by spreading the probability over as many states as possible. The Boltzmann distribution is the compromise between the demand for low energy and the demand for large entropy, and the temperature is what decides where the compromise falls.

The low-temperature and the high-temperature limits can both be read off from the expression for F . At low temperature the factor T in front of the entropy is small, the sum $Z = \sum_i e^{-\beta E_i}$ is dominated by the lowest level, and $F \rightarrow E_0$, the ground-state energy, with the system sitting in that state. At high temperature the exponentials are all close to one, Z approaches the total number of states available, and $F \approx -k_B T \ln(\text{number of states})$, which is $-TS$ with S at its largest value. The entropy term then dominates, and the differences between the energy levels matter very little.

Energy Fluctuations

A system in contact with a reservoir does not have a definite energy, and it is worth asking by how much its energy varies. The variance is $\sigma_E^2 = \overline{E^2} - \bar{E}^2$, and it is obtained from Z by differentiating twice. Starting from $\bar{E} = -(1/Z) \partial Z / \partial \beta$ and differentiating once more with respect to β ,

$$\frac{\partial^2 \ln Z}{\partial \beta^2} = \frac{1}{Z} \sum_i E_i^2 e^{-\beta E_i} - \left(\frac{1}{Z} \sum_i E_i e^{-\beta E_i} \right)^2 = \overline{E^2} - \bar{E}^2, \quad (4.60)$$

where the first term comes from differentiating the exponentials and the second from differentiating $1/Z$. Since $\partial^2 \ln Z / \partial \beta^2 = -\partial \bar{E} / \partial \beta$ and $\partial / \partial \beta = -k_B T^2 \partial / \partial T$,

$$\sigma_E^2 = k_B T^2 \left(\frac{\partial \bar{E}}{\partial T} \right)_{V,N} = k_B T^2 C_V. \quad (4.61)$$

Now let us see how large the spread is for a macroscopic system. Both $\bar{E}$ and C_V are extensive, so each is proportional to N , and

$$\frac{\sigma_E}{\bar{E}} = \frac{\sqrt{k_B T^2 C_V}}{\bar{E}} \propto \frac{1}{\sqrt{N}}, \quad (4.62)$$

the same $1/\sqrt{N}$ that appeared in Sections 4.1 and 4.2. For a mole of gas the energy is therefore fixed so closely that no experiment could resolve the variation, so a system at a fixed temperature has

an energy that is, for all practical purposes, definite. This is why the thermodynamics of Chapter 2, which treats U as a definite function of state, works, and why it does not matter whether we hold the energy fixed or hold the temperature fixed when we calculate a thermodynamic quantity.

Independent Subsystems

Suppose a system is made of two parts whose energies add, so that a microstate of the whole is a microstate j of the first part together with a microstate k of the second, with energy $E_j^{(1)} + E_k^{(2)}$. The sum over microstates of the whole then separates:

$$Z = \sum_{j,k} e^{-\beta(E_j^{(1)} + E_k^{(2)})} = \left(\sum_j e^{-\beta E_j^{(1)}} \right) \left(\sum_k e^{-\beta E_k^{(2)}} \right) = Z_1 Z_2. \quad (4.63)$$

Partition functions of independent parts multiply, so free energies add, as an extensive quantity should. For N identical subsystems that are independent and distinguishable, because each is tied to a different place, as the atoms of a solid are held at different points of a lattice, $Z = z^N$ where z is the partition function of one of them, and $F = -Nk_B T \ln z$.

For N identical particles that move freely through the same volume there are no labels to tell them apart, and the argument of Section 4.3 applies again: each microstate of the whole has been counted $N!$ times in the product z^N . Provided the particles are spread thinly enough that they rarely occupy the same single-particle state,

$$Z = \frac{z^N}{N!}, \quad F = -Nk_B T \ln z + k_B T \ln N! = -Nk_B T \left[\ln \frac{z}{N} + 1 \right], \quad (4.64)$$

where Stirling's approximation of Section 4.1 was used in the last step.

Equipartition

Suppose that a degree of freedom contributes an energy $E = bq^2$ to the total, quadratic in a single coordinate q , where q may be a displacement, a velocity component, or a momentum component, and suppose that the levels associated with q are spaced closely compared with $k_B T$, so that the sum over states may be replaced by an integral over q . The mean energy of that degree of freedom is then

$$\bar{E} = \frac{\int_{-\infty}^{\infty} bq^2 e^{-\beta bq^2} dq}{\int_{-\infty}^{\infty} e^{-\beta bq^2} dq}, \quad (4.65)$$

where any constant factor relating dq to the number of states cancels between the numerator and the denominator. The denominator is the Gaussian integral $\int_{-\infty}^{\infty} e^{-aq^2} dq = \sqrt{\pi} a^{-1/2}$ worked out in Section 1.2, and differentiating that identity under the integral sign with respect to a , as was done in Section 4.4 for the distribution of molecular speeds, brings down a factor $-q^2$ in the integrand, which gives $\int_{-\infty}^{\infty} q^2 e^{-aq^2} dq = \frac{1}{2} \sqrt{\pi} a^{-3/2}$ for the integral in the numerator. Dividing the second of these integrals by the first, the factors of $\sqrt{\pi}$ cancel and leave $\frac{1}{2} a^{-3/2} / a^{-1/2} = 1/2a$, so that with $a = \beta b$,

$$\bar{E} = \frac{b}{2\beta b} = \frac{1}{2} k_B T. \quad (4.66)$$

Every term in the energy that is quadratic in one coordinate or one momentum component contributes $\frac{1}{2} k_B T$ to the mean energy of the system, whatever the constant b multiplying it. This is the equipartition principle quoted in Section 1.2, where it gave $\frac{3}{2} k_B T$ for the three components of translational kinetic energy of an atom. The condition under which it holds is evident from the derivation: the levels must be closely spaced compared with $k_B T$, since otherwise the sum over states cannot be replaced by an integral.

4.6 The Grand Canonical Ensemble

Section 4.4 dealt with a system that exchanges energy with a reservoir and holds a fixed number of particles. Many systems exchange

particles as well: a small region of a gas, the molecules adsorbed on a surface, a solute in contact with a solvent. We now work out the probabilities for a system that exchanges both energy and particles, and they lead to a description in which the chemical potential of Section 3.3 is the variable we control.

Exchanging Particles as Well as Energy

Let the small system be in contact with a reservoir through a wall that passes both energy and particles, and let the pair be isolated with total energy E_{tot} and total particle number N_{tot} . A microstate i of the small system now specifies both its energy E_i and the number of particles N_i it contains. The reservoir then holds $E_{\text{tot}} - E_i$ and $N_{\text{tot}} - N_i$, and by the fundamental assumption of Section 4.2,

$$p_i \propto \Omega_R(E_{\text{tot}} - E_i, N_{\text{tot}} - N_i) . \quad (4.67)$$

Expanding the logarithm of the reservoir multiplicity in both of its arguments, as in Section 4.4,

$$\ln \Omega_R = \ln \Omega_R(E_{\text{tot}}, N_{\text{tot}}) - E_i \frac{\partial \ln \Omega_R}{\partial E} - N_i \frac{\partial \ln \Omega_R}{\partial N} + \dots . \quad (4.68)$$

The two derivatives are $(1/k_B)(\partial S_R/\partial E) = 1/k_B T$ and $(1/k_B)(\partial S_R/\partial N) = -\mu/k_B T$, by the definitions of Section 4.2, with T and μ those of the reservoir. Taking the exponential,

$$p_i \propto e^{-\beta(E_i - \mu N_i)} , \quad (4.69)$$

which is called the Gibbs factor. A microstate holding more particles is favoured when μ is large, because the entropy of the reservoir changes by μ/T for each particle it gives up, by the relation $(\partial S/\partial N)_{U,V} = -\mu/T$ of Section 4.2, and that change is the more favourable the higher the chemical potential.

The Grand Partition Function

Normalising in the same way as in Section 4.4,

$$p_i = \frac{e^{-\beta(E_i - \mu N_i)}}{\mathcal{Z}}, \quad \mathcal{Z} = \sum_i e^{-\beta(E_i - \mu N_i)}, \quad (4.70)$$

where the sum now runs over all microstates with every possible number of particles. The quantity $\mathcal{Z}$ is called the grand partition function, or the grand canonical partition function. The mean number of particles follows by differentiating with respect to μ , since $\partial e^{-\beta(E_i - \mu N_i)} / \partial \mu = \beta N_i e^{-\beta(E_i - \mu N_i)}$:

$$\bar{N} = \sum_i N_i p_i = \frac{1}{\beta} \left(\frac{\partial \ln \mathcal{Z}}{\partial \mu} \right)_{T,V}. \quad (4.71)$$

The Grand Potential

The thermodynamic function that goes with fixed T , V , and μ is called the grand potential

$$\Phi = U - TS - \mu N, \quad (4.72)$$

whose differential follows from the combined law of Section 3.3 as $d\Phi = -S dT - p dV - N d\mu$. The argument that gave $F = -k_B T \ln Z$ in Section 4.5 gives the corresponding result here. Taking the logarithm of the Gibbs probability, $\ln p_i = -\beta(E_i - \mu N_i) - \ln \mathcal{Z}$, and substituting into $S = -k_B \sum_i p_i \ln p_i$,

$$S = k_B \beta (\bar{E} - \mu \bar{N}) + k_B \ln \mathcal{Z}, \quad (4.73)$$

so that, multiplying by $-T$ and rearranging,

$$\Phi = -k_B T \ln \mathcal{Z}. \quad (4.74)$$

From Φ the rest follows by the derivatives implied by its differential: $S = -(\partial \Phi / \partial T)_{V,\mu}$, $p = -(\partial \Phi / \partial V)_{T,\mu}$, and $\bar{N} = -(\partial \Phi / \partial \mu)_{T,V}$. For a single-component system there is a further simplification.

Since G is extensive and depends on the intensive variables T and p together with N , it must be proportional to N , so $G = \mu N$ as Section 3.3 found, and then $\Phi = F - \mu N = G - pV - G = -pV$, where the free energy F has been written as $G - pV$ using the definitions of Section 3.1.

Fluctuations in the Number of Particles

Differentiating $\ln \mathcal{Z}$ twice with respect to μ , in exactly the way we differentiated twice with respect to β in Section 4.5, gives the variance of the particle number,

$$\sigma_N^2 = \overline{N^2} - \bar{N}^2 = \frac{1}{\beta^2} \left(\frac{\partial^2 \ln \mathcal{Z}}{\partial \mu^2} \right)_{T,V} = k_B T \left(\frac{\partial \bar{N}}{\partial \mu} \right)_{T,V}. \quad (4.75)$$

The spread of the particle number is tied to the response of the mean number to a change of chemical potential, just as the spread of the energy was tied to the heat capacity. Both $\bar{N}$ and $\partial \bar{N} / \partial \mu$ are proportional to the size of the system, so the relative spread $\sigma_N / \bar{N}$ is again of order $1/\sqrt{\bar{N}}$.

Molecules on a Surface

As an example, consider a solid surface carrying M sites at which a molecule of a surrounding gas may be held. Each site is either empty or holds exactly one molecule, and a molecule on a site has energy $-\varepsilon$, with $\varepsilon > 0$, since energy is released when it is captured. The surface is in contact with the gas, which acts as a reservoir at temperature T and chemical potential μ , and the sites do not interact with one another.

Since the sites are independent, the grand partition function of the whole surface is the product of the grand partition functions of the individual sites, by the argument of Section 4.5. There are two microstates of a single site: empty, with $E = 0$ and $N = 0$; and occupied, with $E = -\varepsilon$ and $N = 1$. So

$$\mathcal{Z}_1 = 1 + e^{\beta(\mu + \varepsilon)}, \quad \mathcal{Z} = \mathcal{Z}_1^M, \quad (4.76)$$

and the grand potential is $\Phi = -Mk_B T \ln(1 + e^{\beta(\mu+\varepsilon)})$. Differentiating Φ with respect to μ and reversing the sign, since $\bar{N} = -(\partial\Phi/\partial\mu)_{T,V}$, gives the mean number of molecules held, and dividing by M gives the fraction of sites that are occupied,

$$f = \frac{\bar{N}}{M} = \frac{e^{\beta(\mu+\varepsilon)}}{1 + e^{\beta(\mu+\varepsilon)}} = \frac{1}{e^{-\beta(\mu+\varepsilon)} + 1}. \quad (4.77)$$

The fraction rises smoothly from zero to one as μ increases, passing through $\frac{1}{2}$ when $\mu = -\varepsilon$, and the rise takes place over a range of μ of order $k_B T$ (Figure 31). When the chemical potential of the gas is well below $-\varepsilon$ the surface is bare, and when it is well above, almost every site is occupied. Raising the temperature at fixed μ makes the change more gradual, since the width of the rise is set by $k_B T$.

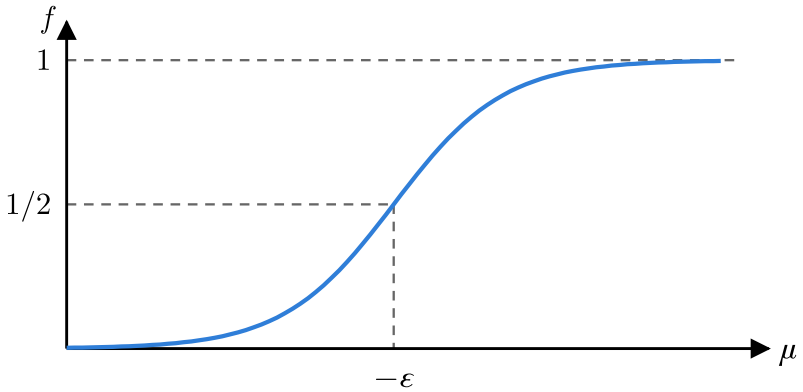


Figure 31. The fraction f of surface sites occupied, as a function of the chemical potential μ of the gas in contact with the surface. The curve passes through $f = \frac{1}{2}$ at $\mu = -\varepsilon$ and rises over a range of μ of order $k_B T$.

The Ensembles Compared

We now have three descriptions of a system in equilibrium. The microcanonical ensemble fixes U , V , and N , and gives every accessible microstate the same probability. The canonical ensemble fixes T , V , and N , and gives microstate i the probability $e^{-\beta E_i}/Z$. The

grand canonical ensemble fixes T , V , and μ , and gives microstate i the probability $e^{-\beta(E_i - \mu N_i)} / \mathcal{Z}$. Ensembles for other choices of the variables held fixed can be built in the same way, and these three are the ones this course uses.

For a macroscopic system the three usually give the same answers for the thermodynamic quantities. The reason is the smallness of the fluctuations. In the canonical ensemble the energy is spread about its mean by a relative amount of order $1/\sqrt{N}$, so fixing the temperature is, for all practical purposes, the same as fixing the energy at $\bar{E}$; in the grand canonical ensemble the particle number is spread by a relative amount of order $1/\sqrt{N}$ as well, so fixing μ is, for all practical purposes, the same as fixing N at $\bar{N}$. Any thermodynamic quantity calculated in one ensemble therefore agrees with the same quantity calculated in another, to within a relative error of order $1/\sqrt{N}$, which is too small to be detected in practice. The argument requires C_V and $(\partial \bar{N} / \partial \mu)_{T,V}$ to be of order N , and where one of them is very much larger the fluctuations are correspondingly larger and the ensembles need not agree.

Since the three descriptions agree, we are free to use whichever makes the sums easiest. Counting the microstates of an isolated system is usually the hardest of the three, because of the constraint of fixed energy, and the canonical and grand canonical sums have no such constraint, which is why they factorise so readily for independent parts.

Problems

§4.1 Probability for Many Particles

1. For a general probability p , apply the operation $p \partial / \partial p$ twice to $\sum_n \binom{N}{n} p^n q^{N-n} = (p+q)^N$, treating p and q as independent until the last step, and hence obtain $\bar{n} = Np$ and $\sigma_n^2 = Npq$.
2. A walker takes N steps of length ℓ , each to the right with probability p and to the left with probability $q = 1 - p$. Find $\bar{x}$ and $\overline{x^2} - \bar{x}^2$. Show that the mean displacement is small compared with the spread when $|p - q| \ll 1/\sqrt{N}$.
3. A more precise form of Stirling's approximation is $N! \approx$

$\sqrt{2\pi N} (N/e)^N$, which keeps the term of order $\ln N$ that the form used in the text drops. Use it to show that the largest value of the symmetric binomial distribution is $P(\frac{1}{2}N) \approx \sqrt{2/\pi N}$, in agreement with the normalised Gaussian found in the text, and check that the Gaussian of width $\sigma = \sqrt{N}/2$ with this peak value has total probability one.

4. A walker takes N independent steps whose lengths are drawn from some distribution with mean zero and variance ℓ^2 , the same for every step. Show that $\overline{x^2} = N\ell^2$ whatever the shape of that distribution, and state what the central limit theorem says about the distribution of x for large N .

§4.2 Microstates, Multiplicity, and Entropy

1. A system has $\Omega \propto U^\alpha N$ with α a constant. Find $S(U)$ and show that $U = \alpha N k_B T$. What is the heat capacity at constant volume?
2. Consider N objects each with two states of energies 0 and ε , so that $\Omega = N!/[n!(N-n)!]$ with $U = n\varepsilon$. Use Stirling's approximation to find S as a function of n , and hence show that $n/N = 1/(e^{\varepsilon/k_B T} + 1)$.
3. Let two systems with $\Omega_i \propto U_i^{\alpha N_i}$ share a fixed total energy U . Find the most probable division U_1^* , and show that the fractional width of the peak in U_1 is of order $1/\sqrt{N_1}$ when the two systems are of comparable size.
4. Let two systems be separated by a wall that passes energy and is free to slide, with $U_1 + U_2$ and $V_1 + V_2$ both fixed. By maximising $S_1 + S_2$ with respect to U_1 and V_1 , show that the temperatures are equal and the pressures are equal in equilibrium.

§4.3 Counting States of an Ideal Gas

1. Differentiate the Sackur–Tetrode entropy with respect to N at fixed U and V to find the chemical potential of a monatomic ideal gas. Using $U = \frac{3}{2}Nk_B T$, show that $\mu = k_B T \ln(n/n_Q)$ where $n = N/V$ and $n_Q = (2\pi m k_B T/h^2)^{3/2}$.

2. Let two different ideal gases at the same temperature and pressure be mixed, with $N = N_A + N_B$ and mole fractions $x_A = N_A/N$ and $x_B = N_B/N$. Show that $\Delta S = -Nk_B(x_A \ln x_A + x_B \ln x_B)$.
3. A system has just two accessible microstates, with probabilities p and $1 - p$. Show from $S = -k_B \sum_i p_i \ln p_i$ that S is largest when $p = \frac{1}{2}$, and find its value there.
4. Suppose the cell volume in the space of position and momentum were taken as $c h^3$ rather than h^3 , with c a constant. Show that the entropy changes by $-Nk_B \ln c$, that T and p are unaffected, and that μ shifts by $k_B T \ln c$.

§4.4 The Boltzmann Factor and the Partition Function

1. A system has three states, of energies 0 , ε , and 2ε . Find Z and $\bar{E}$, and find the limits of $\bar{E}$ at high and at low temperature.
2. Every energy level of a system is raised by the same constant E_0 . Show that Z is multiplied by $e^{-\beta E_0}$, that the probabilities p_i are unchanged, and that $\bar{E}$ increases by E_0 .
3. For the harmonic oscillator, expand $\bar{E} = \frac{1}{2}\hbar\omega + \hbar\omega/(e^{\beta\hbar\omega} - 1)$ for $k_B T \gg \hbar\omega$ as far as the term independent of T , and find the leading behaviour for $k_B T \ll \hbar\omega$.
4. Molecules of mass m in an isothermal atmosphere have potential energy mgz at height z . Use the Boltzmann factor to show that the number density goes as $n(z) = n(0)e^{-mgz/k_B T}$, and find the mean height of a molecule above the ground.

§4.5 Free Energy and Fluctuations

1. Use $\sigma_E^2 = k_B T^2 C_V$ with $C_V = \frac{3}{2}Nk_B$ and $\bar{E} = \frac{3}{2}Nk_B T$ to show that the relative spread of the energy of a monatomic ideal gas at fixed temperature is $\sqrt{2/3N}$.
2. A system consists of N independent distinguishable subsystems, each with partition function z . Write down F , and show that $\bar{E} = -N \partial \ln z / \partial \beta$ and $S = Nk_B (\ln z - \beta \partial \ln z / \partial \beta)$.

3. A degree of freedom has energy $E = b|q|$, with q running over all real values. Show that its mean energy is $k_B T$, and compare with the quadratic case. (The integrals over positive q are elementary.)
4. Starting from $F = -Nk_B T[\ln(z/N) + 1]$ for N indistinguishable particles, show that $\mu = -k_B T \ln(z/N)$, and that the same F for distinguishable particles, $-Nk_B T \ln z$, gives a chemical potential $-k_B T \ln z$ that is not intensive.

§4.6 The Grand Canonical Ensemble

1. For the surface of M independent sites, use $\sigma_N^2 = k_B T (\partial \bar{N} / \partial \mu)_{T,V}$ to show that $\sigma_N^2 = M f(1-f)$. Why is this the same as the variance of the binomial distribution of Section 4.1?
2. The gas above the surface is a monatomic ideal gas of number density n , whose chemical potential was found in Problem 1 of Section 4.3 to be $\mu = k_B T \ln(n/n_Q)$. Show that the fraction of occupied sites is $f = n/(n + n_Q e^{-\varepsilon/k_B T})$, and that at fixed temperature f rises from zero towards one as the pressure of the gas is raised.
3. From $\Phi = -Mk_B T \ln(1 + e^{\beta(\mu+\varepsilon)})$ obtain the entropy $S = -(\partial \Phi / \partial T)_{V,\mu}$, and show that it equals $-Mk_B [f \ln f + (1-f) \ln(1-f)]$, which is what $S = -k_B \sum_i p_i \ln p_i$ gives for M independent sites.
4. Let two systems exchange particles through a wall at a common temperature, with $N_1 + N_2$ fixed. By maximising $S_1 + S_2$ with respect to N_1 , show that equilibrium requires $\mu_1 = \mu_2$, and show that before equilibrium is reached particles move from the system of higher chemical potential to the system of lower chemical potential.

Statistical Mechanics of Ideal Classical Systems

5.1 The Ideal Gas from the Partition Function

In Section 4.3 we counted the states of an ideal gas by hand and obtained its entropy, and in Sections 4.4 and 4.5 we introduced the partition function and showed that the free energy is $F = -k_B T \ln Z$. This section puts the counting of states and the partition function together. Once the partition function of a gas has been written down, the pressure, the entropy, the energy, and the chemical potential all follow by differentiation, without any further physical input. The sections that follow in this part apply the same method first to molecules that can rotate and vibrate, then to spins in a magnetic field, then to the atoms of a solid, and finally to the radiation in a hot cavity.

The Energy Levels of a Particle in a Box

We quote from quantum mechanics the energy levels of a single particle of mass m confined to a cubical box of side L with impenetrable walls. They are

$$\epsilon_{n_x n_y n_z} = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2) , \quad (5.1)$$

where n_x , n_y , and n_z are positive integers. Each triple of integers labels one quantum state of the particle, so the sum over states that defines the partition function is a sum over all three integers independently,

$$Z_1 = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} e^{-\beta \epsilon_{n_x n_y n_z}} . \quad (5.2)$$

Turning the Sum into an Integral

For a box of any ordinary size the prefactor $h^2/8mL^2$ is minute compared with $k_B T$, since h is very small and L is very large on the scale of a molecule, so a great many terms of the sum are of comparable size and the energy changes very little from one state to the next. We may therefore treat each of the three integers as a continuous variable and replace the sum by an integral.

The energy depends on the three integers only through the combination $n = \sqrt{n_x^2 + n_y^2 + n_z^2}$, which we call the mode number. If we think of n_x , n_y , and n_z as Cartesian coordinates in a space of mode numbers, the states sit at the corners of every unit cell of a cubic lattice of unit spacing, so that there is one state per unit cell, and all three coordinates are positive, so the states occupy one octant. The states whose mode number lies between n and $n + dn$ lie in a thin shell, and since one eighth of a complete spherical shell lies in the octant, their number is

$$\frac{1}{8} \cdot 4\pi n^2 dn = \frac{1}{2} \pi n^2 dn . \quad (5.3)$$

This is drawn in Figure 32.

The single-particle partition function is therefore

$$Z_1 = \int_0^{\infty} \frac{\pi n^2}{2} e^{-\beta h^2 n^2 / 8mL^2} dn . \quad (5.4)$$

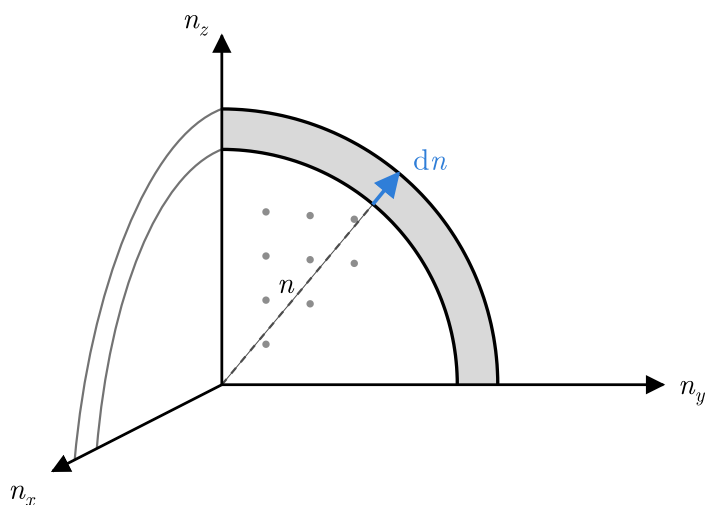


Figure 32. The states of a particle in a cubical box, plotted in the space whose coordinates are the mode numbers n_x , n_y , and n_z . There is one state per unit cell of a cubic lattice, all three coordinates are positive, and the energy depends only on the distance n from the origin. The states whose mode number lies between n and $n+dn$ fill the octant of a spherical shell, whose cross-section in the plane of n_y and n_z is shaded here.

We need $\int_0^\infty n^2 e^{-an^2} dn$, which we obtain by parts from the Gaussian integral worked out in Section 1.2. Integrating by parts, with the integrand written as $n \times ne^{-an^2}$ so that the second of the two factors is the one integrated,

$$\int_0^\infty n^2 e^{-an^2} dn = \left[-\frac{n}{2a} e^{-an^2} \right]_0^\infty + \frac{1}{2a} \int_0^\infty e^{-an^2} dn = \frac{1}{4a} \sqrt{\frac{\pi}{a}}. \quad (5.5)$$

With $a = \beta h^2 / 8mL^2$ and $L^3 = V$, the result is

$$Z_1 = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} = \frac{V}{\lambda_{\text{th}}^3}, \quad \lambda_{\text{th}} = \frac{h}{\sqrt{2\pi m k_B T}}. \quad (5.6)$$

The Thermal Wavelength

The length λ_{th} is called the thermal wavelength. It is the de Broglie wavelength h/p of a particle whose momentum is $p = \sqrt{2\pi m k_B T}$, and a particle with that momentum has a kinetic energy of order $k_B T$. So λ_{th} is, to within a factor of order one, the wavelength of a typical atom or molecule of the gas, and it shrinks as the temperature rises.

The partition function V/λ_{th}^3 can then be given a direct interpretation: it is the number of cubes of side λ_{th} that fit into the box, and therefore a measure of how many single-particle states are accessible at temperature T . A larger box or a higher temperature makes more of them available.

The Gas of N Particles

If the atoms or molecules of the gas do not interact, the energy of the gas is the sum of their separate energies, and the exponential of a sum is a product, so a sum over all the states of the gas would give Z_1^N . That expression counts too many states. Constituents of the same kind cannot be told apart, so a list saying which molecule occupies which single-particle state is not a distinct state of the gas; only the set of occupied states matters. Each set is counted $N!$

times, once for every way of permuting the labels, and we correct for this by writing

$$Z_N = \frac{Z_1^N}{N!}. \quad (5.7)$$

The correction by exactly $N!$ is valid as long as it is very unlikely that two molecules are in the same single-particle state, since a permutation of two molecules in the same state is not a genuine rearrangement. We shall see at the end of this section that the requirement that two molecules very rarely occupy the same state is also the condition that makes the classical treatment valid. The factor of $N!$ is the same one that was needed in Section 4.3 to make the entropy proportional to the size of the sample.

The Free Energy and What Follows from It

Taking the logarithm and using Stirling's approximation $\ln N! \approx N \ln N - N$ from Section 4.1,

$$\ln Z_N = N \ln \frac{V}{\lambda_{\text{th}}^3} - N \ln N + N, \quad (5.8)$$

so that

$$F = -k_B T \ln Z_N = -N k_B T \left[\ln \left(\frac{V}{N \lambda_{\text{th}}^3} \right) + 1 \right]. \quad (5.9)$$

From Section 3.1 the pressure, the entropy, and the chemical potential all follow from first derivatives of F . Taking them in turn, and remembering that λ_{th} depends on T but not on V or N , the pressure is

$$p = - \left(\frac{\partial F}{\partial V} \right)_{T,N} = \frac{N k_B T}{V}, \quad (5.10)$$

which is the ideal gas law. For the entropy we write $\lambda_{\text{th}}^{-3} \propto T^{3/2}$, so that $F = -N k_B T [\ln(V/N) + \frac{3}{2} \ln T + \text{constant} + 1]$. Differentiating by the product rule, the factor of T in front of the bracket

returns the bracket itself, and the $\frac{3}{2} \ln T$ inside it contributes a further $Nk_B T \times \frac{3}{2T}$, so that

$$\begin{aligned} S = - \left(\frac{\partial F}{\partial T} \right)_{V,N} &= Nk_B \left[\ln \left(\frac{V}{N\lambda_{\text{th}}^3} \right) + 1 \right] + \frac{3}{2} Nk_B \\ &= Nk_B \left[\ln \left(\frac{V}{N\lambda_{\text{th}}^3} \right) + \frac{5}{2} \right]. \end{aligned} \quad (5.11)$$

This is called the Sackur–Tetrode entropy, and it is the same expression that came out of the direct count of states in Section 4.3. The energy follows from $U = F + TS$, in which the two logarithms, one from F and one from TS , cancel against each other:

$$\begin{aligned} U = -Nk_B T \left[\ln \left(\frac{V}{N\lambda_{\text{th}}^3} \right) + 1 \right] + Nk_B T \left[\ln \left(\frac{V}{N\lambda_{\text{th}}^3} \right) + \frac{5}{2} \right] \\ = \frac{3}{2} Nk_B T, \end{aligned} \quad (5.12)$$

so $C_V = \frac{3}{2} Nk_B$, in agreement with equipartition applied to the three components of the momentum, as in Section 4.5. Finally we differentiate F with respect to N at fixed T and V . Again by the product rule, the factor of N in front of the bracket and the $-\ln N$ inside it both contribute, so that

$$\begin{aligned} \mu = \left(\frac{\partial F}{\partial N} \right)_{T,V} &= -k_B T \left[\ln \left(\frac{V}{N\lambda_{\text{th}}^3} \right) + 1 \right] + k_B T \\ &= k_B T \ln \left(\frac{N\lambda_{\text{th}}^3}{V} \right), \end{aligned} \quad (5.13)$$

where the last step uses the fact that a logarithm changes sign when its argument is inverted.

When the Classical Treatment Applies

The quantity $N\lambda_{\text{th}}^3/V$ that appears in the chemical potential is the number of molecules in a cube of side λ_{th} . The condition for the results of this section to hold is

$$\frac{N\lambda_{\text{th}}^3}{V} \ll 1, \quad (5.14)$$

which says that the mean spacing between molecules, $(V/N)^{1/3}$, is much larger than the thermal wavelength. When that is so, the molecules are spread thinly over the available single-particle states and the chance of two of them being in the same state is negligible, which is what the factor $N!$ required. It also makes μ large and negative, so that in the language of Section 4.6 the factor $e^{\beta\mu} = N\lambda_{\text{th}}^3/V$ is small and any particular single-particle state is nearly always empty.

The condition is broken by making the gas dense or the temperature low, since λ_{th} grows as $T^{-1/2}$. For the gases of the atmosphere at room temperature it holds by a very wide margin. For liquid helium near its boiling point, or for the conduction electrons in a metal, where the mass is small and the density high, it fails completely, and the results of this section do not apply to them. The statistical mechanics of a gas in that condition is not treated in this course.

5.2 Internal Degrees of Freedom

A molecule has more structure than an atom and cannot always be approximated as a point particle. A pair of atoms joined by a bond can turn end over end and can stretch along the bond, and each of those motions contributes to the energy of the molecule. Section 1.2 quoted that the measured heat capacity of a diatomic gas at room temperature is close to $\frac{5}{2}Nk_B$, which is neither the $\frac{3}{2}Nk_B$ of a gas of atoms nor the $\frac{7}{2}Nk_B$ that equipartition gives when the rotation and the vibration are included, and that the measured value changes with temperature. We are now in a position to explain this.

Factorising the Partition Function

To a good approximation the energy of a molecule separates into a part belonging to the motion of its centre of mass, a part belonging to its rotation, and a part belonging to its vibration,

$$\epsilon = \epsilon_{\text{tr}} + \epsilon_{\text{rot}} + \epsilon_{\text{vib}}. \quad (5.15)$$

A state of the molecule is fixed by giving the state of each of the three motions, and the sum over all states of the molecule runs over all combinations of them. Since the exponential of a sum is a product, the sum factorises:

$$Z_1 = \sum_{\text{tr}} \sum_{\text{rot}} \sum_{\text{vib}} e^{-\beta\epsilon_{\text{tr}}} e^{-\beta\epsilon_{\text{rot}}} e^{-\beta\epsilon_{\text{vib}}} = Z_{\text{tr}} Z_{\text{rot}} Z_{\text{vib}}. \quad (5.16)$$

The logarithm of a product is a sum of logarithms, so the free energies add, and with them the energies, the entropies, and the heat capacities. The translational factor is the V/λ_{th}^3 of Section 5.1, and it alone depends on the volume. The rotational and vibrational factors depend on the temperature and not on the volume, so they drop out of $p = -(\partial F/\partial V)_T$ and the equation of state remains $pV = Nk_B T$. Internal structure changes the heat capacity of a gas without changing its equation of state.

The Vibrational Partition Function

The stretching of the bond is, for small displacements, a harmonic oscillation of some angular frequency ω , since a system held near a minimum of its potential energy moves harmonically, as we saw when studying oscillations in the mechanics notes. The energies of a harmonic oscillator are taken from quantum mechanics,

$$\epsilon_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad n = 0, 1, 2, \dots. \quad (5.17)$$

There is one state for each value of n , so the partition function is

$$Z_{\text{vib}} = \sum_{n=0}^{\infty} e^{-\beta(n+\frac{1}{2})\hbar\omega} = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} r^n, \quad r = e^{-\beta\hbar\omega}. \quad (5.18)$$

The sum is a geometric series, and its common ratio r is less than one at any positive temperature. Let s be the value of the sum, so that $s = 1 + r + r^2 + \dots$; multiplying by r shifts the series along by one term, giving $rs = r + r^2 + r^3 + \dots = s - 1$, and solving for s gives $s = 1/(1 - r)$. The partition function is therefore

$$Z_{\text{vib}} = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}. \quad (5.19)$$

The mean energy follows from $\bar{\epsilon} = -\partial \ln Z / \partial \beta$, which was obtained in Section 4.4. Here $\ln Z_{\text{vib}} = -\frac{1}{2}\beta\hbar\omega - \ln(1 - e^{-\beta\hbar\omega})$, and the derivative of the second logarithm with respect to β is $\hbar\omega e^{-\beta\hbar\omega} / (1 - e^{-\beta\hbar\omega})$, so that

$$\bar{\epsilon}_{\text{vib}} = \frac{1}{2}\hbar\omega + \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} = \frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1}, \quad (5.20)$$

where the second form comes from multiplying the numerator and the denominator by $e^{\beta\hbar\omega}$.

The first term is the energy of the oscillator in its lowest state and does not depend on the temperature. Writing $\theta_v = \hbar\omega/k_B$ for the characteristic temperature of the vibration and differentiating the second term with respect to T , the vibrational contribution to the heat capacity of N molecules is

$$C_{\text{vib}} = Nk_B \left(\frac{\theta_v}{T} \right)^2 \frac{e^{\theta_v/T}}{(e^{\theta_v/T} - 1)^2}. \quad (5.21)$$

When $T \gg \theta_v$ the exponent θ_v/T is small, so that $e^{\theta_v/T} - 1 \approx \theta_v/T$, and $C_{\text{vib}} \rightarrow Nk_B$. That is the value equipartition leads to for an oscillator, whose mean energy is $\frac{1}{2}k_B T$ from the kinetic term and $\frac{1}{2}k_B T$ from the potential term, as in Section 4.5. When $T \ll \theta_v$ the exponential in the denominator dominates and

$$C_{\text{vib}} \approx Nk_B \left(\frac{\theta_v}{T} \right)^2 e^{-\theta_v/T}, \quad (5.22)$$

which falls to zero very rapidly as the temperature drops. The reason is that the lowest excited state of the oscillator lies an energy $\hbar\omega$ above the ground state, and the Boltzmann factor $e^{-\theta_v/T}$ for reaching it is very small, so almost every molecule is vibrating in its ground state and a small rise in temperature changes the energy of the gas hardly at all. A degree of freedom in this condition is said to be frozen out.

The Rotational Partition Function

For the rotation, quantum mechanics gives the energy levels of a rigid rotor as

$$\epsilon_\ell = \frac{\hbar^2}{2I} \ell(\ell + 1), \quad \ell = 0, 1, 2, \dots, \quad (5.23)$$

where I is the moment of inertia about an axis through the centre of mass and perpendicular to the bond, defined as it was for rigid bodies in the mechanics notes. The level labelled by ℓ is degenerate: there are $2\ell + 1$ distinct states with that energy, and each of them contributes a term to the sum. Writing $\theta_r = \hbar^2/2Ik_B$,

$$Z_{\text{rot}} = \sum_{\ell=0}^{\infty} (2\ell + 1) e^{-\theta_r \ell(\ell+1)/T}. \quad (5.24)$$

When $T \gg \theta_r$ successive terms differ very little and the sum may be replaced by an integral over ℓ . The substitution $u = \ell(\ell + 1)$, for which $du = (2\ell + 1) d\ell$, removes the degeneracy factor and leaves

$$Z_{\text{rot}} \approx \int_0^{\infty} e^{-\theta_r u/T} du = \frac{T}{\theta_r}. \quad (5.25)$$

Then $\ln Z_{\text{rot}} = \ln T - \ln \theta_r$, which differs by a constant from $-\ln \beta$, since $T = 1/k_B\beta$, so that differentiating with respect to β gives $\bar{\epsilon}_{\text{rot}} = k_B T$ and $C_{\text{rot}} = Nk_B$. This is the equipartition result for a body that can turn about two axes, since a linear molecule has no appreciable moment of inertia about the bond itself. For a molecule

whose two atoms are the same, the high-temperature partition function carries an extra factor of $\frac{1}{2}$; since that factor is a constant it shifts the entropy and leaves the heat capacity unchanged.

When $T \ll \theta_r$ the exponentials fall away quickly and we keep the first two terms of the sum,

$$Z_{\text{rot}} \approx 1 + 3e^{-2\theta_r/T}. \quad (5.26)$$

Since the second term is small, the expansion $\ln(1 + \delta) \approx \delta$ for small δ gives $\ln Z_{\text{rot}} \approx 3e^{-2\theta_r/T}$, and differentiating with respect to β leaves $\bar{\epsilon}_{\text{rot}} \approx 6k_B\theta_r e^{-2\theta_r/T}$ and

$$C_{\text{rot}} \approx 12Nk_B \left(\frac{\theta_r}{T}\right)^2 e^{-2\theta_r/T}, \quad (5.27)$$

so the rotation freezes out as the vibration does, because at low temperature the Boltzmann factor for reaching the first excited level of the rotor is again very small.

The Heat Capacity of a Diatomic Gas

Adding the three contributions, the heat capacity at constant volume of a diatomic gas is

$$\frac{C_V}{Nk_B} = \frac{3}{2} + \frac{C_{\text{rot}}}{Nk_B} + \frac{C_{\text{vib}}}{Nk_B}, \quad (5.28)$$

and Figure 33 shows it as a function of temperature. The curve has two steps. At temperatures well below θ_r only the translation carries energy and $C_V = \frac{3}{2}Nk_B$, as for a gas of points. As T passes θ_r the rotation begins to take up energy, and its contribution to the heat capacity rises a little above Nk_B before falling back to that value, so that the heat capacity reaches $\frac{5}{2}Nk_B$. It stays there over a wide range, because θ_v is very much larger than θ_r , and then rises to $\frac{7}{2}Nk_B$ as T passes θ_v .

For most diatomic gases θ_r is a few tens of kelvin and θ_v a few thousand, so room temperature falls in the middle range and the measured heat capacity is close to $\frac{5}{2}Nk_B$. That is the value quoted

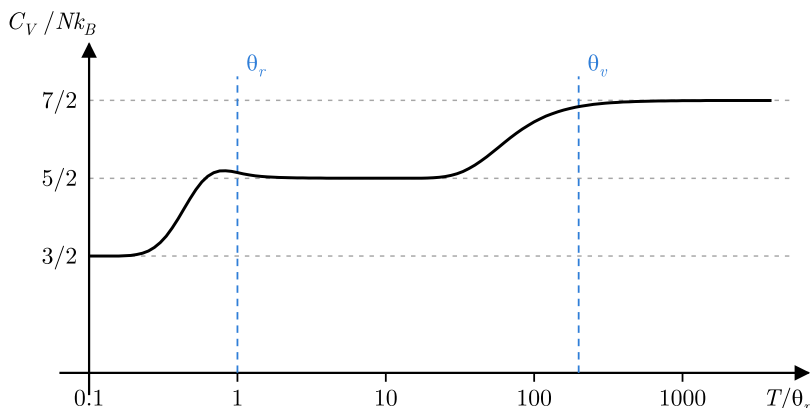


Figure 33. The heat capacity of a diatomic gas against temperature on a logarithmic scale, drawn for $\theta_v = 200 \theta_r$. The rotational contribution appears near $T = \theta_r$ and the vibrational contribution near $T = \theta_v$, giving the two steps between $\frac{3}{2}Nk_B$, $\frac{5}{2}Nk_B$, and $\frac{7}{2}Nk_B$.

in Section 1.2, and we now understand why this is the case. Equipartition assumes that energy can be shared out in arbitrarily small amounts, and it is a valid approximation for a given degree of freedom only when $k_B T$ is large compared with the spacing of that degree of freedom's energy levels. The rotational levels are closely spaced because I is large, and the vibrational levels are widely spaced because the bond is stiff, so at room temperature $k_B T$ is large compared with the rotational spacing and small compared with the vibrational one.

5.3 Two-Level Systems and Paramagnetism

The simplest system whose partition function can be written down in a single line is one with two energy levels and nothing else, and it was the first example worked out in Section 4.4, where the partition function of levels at 0 and ε came to $Z = 1 + e^{-\beta\varepsilon}$. Systems of this kind are common: an atom with a low-lying excited state, a defect in a crystal that can sit in either of two positions, and a spin in a magnetic field are all described by two levels. They behave in ways that have no counterpart in a gas; the energy of such a system cannot be raised without limit, and a set of spins can be prepared

in a state whose temperature comes out negative, which takes some care to interpret correctly.

The Two-Level System

Let the two levels have energies 0 and Δ , with one state at each. The partition function is

$$Z = 1 + e^{-\beta\Delta}, \quad (5.29)$$

and the probabilities of the two levels are $1/Z$ and $e^{-\beta\Delta}/Z$. The mean energy of one such system is therefore

$$\bar{\epsilon} = \frac{\Delta e^{-\beta\Delta}}{1 + e^{-\beta\Delta}} = \frac{\Delta}{e^{\Delta/k_B T} + 1}. \quad (5.30)$$

If a solid contains N of these two-level systems, fixed at distinct sites so that they can be told apart, then $Z_N = Z^N$ with no factor of $N!$, and the energy of the solid is $U = N\bar{\epsilon}$. At low temperature $U \rightarrow 0$, since every system sits in its lower level. At high temperature the two Boltzmann factors become equal, each level is occupied half the time, and $U \rightarrow N\Delta/2$. However hot the solid is made, the energy never rises above half of $N\Delta$ while the temperature is positive.

Differentiating U with respect to T , and writing $x = \Delta/k_B T$,

$$C = Nk_B \frac{x^2 e^x}{(e^x + 1)^2}, \quad (5.31)$$

which is drawn in Figure 34. The heat capacity vanishes at both ends of the temperature range and has a maximum in between, near $k_B T = 0.42 \Delta$. A peak of this shape is called a Schottky anomaly, and it appears in the heat capacity of a paramagnetic salt at low temperature.

The vanishing of the heat capacity at each end of the temperature range has a simple explanation. At low temperature the gap is large compared with $k_B T$, so heat supplied to the solid can excite almost nothing. At high temperature the two levels are already almost equally occupied, so further heating can do little to change

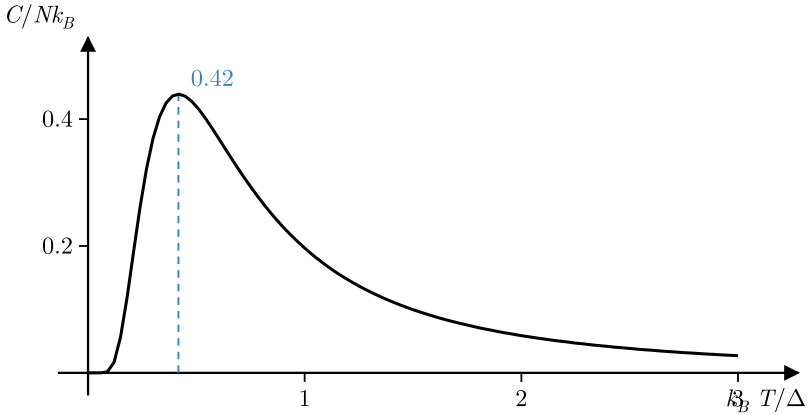


Figure 34. The heat capacity of N two-level systems with splitting Δ , against $k_B T / \Delta$. The maximum lies near $k_B T = 0.42 \Delta$, and the heat capacity falls to zero on both sides of it.

the occupations. In either case the energy hardly changes, and only when $k_B T$ is comparable with Δ does a small rise in temperature move an appreciable number of systems from the lower level to the upper one.

The entropy is most easily obtained from $S = (U - F)/T$, using $F = -Nk_B T \ln(1 + e^{-\beta\Delta})$ together with the energy $U = N\Delta/(e^{\beta\Delta} + 1)$ found above. Dividing each of those two expressions by T , and using $\Delta/T = k_B \beta \Delta$,

$$S = Nk_B \left[\ln(1 + e^{-\beta\Delta}) + \frac{\beta\Delta}{e^{\beta\Delta} + 1} \right], \quad (5.32)$$

which rises from zero at $T = 0$ to $Nk_B \ln 2$ at high temperature. The high-temperature value agrees with the direct count of Section 4.2; when both levels are equally likely there are 2^N arrangements of comparable probability, and $k_B \ln 2^N = Nk_B \ln 2$.

The Spin-Half Paramagnet

Quantum mechanics tells us that a particle of spin one half with a magnetic moment of magnitude μ_B , placed in a magnetic field B , has two energy levels, $-\mu_B B$ for the moment along the field and

$+\mu_B B$ for the moment against it. This is a two-level system with $\Delta = 2\mu_B B$, and

$$Z = e^{\beta\mu_B B} + e^{-\beta\mu_B B} = 2 \cosh \frac{\mu_B B}{k_B T}. \quad (5.33)$$

Let us write $y = \mu_B B/k_B T$ for the ratio of the magnetic energy $\mu_B B$ to the thermal energy $k_B T$, so that $Z = 2 \cosh y$. The letter x has already been used in this section for the ratio $\Delta/k_B T$ of the two-level system, and since $\Delta = 2\mu_B B$ here, the ratios x and y differ by a factor of two.

For N such moments on the sites of a solid, $F = -Nk_B T \ln [2 \cosh(\mu_B B/k_B T)]$. The work done on the system when the field is changed by dB is $-M dB$, where M is the total magnetic moment along the field, so M stands in the same relation to B as p does to V in Section 3.1, and

$$M = -\left(\frac{\partial F}{\partial B}\right)_T = Nk_B T \frac{\partial}{\partial B} \ln \left[2 \cosh \frac{\mu_B B}{k_B T} \right] = N\mu_B \tanh \frac{\mu_B B}{k_B T}, \quad (5.34)$$

where the last step uses the derivative of $\ln \cosh y$, which is $\tanh y$, together with $\partial y/\partial B = \mu_B/k_B T$.

Figure 35 shows this magnetisation. When $y \gg 1$ the hyperbolic tangent approaches one and $M \rightarrow N\mu_B$; every moment lies along the field, and the magnetisation cannot grow beyond that. This is called saturation. When $y \ll 1$ we use $\tanh y \approx y$ and obtain

$$M \approx \frac{N\mu_B^2 B}{k_B T}. \quad (5.35)$$

This is Curie's law, and it was known from measurement long before it was derived from first principles. The picture behind saturation and behind Curie's law is the same one: the field tends to line the moments up, thermal agitation tends to disorder them, and the ratio $\mu_B B/k_B T$ measures which of those two tendencies prevails.

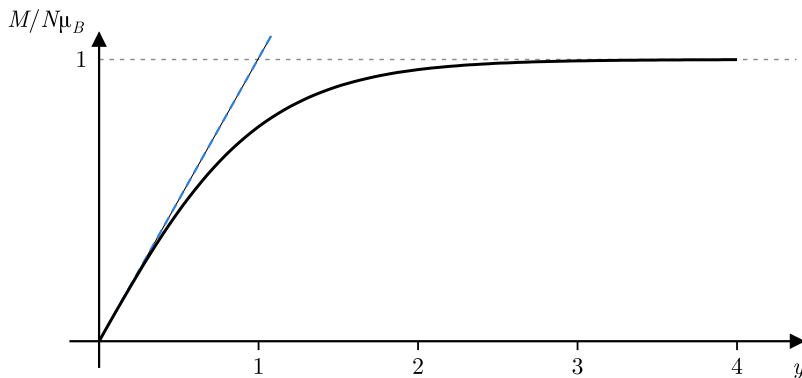


Figure 35. The magnetisation of a spin-half paramagnet against $y = \mu_B B/k_B T$. The dashed straight line is the Curie law $M = N\mu_B y$, which the curve follows at small y , and the magnetisation saturates at $N\mu_B$.

Negative Temperature

Let us now take the spins to be isolated and ask what temperature means for them. The energy of the system is $E = -MB$, and it runs from $-N\mu_B B$, with every moment along the field, to $+N\mu_B B$, with every moment against it. If $N_\uparrow$ moments lie along the field and $N_\downarrow = N - N_\uparrow$ against it, the number of arrangements is $\Omega = N!/(N_\uparrow! N_\downarrow!)$. Stirling's approximation from Section 4.1 replaces each factorial, leaving $\ln \Omega = N \ln N - N_\uparrow \ln N_\uparrow - N_\downarrow \ln N_\downarrow$, since the terms $-N$, $+N_\uparrow$, and $+N_\downarrow$ add to nothing. Writing $P = N_\uparrow/N$ for the fraction of the moments that lie along the field, and $S = k_B \ln \Omega$,

$$\frac{S}{Nk_B} = -P \ln P - (1 - P) \ln(1 - P), \quad P = \frac{1}{2} \left(1 - \frac{E}{N\mu_B B} \right). \quad (5.36)$$

Figure 36 shows this entropy as a function of the energy. It is zero at the lowest energy, where there is one arrangement; it rises to $Nk_B \ln 2$ at $E = 0$, where the moments are equally divided and the number of arrangements is largest; and it falls back to zero at the highest energy, where again there is one arrangement.

The statistical definition of temperature from Section 4.2 is $1/T = (\partial S / \partial E)$. On the left half of the curve the slope is positive and the

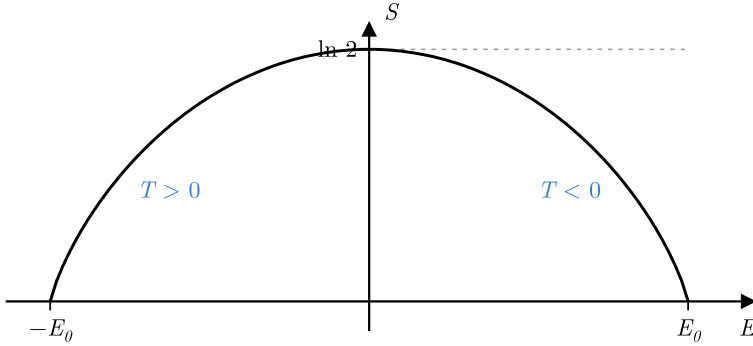


Figure 36. The entropy of N isolated spins against their energy, with $E_0 = N\mu_B B$. On the left half the slope is positive and the temperature is positive; at the maximum the slope is zero and the temperature is infinite; on the right half the slope is negative and the temperature is negative.

temperature is positive, as we are used to. At the maximum the slope is zero and $1/T = 0$, which is to say that the temperature is infinite. On the right half the slope is negative, and so is the temperature.

It seems at first as though a negative temperature must be colder than absolute zero. This turns out not to be the case, and the reason is the way the temperature runs as energy is added: it starts at 0^+ , passes through large positive values, becomes infinite at $E = 0$, and then comes back from minus infinity towards 0^- at the highest energy. The quantity that changes smoothly throughout is $\beta = 1/k_B T$, which decreases steadily from $+\infty$ through zero to $-\infty$. A system at a negative temperature is hotter than any system at a positive temperature, in the sense that if the two systems are put in thermal contact, energy flows out of the system at negative temperature.

A system must meet two conditions before a negative temperature is possible for it. The first is that the energy of the system must be bounded above, so that the entropy can start to fall as the energy rises. A gas has no such bound, because the kinetic energy of its molecules can be made as large as we please, and the entropy of a gas rises with energy without limit; negative temperatures are

impossible for it. The second is isolation. The spins sit in a crystal whose atoms can vibrate, and the vibrations have no upper bound on their energy, so a state of the spins with $E > 0$ lasts only as long as it takes the spins to exchange energy with the rest of the crystal. In practice such a state can be prepared by reversing the applied field faster than the moments can follow, and it then relaxes to a positive temperature over that exchange time.

5.4 The Einstein Solid

The atoms of a crystal sit near fixed sites and move only a little way from them. A small displacement from a site meets a restoring force that is, to a first approximation, proportional to the displacement, so each atom behaves like a three-dimensional harmonic oscillator. Einstein's model of a solid takes this picture at its simplest: a crystal of N atoms is treated as $3N$ independent harmonic oscillators, one for each atom and each direction, all with the same angular frequency ω . We will work the model out in two ways, first by counting arrangements and then through the partition function, because each of the two routes provides a complementary perspective.

Counting the Ways of Sharing the Quanta

Measuring the energy of each oscillator from its lowest level, the energies of an oscillator are $0, \hbar\omega, 2\hbar\omega, \dots$, so the state of the solid is fixed by saying how many quanta each oscillator holds. Let the total number of quanta be q , so that the energy above the lowest state is $E = q\hbar\omega$.

The multiplicity Ω is the number of ways of sharing q identical quanta among the $3N$ oscillators, no two of which are alike, since each one belongs to a particular atom and a particular direction of motion.

We count the ways by laying out a row of q dots for the quanta together with $3N - 1$ dividers, as in Figure 37, and taking each arrangement of the row to be a distribution: the dots before the first divider go to the first oscillator, those between the first and second dividers to the second, and so on. Every arrangement gives one distribution and every distribution comes from one arrangement,

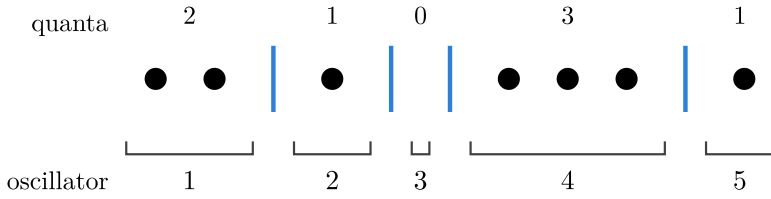


Figure 37. One arrangement of a row of dots and dividers, drawn for $q = 7$ quanta shared among five oscillators. The dots lying between one divider and the next belong to one oscillator, so this arrangement gives two quanta to the first oscillator, one to the second, none to the third, three to the fourth, and one to the fifth. Every ordering of the row corresponds to one way of sharing the quanta out.

and there are $(q + 3N - 1)!$ orderings of the row, of which the $q!$ permutations of the dots and the $(3N - 1)!$ permutations of the dividers change nothing. So

$$\Omega = \frac{(q + 3N - 1)!}{q! (3N - 1)!}. \quad (5.37)$$

With q and N both very large the difference between $3N$ and $3N - 1$ is of no consequence, and Stirling's approximation from Section 4.1 gives

$$\ln \Omega \approx (q + 3N) \ln(q + 3N) - q \ln q - 3N \ln 3N. \quad (5.38)$$

The Temperature and the Heat Capacity

The entropy is $S = k_B \ln \Omega$ from Section 4.2, and the temperature is defined by $1/T = (\partial S / \partial E)$ from the same section. Since $E = q\hbar\omega$, differentiating with respect to E is the same as differentiating with respect to q and dividing by $\hbar\omega$. The derivative of the expression above is

$$\frac{\partial \ln \Omega}{\partial q} = \ln(q + 3N) + 1 - \ln q - 1 = \ln \left(1 + \frac{3N}{q} \right), \quad (5.39)$$

where the two terms in 1 come from differentiating the logarithms and cancel. So

$$\frac{1}{T} = \frac{k_B}{\hbar\omega} \ln \left(1 + \frac{3N}{q} \right). \quad (5.40)$$

We can rearrange this expression to give q . Exponentiating gives $1 + 3N/q = e^{\hbar\omega/k_B T}$, so that

$$q = \frac{3N}{e^{\hbar\omega/k_B T} - 1}, \quad E = \frac{3N\hbar\omega}{e^{\hbar\omega/k_B T} - 1}. \quad (5.41)$$

Writing $\theta_E = \hbar\omega/k_B$ for the characteristic temperature of the model, which is called the Einstein temperature, and differentiating the energy with respect to T as was done for the vibration of a molecule in Section 5.2,

$$C = 3Nk_B \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2}. \quad (5.42)$$

The Same Result from the Partition Function

The oscillators are independent, and each of them is attached to a particular atom and a particular direction, so the partition function of the solid is the product of $3N$ single-oscillator partition functions, with no factor of $N!$ to correct for indistinguishability:

$$Z = (Z_{\text{vib}})^{3N}, \quad Z_{\text{vib}} = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}, \quad (5.43)$$

where Z_{vib} was worked out in Section 5.2. The mean energy of one oscillator was found there to be $\frac{1}{2}\hbar\omega + \hbar\omega/(e^{\beta\hbar\omega} - 1)$, so

$$U = 3N \left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} \right], \quad (5.44)$$

which differs from the energy found by counting the arrangements only by the constant $\frac{3}{2}N\hbar\omega$, the energy of the lowest state, from which the energy was measured in the count of arrangements. The heat capacity is the same. The route through the partition function is much the shorter of the two, and it is usually the route taken in practice; the route through the count of arrangements shows where the temperature comes from, and it applies to a solid that is isolated.

High and Low Temperature

When $T \gg \theta_E$ we expand the exponential, $e^{\theta_E/T} - 1 \approx \theta_E/T$, and the heat capacity becomes

$$C \rightarrow 3Nk_B, \quad (5.45)$$

which is $3k_B$ per atom. This is the law of Dulong and Petit, found by measurement in 1819, that the heat capacity per atom is the same for all simple solids, as it is for copper and for aluminium at room temperature. It is what equipartition gives, since each atom has three kinetic and three potential quadratic terms in its energy, contributing $6 \times \frac{1}{2}k_B T$ to the mean energy.

When $T \ll \theta_E$ the exponential in the denominator is large and

$$C \approx 3Nk_B \left(\frac{\theta_E}{T} \right)^2 e^{-\theta_E/T}, \quad (5.46)$$

which vanishes as $T \rightarrow 0$, as it must if the entropy is to go to zero there. The whole curve is drawn in Figure 38. The model therefore accounts for something that classical physics could not; the measured heat capacities of solids fall well below the Dulong–Petit value when the solid is cold, and the reason is that the oscillators are quantised, so that at low temperature almost all of them are in their lowest state and cannot take up the energy supplied.

How the Model Compares with Measurement

The agreement with measurement is good at high temperature and qualitatively right through the middle of the temperature range,

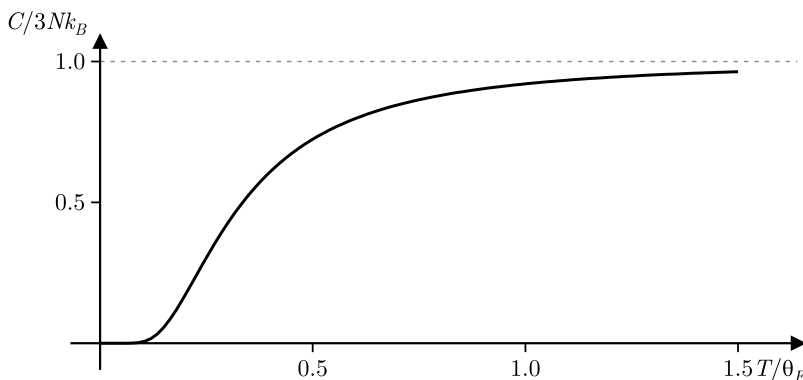


Figure 38. The heat capacity of the Einstein solid against T/θ_E , in units of $3Nk_B$. The curve approaches the Dulong–Petit value at high temperature and falls exponentially at low temperature.

while at low temperature the model gives the wrong answer. Measured heat capacities of insulating crystals do fall towards zero, but they go as T^3 , and an exponential falls very much faster than a power, so that well below θ_E the model lies far under the measured curve. The assumption at fault is the one that gives every oscillator the same frequency. An atom in a crystal is bound to its neighbours rather than to a fixed point in space, so the displacements that oscillate at a definite frequency involve many atoms moving together, which are the normal modes of coupled oscillators studied in the mechanics notes, and among those collective displacements are some of very long wavelength, whose frequencies are as low as we please. Oscillators of low frequency are not frozen out at low temperature, and they are what the measured heat capacity is made of there.

5.5 Phonons and the Debye Model

Section 5.4 ended with a discrepancy: the Einstein solid gives a heat capacity that falls exponentially at low temperature, while measured heat capacities of insulating crystals go as T^3 . The assumption at fault was that every oscillator has the same frequency. Debye’s model keeps the idea that a solid is a collection of oscillators and changes the assumption about their frequencies, taking them to be the frequencies of sound waves in the solid.

Sound Waves as Modes

A displacement of long wavelength in a crystal is a sound wave, and its frequency and wavevector are related by $\omega = c_s k$, where c_s is the speed of sound. Sound in a solid has three independent polarisations, one in which the displacement lies along the direction of travel and two in which it lies across that direction, and we take all three to travel at the same speed.

Let the solid be a cube of side L , with the displacement held fixed at its surfaces. A standing wave must then have a node at each face, which means that the cube holds a whole number of half-wavelengths along each of the three directions:

$$k_x = \frac{\pi n_x}{L}, \quad k_y = \frac{\pi n_y}{L}, \quad k_z = \frac{\pi n_z}{L}, \quad (5.47)$$

with n_x , n_y , and n_z positive integers. The frequency of the mode labelled by the three integers is therefore

$$\omega = \frac{\pi c_s}{L} \sqrt{n_x^2 + n_y^2 + n_z^2} = \frac{\pi c_s n}{L}, \quad (5.48)$$

with n defined as in Section 5.1.

The counting of modes is now exactly the counting of states in Section 5.1. The states sit at the corners of every unit cell of a cubic lattice of unit spacing in the space of mode numbers, so that there is one state per unit cell; only the octant with all three positive is used, and with three polarisations for every triple of integers the number of modes whose mode number lies between n and $n + dn$ is $3 \times \frac{1}{8} \times 4\pi n^2 dn = \frac{3}{2}\pi n^2 dn$. Substituting $n = L\omega/\pi c_s$ and $dn = L d\omega/\pi c_s$, the number of modes with frequency in $d\omega$ is

$$g(\omega) d\omega = \frac{3V\omega^2}{2\pi^2 c_s^3} d\omega, \quad (5.49)$$

where $V = L^3$. A quantum of a vibrational mode of a crystal is called a phonon, so that a mode is said to hold a whole number of phonons, one for each $\hbar\omega$ of energy it carries above its lowest state.

The Debye Cut-Off

A crystal of N atoms has $3N$ degrees of freedom, since each of its atoms can move in each of three directions, so it has exactly $3N$ modes of vibration. The count above, which grows without limit, cannot be right at high frequency, and Debye's assumption is to keep it up to a maximum frequency ω_D chosen so that the total comes out correct:

$$\int_0^{\omega_D} \frac{3V\omega^2}{2\pi^2 c_s^3} d\omega = \frac{V\omega_D^3}{2\pi^2 c_s^3} = 3N, \quad (5.50)$$

from which

$$\omega_D = c_s \left(\frac{6\pi^2 N}{V} \right)^{1/3}, \quad \theta_D = \frac{\hbar\omega_D}{k_B}. \quad (5.51)$$

The frequency ω_D is called the Debye frequency and θ_D the Debye temperature. The wavelength that goes with ω_D is $2\pi c_s/\omega_D$, which is of the order of the spacing between atoms, and that is as it should be: a wave whose wavelength is shorter than the spacing between the atoms describes no motion that the crystal can perform.

The Energy and the Heat Capacity

Each mode is an oscillator whose energy is quantised in units of $\hbar\omega$, so the mean energy of a mode of frequency ω is once more the expression found in Section 5.2. With the energy of the lowest state again set aside, since it does not change with temperature, the energy of the solid is

$$U = \int_0^{\omega_D} \frac{3V\omega^2}{2\pi^2 c_s^3} \frac{\hbar\omega d\omega}{e^{\hbar\omega/k_B T} - 1}. \quad (5.52)$$

Substituting $x = \hbar\omega/k_B T$ and using $3V/2\pi^2 c_s^3 = 9N/\omega_D^3$ from the condition that fixed the cut-off,

$$U = 9Nk_B T \left(\frac{T}{\theta_D} \right)^3 \int_0^{\theta_D/T} \frac{x^3 dx}{e^x - 1}. \quad (5.53)$$

The integral cannot be done in closed form, and the model is usually solved numerically, but the high-temperature and low-temperature limits can both be worked out by hand.

When $T \gg \theta_D$ the upper limit is small, so x is small throughout the range of integration, $e^x - 1 \approx x$, and the integrand is approximately x^2 . The integral is then $\frac{1}{3}(\theta_D/T)^3$, and

$$U = 3Nk_B T, \quad C = 3Nk_B, \quad (5.54)$$

which is the Dulong–Petit value, as it must be, since at high temperature every one of the $3N$ oscillators has a mean energy $k_B T$, the equipartition value, whatever its frequency.

When $T \ll \theta_D$ the upper limit is very large and may be replaced by infinity, since the integrand falls exponentially. The integral that remains is evaluated by turning the integrand into a series whose terms can each be integrated separately. Since e^{-x} is less than one for $x > 0$, the denominator can be expanded as a geometric series in the manner of Section 5.2,

$$\frac{1}{e^x - 1} = \frac{e^{-x}}{1 - e^{-x}} = \sum_{j=1}^{\infty} e^{-jx}, \quad (5.55)$$

and integrating each term of the series in turn, by parts three times over, gives $\int_0^{\infty} x^3 e^{-jx} dx = 6/j^4$, so that

$$\int_0^{\infty} \frac{x^3 dx}{e^x - 1} = 6 \sum_{j=1}^{\infty} \frac{1}{j^4}. \quad (5.56)$$

The sum of the inverse fourth powers of the integers is $\pi^4/90$, which is a piece of mathematics beyond this course, so we quote it and take the value of the integral to be $\pi^4/15$. With that value,

$$U = \frac{3\pi^4}{5} Nk_B T \left(\frac{T}{\theta_D} \right)^3, \quad C = \frac{12\pi^4}{5} Nk_B \left(\frac{T}{\theta_D} \right)^3, \quad (5.57)$$

which is the T^3 law. The reason for the power can be seen without doing the integral. At temperature T the modes that hold an appreciable energy are those with $\hbar\omega \lesssim k_B T$, and since the number of modes below a frequency ω is proportional to ω^3 , the number of excited modes is proportional to T^3 . Each of them holds an energy of order $k_B T$, so the energy is of order T^4 and the heat capacity of order T^3 . In the Einstein solid every oscillator has the same frequency ω , so there are no low-frequency oscillators available to take up energy, and almost none of them is excited until $k_B T$ becomes comparable with $\hbar\omega$, which is why the energy there falls exponentially instead.

Comparison with the Einstein Model and with Measurement

Figure 39 shows the Debye and Einstein models over the low-temperature part of the range. The Einstein temperature has been chosen to make the two curves agree as well as possible over the upper part of the range, and even so the Einstein curve lies well below the Debye curve at the bottom.

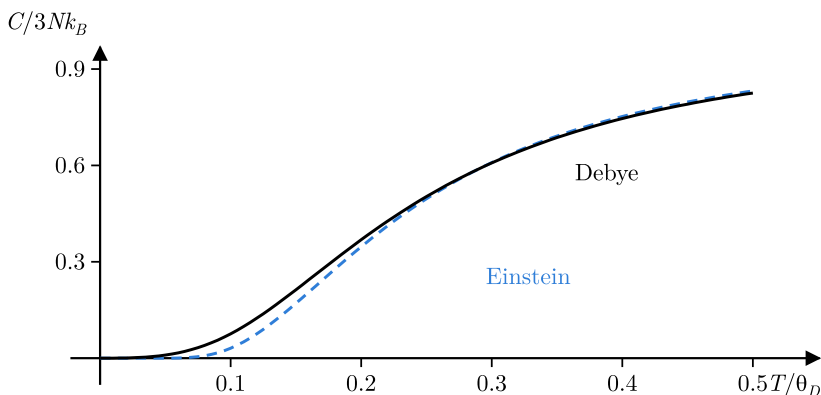


Figure 39. The heat capacity of a solid at low temperature, in units of $3Nk_B$, against T/θ_D . The solid curve is the Debye model and the dashed curve the Einstein model of Section 5.4, drawn with $\theta_E = 0.75\theta_D$. The Einstein curve lies under the Debye curve below about $T = 0.3\theta_D$, and falls further under it as the temperature drops.

The model has the right form at both ends of the temperature

range. At high temperature it gives the Dulong–Petit value, which any model of $3N$ oscillators would give. At low temperature it gives the right power, and the right coefficient as well once c_s is understood as a suitable average over the three polarisations, because the only modes excited there are of long wavelength, and a wave whose wavelength covers many atoms really is a sound wave of speed c_s , so the assumption that $\omega = c_s k$ is accurate exactly where it is being used.

In between the model is an approximation, for two reasons in particular. The first is that the relation $\omega = c_s k$ holds only while the wavelength is long; as the wavelength approaches the atomic spacing the frequency stops rising in proportion to k and flattens off, so the true distribution of frequencies near the top of the range departs from the ω^2 of the model. The second is that the sharp cut-off at ω_D replaces the real boundary of the region of allowed wavevectors, whose shape is set by the lattice rather than being a sphere. The consequence is that around $T \sim \theta_D/5$ the measured heat capacity departs from the Debye curve by a few per cent, and a value of θ_D fitted to low-temperature measurements differs somewhat from one fitted higher up. In a metal there is a further contribution to the heat capacity at very low temperature, coming from the conduction electrons. That contribution is not treated in this course, and the Debye model does not describe it.

5.6 Photons and Blackbody Radiation

A cavity cut into a block of material and held at temperature T contains electromagnetic radiation in equilibrium with its walls. The walls absorb and emit continually, and after a while the radiation inside reaches a state that depends on the temperature alone, whatever the walls are made of. Radiation in this state is called blackbody radiation, and the object of this section is to work out how its energy is distributed over frequency and how much of it there is.

The Modes of the Field in a Cavity

Let the cavity be a cube of side L with conducting walls. The field inside can be written as a sum of standing waves, and a standing

wave must have a node at each wall, which means that the cavity holds a whole number of half-wavelengths along each of the three directions:

$$k_x = \frac{\pi n_x}{L}, \quad k_y = \frac{\pi n_y}{L}, \quad k_z = \frac{\pi n_z}{L}, \quad (5.58)$$

with n_x , n_y , and n_z positive integers. Since $\omega = ck$ for light, the frequency of the mode labelled by the three integers is

$$\omega = \frac{\pi c}{L} \sqrt{n_x^2 + n_y^2 + n_z^2} = \frac{\pi c n}{L}, \quad (5.59)$$

with n defined as in Section 5.1. For each triple of integers there are two modes, since light has two independent polarisations, left-circular and right-circular.

The counting of modes is the same as the counting of the sound modes of Section 5.5, with two polarisations in place of three and the speed of light in place of the speed of sound. The number of modes whose mode number lies between n and $n + dn$ is $2 \times \frac{1}{8} \times 4\pi n^2 dn = \pi n^2 dn$, and substituting $n = L\omega/\pi c$ and $dn = L d\omega/\pi c$, the number of modes with frequency in $d\omega$ is

$$g(\omega) d\omega = \frac{V\omega^2}{\pi^2 c^3} d\omega, \quad (5.60)$$

where $V = L^3$. The number of modes per unit frequency grows as ω^2 and never stops growing, which is a point we return to when the spectrum is integrated over all frequencies.

Each Mode as an Oscillator

A single standing-wave mode of the field oscillates harmonically at the frequency ω set by its mode numbers, and its energy is quantised in units of $\hbar\omega$: measuring from the state with no quanta, the energies of the mode are $0, \hbar\omega, 2\hbar\omega, \dots$, exactly as for the vibrating molecule of Section 5.2. A quantum of the electromagnetic field is called a photon, in the same way that a quantum of a vibrational mode of a crystal is called a phonon, so that a mode is said to hold a whole number of photons. Leaving out the energy of the lowest

state, which does not depend on the temperature, the mean energy of a mode at temperature T is the result obtained in Section 5.2,

$$\bar{\epsilon}(\omega) = \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1}. \quad (5.61)$$

The number of quanta in a mode is not fixed; the walls create and destroy them freely, and what the temperature fixes is the mean energy of each mode. For that reason no chemical potential appears anywhere in this section, and the grand canonical description of Section 4.6 is not needed.

The Planck Spectrum

Multiplying the number of modes per unit frequency by the mean energy of a mode, and dividing by the volume, the energy per unit volume in the frequency range $d\omega$ is $u_\omega d\omega$, with

$$u_\omega = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3 d\omega}{e^{\hbar\omega/k_B T} - 1}. \quad (5.62)$$

This is called the Planck spectrum, and it is drawn at two temperatures in Figure 40. At low frequencies, where $\hbar\omega \ll k_B T$, the denominator is approximately $\hbar\omega/k_B T$ and

$$u_\omega \approx \frac{\omega^2 k_B T}{\pi^2 c^3}, \quad (5.63)$$

which is the number of modes per unit volume per unit frequency multiplied by $k_B T$, the mean energy that equipartition assigns to an oscillator. Had every mode held $k_B T$ at every frequency, the total energy density would be $\int \omega^2 k_B T d\omega / \pi^2 c^3$, which diverges, since the modes go on being counted however high the frequency. What happens instead is that at high frequencies the exponential in the denominator becomes enormous and u_ω falls away exponentially. A mode whose quantum $\hbar\omega$ is much larger than $k_B T$ is almost always in its lowest state, for the same reason that the vibration of a molecule freezes out below θ_v , and it holds essentially no energy.

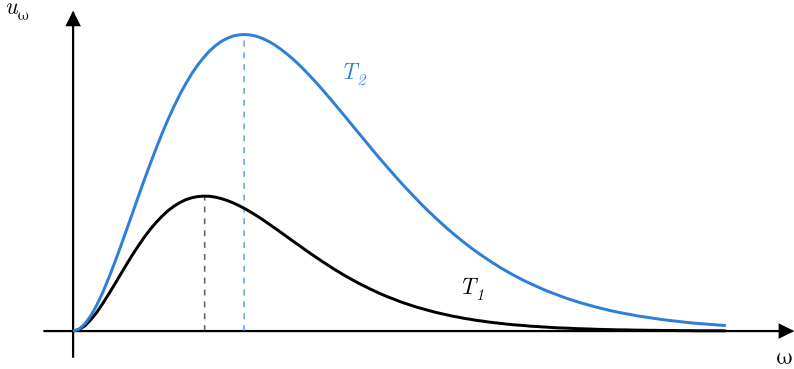


Figure 40. The spectrum of blackbody radiation at two temperatures, with T_2 the higher temperature. The dashed lines mark the frequencies at which the curves peak. Raising the temperature raises the curve everywhere and moves the peak to higher frequency in proportion to T .

Wien's Displacement Law

To find where the spectrum peaks we write $x = \hbar\omega/k_B T$, so that $u_\omega \propto x^3/(e^x - 1)$, and set the derivative with respect to x to zero. Differentiating,

$$\frac{3x^2}{e^x - 1} - \frac{x^3 e^x}{(e^x - 1)^2} = 0 \quad \Rightarrow \quad 3(1 - e^{-x}) = x, \quad (5.64)$$

whose root, apart from $x = 0$, is $x \approx 2.82$. So the peak sits at $\hbar\omega_{\max} \approx 2.82 k_B T$, and its frequency is proportional to the absolute temperature. This is Wien's displacement law, and it is why a body glows first dull red and then white as it is heated.

The Stefan–Boltzmann Law

The total energy density is obtained by integrating the spectrum over all frequencies. Substituting $x = \hbar\omega/k_B T$ throughout,

$$u = \int_0^\infty u_\omega d\omega = \frac{(k_B T)^4}{\pi^2 c^3 \hbar^3} \int_0^\infty \frac{x^3 dx}{e^x - 1}. \quad (5.65)$$

The integral was worked out in Section 5.5, where its value came to $\pi^4/15$, so that

$$u = \frac{\pi^2 k_B^4}{15c^3 \hbar^3} T^4. \quad (5.66)$$

What is usually measured is the radiation leaving a small hole in the wall of the cavity, which is one standard way of producing blackbody radiation in the laboratory. The energy flux through the hole is worked out by exactly the argument used for effusion in Section 1.3: the energy carried in a given direction travels at speed c , the directions are distributed evenly over the sphere, and averaging the outward component over the hemisphere of directions that point at the hole gives one quarter of the product of the energy density and the speed. So the flux is

$$J = \frac{1}{4}uc = \sigma T^4, \quad \sigma = \frac{\pi^2 k_B^4}{60c^2 \hbar^3}, \quad (5.67)$$

which is the Stefan–Boltzmann law, with σ the Stefan constant. Stefan found the law by measurement and Boltzmann derived it from thermodynamics before Planck wrote down the spectrum, and the calculation above gives the constant in terms of k_B , c , and $\hbar$.

Free Energy, Entropy, and Radiation Pressure

With the energy of a mode measured from its lowest state, its partition function is $Z_{\text{mode}} = 1/(1 - e^{-\beta\hbar\omega})$, by the geometric series summed in Section 5.2, and the mode contributes $k_B T \ln(1 - e^{-\beta\hbar\omega})$ to the free energy. Adding the contributions of all the modes,

$$\begin{aligned} F &= k_B T \int_0^\infty \frac{V\omega^2}{\pi^2 c^3} \ln(1 - e^{-\beta\hbar\omega}) d\omega \\ &= \frac{V(k_B T)^4}{\pi^2 c^3 \hbar^3} \int_0^\infty x^2 \ln(1 - e^{-x}) dx. \end{aligned} \quad (5.68)$$

We integrate by parts, with $\ln(1 - e^{-x})$ as the part to be differentiated and x^2 as the part to be integrated. The boundary terms

vanish at both ends, and since the derivative of $\ln(1 - e^{-x})$ is $e^{-x}/(1 - e^{-x}) = 1/(e^x - 1)$,

$$\int_0^\infty x^2 \ln(1 - e^{-x}) dx = -\frac{1}{3} \int_0^\infty \frac{x^3 dx}{e^x - 1} = -\frac{\pi^4}{45}. \quad (5.69)$$

Comparing this free energy with the energy density found above gives $F = -U/3$, and since F is proportional to V at fixed temperature, the pressure is

$$p = -\left(\frac{\partial F}{\partial V}\right)_T = \frac{u}{3}, \quad (5.70)$$

which is called the radiation pressure. The same relation between pressure and energy density was obtained for an ultra-relativistic gas in the problems of Section 5.1, and the explanation is the same for the radiation as for the gas: a quantum of energy ϵ travelling at speed c carries momentum ϵ/c . Differentiating F with respect to temperature gives the entropy, $S = 4U/3T$.

Problems

§5.1 The Ideal Gas from the Partition Function

1. A gas of N molecules is confined to a square box of side L in two dimensions, with levels $\epsilon = (h^2/8mL^2)(n_x^2 + n_y^2)$. Counting states in rings rather than shells, show that $Z_1 = A/\lambda_{\text{th}}^2$ with $A = L^2$, and find the energy, the entropy, and the force per unit length exerted on the edge of the box.
2. For an ultra-relativistic gas the levels in a cubical box are $\epsilon = hcn/2L$, with n defined as above. Show that $Z_1 = 8\pi V(k_B T/hc)^3$, using $\int_0^\infty n^2 e^{-bn} dn = 2/b^3$. Hence show that $U = 3Nk_B T$ and that the equation of state is still $pV = Nk_B T$, so that $pV = U/3$.
3. Suppose the factor $1/N!$ were omitted. Show that the entropy would then contain $Nk_B \ln V$ rather than $Nk_B \ln(V/N)$, and that combining two equal samples of the same gas at the same temperature and density, by removing the partition between

them, would appear to increase the entropy by $2Nk_B \ln 2$ even though nothing has changed.

- Using the expressions above for F , p , and μ , verify that $F = -pV + \mu N$, as required by the general relation of Section 3.3.

§5.2 Internal Degrees of Freedom

- Show that the mean number of vibrational quanta in a molecule is $\bar{n} = 1/(e^{\beta\hbar\omega} - 1)$, so that $\bar{\epsilon}_{\text{vib}} = (\bar{n} + \frac{1}{2})\hbar\omega$, and find $\bar{n}$ when $T \gg \theta_v$.
- Show that the vibrational contribution to the entropy of a gas of N molecules is $S/Nk_B = x/(e^x - 1) - \ln(1 - e^{-x})$ with $x = \theta_v/T$. Check that it tends to zero as $T \rightarrow 0$ and that it becomes $1 + \ln(T/\theta_v)$ when $T \gg \theta_v$.
- A molecule constrained to rotate in a plane has levels $\epsilon_m = \hbar^2 m^2 / 2I$ with $m = 0, \pm 1, \pm 2, \dots$, each state distinct. Show that when $T \gg \theta_r$ its partition function is $\sqrt{\pi T / \theta_r}$ and its contribution to the heat capacity is $\frac{1}{2}Nk_B$.
- Using $C_p - C_V = Nk_B$ for an ideal gas, find the ratio $\gamma = C_p/C_V$ for a diatomic gas in each of the three ranges of Figure 33.

§5.3 Two-Level Systems and Paramagnetism

- Show that the maximum of the heat capacity of a two-level system occurs where $x \tanh(x/2) = 2$, with $x = \Delta/k_B T$, and check that this condition is consistent with the position of the peak in Figure 34.
- A system of N moments of magnitude μ has three states in a field, with energies $-\mu B$, 0 , and $+\mu B$. Find Z , show that $M = N\mu \frac{2 \sinh(\mu B/k_B T)}{1 + 2 \cosh(\mu B/k_B T)}$, and obtain Curie's law from it at high temperature.
- Show that the entropy of the spin-half paramagnet depends on B and T only through the ratio B/T . Hence show that if the field is reduced slowly while the spins exchange no heat with

their surroundings, the temperature falls in proportion to the field.

4. For the two-level system, find the energy and the heat capacity when $\Delta \ll k_B T$, keeping the first correction in $\Delta/k_B T$, and confirm that the heat capacity goes as T^{-2} at high temperature.

§5.4 The Einstein Solid

1. For a system of $\mathcal{N}$ oscillators holding q quanta with $q \gg \mathcal{N}$, show that $\ln \Omega \approx \mathcal{N} [\ln(q/\mathcal{N}) + 1]$, and hence that $E = \mathcal{N} k_B T$. Show that this is the high-temperature limit of the energy of the Einstein solid derived in the text.
2. Find the entropy of the Einstein solid from $F = -k_B T \ln Z$, and show that it tends to zero as $T \rightarrow 0$ and to $3Nk_B [1 + \ln(T/\theta_E)]$ at high temperature.
3. At low temperature only the lowest excited state of each oscillator is appreciably occupied. Show that the heat capacity of the Einstein solid in that limit has the same form as that of $3N$ two-level systems with splitting $\Delta = \hbar\omega$, found in Section 5.3.
4. Show that the mean number of quanta per oscillator at low temperature is $e^{-\theta_E/T}$, and hence that the energy of the solid is $3N\hbar\omega e^{-\theta_E/T}$. Interpret this expression as counting the oscillators that are excited.

§5.5 Phonons and the Debye Model

1. Expand the integrand using $x/(e^x - 1) = 1 - x/2 + x^2/12 - \dots$ and show that at high temperature $C = 3Nk_B [1 - \frac{1}{20}(\theta_D/T)^2 + \dots]$.
2. Show that the total energy of the lowest state of a Debye solid, which was left out of the calculation above, is $\frac{9}{8}N\hbar\omega_D$, and check that it makes no contribution to the heat capacity.
3. A solid film in which the atoms move only in the plane has two polarisations and a mode count proportional to ω rather than ω^2 . Show that its heat capacity is proportional to T^2 at

low temperature, and find the Debye frequency in terms of the number of atoms per unit area.

4. Show that the number of modes with frequency below ω is $V\omega^3/2\pi^2c_s^3$, and use it to give the argument for the T^3 law without evaluating any integral.

§5.6 Photons and Blackbody Radiation

1. Carry out the differentiation that leads to Wien's law and show that the equation for the peak is $3(1 - e^{-x}) = x$. Show that the root is close to 3, and explain why.
2. The mean number of quanta in a mode of frequency ω is $1/(e^{\beta\hbar\omega} - 1)$. Show that the number of quanta per unit volume in the cavity is proportional to T^3 , using $\int_0^\infty x^2 dx/(e^x - 1) \approx 2.40$, and hence that the mean energy per quantum is about $2.7 k_B T$.
3. Show that $S = 4U/3T$ by differentiating F , and hence that radiation expanded slowly with no heat exchanged keeps VT^3 constant. Show that this implies $pV^{4/3}$ is constant.
4. Show that the energy density in the range $d\omega$ at low frequency does not contain $\hbar$, and explain which frequencies are responsible for the divergence that would follow if the low-frequency form held everywhere.

Interacting Systems, Real Gases, and Phase Equilibria

6.1 Phase Coexistence

A substance held at a fixed temperature and pressure is usually found in one homogeneous form, with a definite density and a definite arrangement of its constituent atoms and molecules. Such a form is called a phase, and the familiar phases are the solid, the liquid, and the gas. At certain temperatures and pressures, however, two phases sit side by side in the same vessel, neither one growing while the other shrinks, as water and its vapour do in a closed vessel, with the liquid at the bottom and the vapour above it, and the substance is then said to be in phase coexistence. In this section we find the condition for two phases to coexist, and use it to work out the shape of the curves along which coexistence occurs. Following that, we allow the substance to be a mixture of two components, and ask when a uniform mixture separates into two phases of different composition.

The Phase Diagram of a Pure Substance

The behaviour of a pure substance is summarised by a plot of pressure against temperature, divided into regions by curves, as in Fig-

ure 41. Let us take carbon dioxide as the example. Inside each region one phase is stable, and carbon dioxide held there is found entirely in that phase; along the curves two phases coexist. Each curve is named after the change that takes place across it. The lowest of the three separates the solid from the gas, and crossing it turns solid carbon dioxide directly into gas, which is what happens when a block of it is left in the open air, so the curve is called the sublimation curve. The curve that rises steeply on the left separates the solid from the liquid, and it is called the melting curve. The curve that runs up to the right separates the liquid from the gas, and it is called the vaporisation curve. The three curves meet at the triple point, the single temperature and pressure at which solid, liquid, and gas coexist.

A substance may have more phases than these, since a solid can often be arranged in more than one crystal structure, and each arrangement is a separate phase, so that the diagram of such a substance carries further regions divided by further curves. The solid, the liquid, and the gas are as many as we shall need here.

The vaporisation curve does not continue upwards indefinitely. It ends at a point called the critical point, whose temperature and pressure are called the critical temperature and the critical pressure. Below the critical temperature a liquid and its vapour are two distinguishable phases of different density, with a visible meniscus between them; above it, compressing the fluid produces no sudden change of density and no meniscus appears. The behaviour of a fluid near the critical point is not treated in this course, and the critical point is used here only as the end of the vaporisation curve.

The melting curve is much steeper than the sublimation curve and the vaporisation curve, because the solid and the liquid differ little in volume while the gas is far less dense than either. For most substances the melting curve leans to the right, so that raising the pressure raises the melting temperature. Water is the exception, since ice is less dense than liquid water, and its melting curve leans to the left.

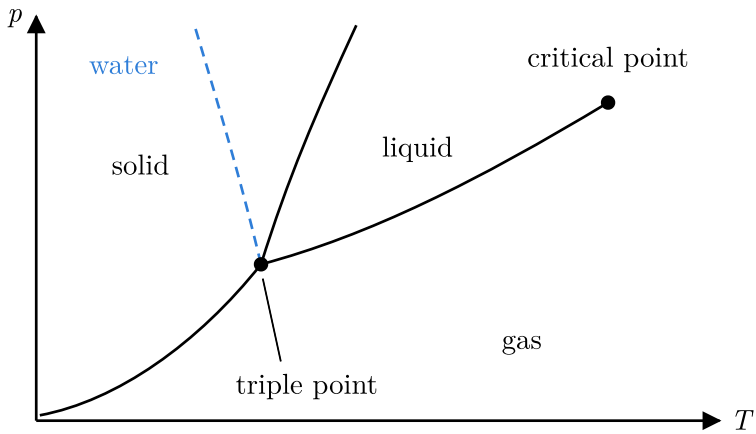


Figure 41. The phase diagram of a pure substance. The three coexistence curves meet at the triple point, and the vaporisation curve ends at the critical point. The melting curve is steep and leans to the right for most substances; the dashed line shows the melting curve of water, which leans to the left.

Equilibrium Between Two Phases

We found in Section 3.1 that a system held at fixed temperature and pressure comes to equilibrium at the minimum of its Gibbs function G , and in Section 3.3 that the Gibbs function of a pure substance is $G = \mu N$, where μ is the chemical potential and N the number of molecules. Suppose that a vessel held at temperature T and pressure p contains N molecules of one pure substance, of which N_1 are in phase 1 and $N_2 = N - N_1$ in phase 2. The Gibbs function of the whole vessel is the sum of the Gibbs functions of the two parts,

$$G = \mu_1 N_1 + \mu_2 N_2, \quad (6.1)$$

where μ_1 and μ_2 are the chemical potentials of the two phases. We now move a small number dN_1 of molecules from phase 2 into phase 1, so that $dN_2 = -dN_1$. The change in the Gibbs function is

$$dG = (\mu_1 - \mu_2) dN_1. \quad (6.2)$$

At a minimum of G this change vanishes for a transfer in either direction, so the condition for the two phases to coexist in equilibrium is

$$\mu_1(T, p) = \mu_2(T, p). \quad (6.3)$$

If instead μ_1 is smaller than μ_2 , then dG is negative for a transfer into phase 1, and molecules keep moving that way until none of phase 2 remains. The stable phase is therefore the one with the lower chemical potential, and the coexistence curves are the places where the two chemical potentials are equal.

The Chemical Potential of Each Phase

To use this condition we need to know how μ depends on temperature and pressure. From Section 3.1 the differential of the Gibbs function is $dG = -S dT + V dp + \mu dN$. Setting $\mu = G/N$ and holding N fixed, and writing $s = S/N$ and $v = V/N$ for the entropy and the volume per molecule,

$$d\mu = -s dT + v dp, \quad (6.4)$$

so that $(\partial\mu/\partial T)_p = -s$ and $(\partial\mu/\partial p)_T = v$.

Figure 42 shows μ against T at a fixed pressure for a liquid and for its vapour. The curve belonging to the liquid and the curve belonging to the gas both fall as the temperature rises, because the entropy per molecule is positive, and each of them bends downwards, because the entropy per molecule grows with temperature. The gas has a much larger entropy per molecule than the liquid, since its molecules are spread through the whole of the vessel rather than held close to one another, so the curve belonging to the gas falls much more steeply, and the two curves cross at a temperature we call T_0 . Below T_0 the liquid has the lower chemical potential and the substance is a liquid; above T_0 the gas has the lower chemical potential and the substance is a gas; and at T_0 the chemical potentials of the liquid and of the gas are equal and the two phases coexist. The crossing of the two curves is the transition itself, and the difference between their slopes there is the difference between the entropy per molecule of the gas and the entropy per molecule of the liquid.

Raising the pressure lifts both curves of Figure 42, since $(\partial\mu/\partial p)_T = v$ is positive, and it lifts the curve belonging to the gas much further, since a gas has by far the larger volume per molecule. The crossing therefore moves to a higher temperature, which is why water boils at a lower temperature at higher altitudes than at sea level.

The Clausius–Clapeyron Equation

The slope of a coexistence curve follows from the equality of the chemical potentials. Let T and p be a point on the curve, where $\mu_1 = \mu_2$, and move a short distance along it to $T + dT$ and $p + dp$. The two chemical potentials must again be equal at the new point, so their changes must be equal, $d\mu_1 = d\mu_2$, which gives

$$-s_1 dT + v_1 dp = -s_2 dT + v_2 dp. \quad (6.5)$$

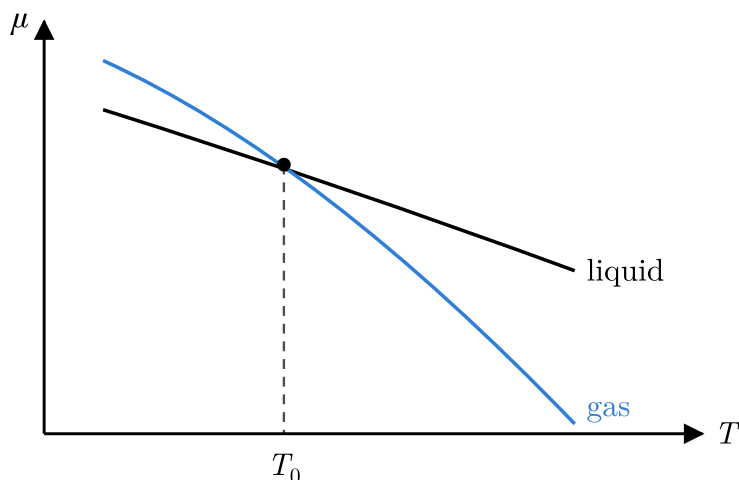


Figure 42. The chemical potential of the liquid and of the gas against temperature, at a fixed pressure. The gas has the larger entropy per molecule and so the steeper slope. The two curves cross at T_0 , and the stable phase at each temperature is the one whose curve lies lower.

Bringing the terms in dT to one side of this equation and the terms in dp to the other,

$$(s_2 - s_1) dT = (v_2 - v_1) dp, \quad (6.6)$$

and dividing both sides by $(v_2 - v_1) dT$,

$$\frac{dp}{dT} = \frac{s_2 - s_1}{v_2 - v_1}. \quad (6.7)$$

Multiplying the numerator and the denominator by N turns the entropy and the volume per molecule into the entropy and the volume of the whole sample, since $N(s_2 - s_1) = S_2 - S_1 = \Delta S$ and $N(v_2 - v_1) = V_2 - V_1 = \Delta V$ are the changes in entropy and in volume when the whole sample passes from phase 1 to phase 2, so that

$$\frac{dp}{dT} = \frac{\Delta S}{\Delta V}. \quad (6.8)$$

The transition happens at a fixed temperature, so by the result of Section 2.6 that a reversible transfer of heat Q at a constant temperature T changes the entropy by Q/T , the heat absorbed is $L = T\Delta S$, which is the latent heat familiar from school, the heat taken in by a block of ice at its melting temperature as it becomes water, for example. In terms of it,

$$\frac{dp}{dT} = \frac{L}{T \Delta V}, \quad (6.9)$$

which is called the Clausius–Clapeyron equation. It accounts for the shapes drawn in Figure 41. Melting absorbs heat and changes the volume only slightly, so ΔV is small and the slope is large, which is why the melting curve is steep. For most substances the liquid is less dense than the solid, so ΔV is positive and the curve leans to the right; for water ΔV is negative and it leans to the left.

The Vapour Pressure of a Liquid

Let us apply the Clausius–Clapeyron equation to the vaporisation curve, along which one of the two phases is a gas, whose equation of state we already know. The pressure on this curve at a given temperature, which is the pressure of the vapour in equilibrium with its liquid, is called the vapour pressure of the liquid at that temperature. We take phase 1 to be the liquid and phase 2 the vapour. The volume of the vapour is much the larger, so $\Delta V \approx V_{\text{gas}} = Nk_B T/p$ by the ideal gas law of Section 5.1. Writing $\ell = L/N$ for the latent heat per molecule and putting $L = N\ell$ together with $\Delta V = Nk_B T/p$ into $dp/dT = L/(T \Delta V)$,

$$\frac{dp}{dT} = \frac{N\ell}{T} \frac{p}{Nk_B T} = \frac{\ell p}{k_B T^2}. \quad (6.10)$$

Taking the latent heat per molecule to be constant, which is valid over a moderate range of temperature, and separating the variables,

$$\int \frac{dp}{p} = \frac{\ell}{k_B} \int \frac{dT}{T^2}. \quad (6.11)$$

The integral on the left is $\ln p$, and the integral on the right is $\int T^{-2} dT = -1/T$, so that

$$\ln p = -\frac{\ell}{k_B T} + \text{constant}, \quad (6.12)$$

and taking the exponential of both sides,

$$p = p_0 e^{-\ell/k_B T}, \quad (6.13)$$

where p_0 is the exponential of the constant of integration. The vapour pressure therefore rises very steeply with temperature. The exponential is a Boltzmann factor of the kind studied in Section 4.4, and it can be interpreted in the same way; ℓ is the energy needed to take one molecule out of the liquid and put it into the vapour, and $e^{-\ell/k_B T}$ is the Boltzmann factor for a state whose energy lies that far above the state in the liquid. A liquid in an open vessel boils when its vapour pressure reaches the pressure of the surrounding air, so this relation also gives the boiling temperature.

First-Order Transitions

We can now say what happens to the Gibbs function itself as the substance is carried through a transition. Since $\mu_1 = \mu_2$ along the coexistence curve, the Gibbs function is continuous there; it is the lower of the two branches of Figure 42, multiplied by N , and the branches meet at T_0 without a jump. The Gibbs function does, however, have a kink at T_0 , since the branch belonging to the liquid and the branch belonging to the gas meet there at different slopes. From Section 3.1 the first derivatives of G are

$$S = -\left(\frac{\partial G}{\partial T}\right)_p, \quad V = \left(\frac{\partial G}{\partial p}\right)_T, \quad (6.14)$$

so a kink in G means that the entropy and the volume jump as the substance crosses the curve, as drawn in Figure 43. A transition

in which the first derivatives of the Gibbs function jump is called a first-order transition, and melting, boiling, and sublimation are all of this kind. The entropy jump is what the latent heat measures, $L = T\Delta S$, and since the transition happens at constant pressure the latent heat is also the change in the enthalpy of Section 3.1, $L = \Delta H$.

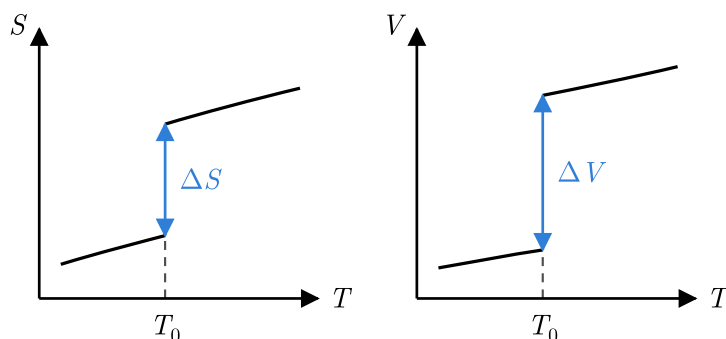


Figure 43. The entropy and the volume of a substance heated at constant pressure through a first-order transition at T_0 . The entropy and the volume both jump, by ΔS and by ΔV , while the Gibbs function itself is continuous.

As the transition temperature is raised along the vaporisation curve, the liquid and the gas grow more alike in density and in entropy per molecule, and ΔV , ΔS , and the latent heat all go to zero at the critical point. The vanishing of those three quantities together is what is meant by saying that the vaporisation curve ends at the critical point rather than stopping abruptly.

Superheating and Supercooling

For a liquid to boil, a bubble of vapour has to appear somewhere in it first, and a very small bubble is almost all surface. The molecules at a surface are held less tightly than those surrounded on all sides, so forming a small bubble raises the Gibbs function even when the vapour has the lower chemical potential, and the bubble tends to collapse again rather than grow. The change therefore needs somewhere to begin: a scratch on the wall of the vessel, a speck of dust,

a little dissolved gas. A clean liquid in a smooth vessel can be heated above its boiling temperature and remain liquid, which is called superheating, or cooled below its freezing temperature and remain liquid, which is called supercooling, and in both the superheated liquid and the supercooled liquid a small disturbance sets the change off suddenly.

6.2 Mixtures and Phase Separation

Everything in Section 6.1 concerned a single pure substance. Most materials of practical interest are mixtures: an alloy of two metals, salt dissolved in water, the air in this room. A mixture raises a further question, because the composition is now something the system can adjust. A uniform mixture of two metals, cooled slowly, may stay uniform, or it may separate into two solids of different composition. We work out below the Gibbs function of a binary mixture, then find the condition that decides between these two outcomes, and finally draw the phase diagram that follows from it.

The Entropy of Mixing

We describe the mixture by a lattice model, which keeps the essential physics while remaining simple enough to work with. Consider N sites arranged on a lattice, with one atom or molecule at each site, of which xN are of species B and $(1 - x)N$ are of species A. The number of distinct arrangements of the two species over the sites is the binomial coefficient met in Section 4.1,

$$\Omega = \frac{N!}{(xN)! [(1 - x)N]!} . \quad (6.15)$$

The entropy is $S = k_B \ln \Omega$ by the definition of Section 4.2. Taking the logarithm of the expression for Ω turns the quotient into a difference,

$$\ln \Omega = \ln N! - \ln(xN)! - \ln[(1 - x)N]! , \quad (6.16)$$

and applying Stirling's approximation of Section 4.1, $\ln N! \approx N \ln N - N$, which holds when the number concerned is large, to

each of the three factorials in turn,

$$\begin{aligned} \ln \Omega \approx & (N \ln N - N) - (xN \ln(xN) \\ & - xN) - \left((1-x)N \ln [(1-x)N] - (1-x)N \right). \end{aligned} \quad (6.17)$$

The terms that are linear in N add up to $-N + xN + (1-x)N = 0$, so they cancel. What is left contains logarithms of products, and since $\ln(xN) = \ln x + \ln N$ and $\ln[(1-x)N] = \ln(1-x) + \ln N$,

$$\begin{aligned} \ln \Omega \approx & N \ln N - xN \ln x - xN \ln N \\ & - (1-x)N \ln(1-x) - (1-x)N \ln N. \end{aligned} \quad (6.18)$$

Now we collect the terms in $\ln N$, which are $N \ln N - xN \ln N - (1-x)N \ln N$ and therefore add up to zero as well, and what remains is

$$\ln \Omega \approx -N[x \ln x + (1-x) \ln(1-x)]. \quad (6.19)$$

Multiplying by k_B gives the entropy of the mixture,

$$S = -Nk_B[x \ln x + (1-x) \ln(1-x)]. \quad (6.20)$$

Before mixing there is one way to arrange the A molecules among themselves and one way to arrange the B molecules among themselves, so the separated state has $\Omega = 1$ and an entropy of zero. The expression above is therefore also the change in entropy on mixing, and we call it the entropy of mixing ΔS_{mix} . The logarithm of x and the logarithm of $1-x$ are both negative for $0 < x < 1$, so ΔS_{mix} is positive: mixing always raises the entropy. It is largest at $x = \frac{1}{2}$, where it takes the value $Nk_B \ln 2$.

The Energy of Mixing

Mixing also changes the energy, because the energy of a molecule depends on which species sit next to it. We assign to each neighbouring pair an energy that depends on the two species involved,

writing u_{AA} , u_{BB} , and u_{AB} for the three possibilities, and let each site have z neighbours, so that the lattice holds $Nz/2$ pairs in all. If the two species are arranged at random, the probability that a given pair is A–A is $(1-x)^2$, that it is B–B is x^2 , and that it is A–B is $2x(1-x)$, the factor of two counting the two ways round. The energy of the mixture is therefore

$$U_{\text{mix}} = \frac{Nz}{2} \left[(1-x)^2 u_{AA} + x^2 u_{BB} + 2x(1-x) u_{AB} \right], \quad (6.21)$$

while the separated state, in which the $(1-x)N$ A molecules form one pure piece and the xN B molecules another, has

$$U_{\text{sep}} = \frac{Nz}{2} \left[(1-x) u_{AA} + x u_{BB} \right]. \quad (6.22)$$

Subtracting the energy of the separated state from the energy of the mixture, and collecting the terms in each of the three pair energies,

$$\Delta U_{\text{mix}} = \frac{Nz}{2} \left[[(1-x)^2 - (1-x)] u_{AA} + [x^2 - x] u_{BB} + 2x(1-x) u_{AB} \right]. \quad (6.23)$$

Each of the inner square brackets is equal to $-x(1-x)$, since $(1-x)^2 - (1-x) = (1-x)[(1-x) - 1] = -x(1-x)$ and $x^2 - x = x(x-1) = -x(1-x)$, so a common factor of $x(1-x)$ appears in every term and

$$\begin{aligned} \Delta U_{\text{mix}} &= \frac{Nz}{2} x(1-x) \left[2u_{AB} - u_{AA} - u_{BB} \right] \\ &= N\varepsilon x(1-x), \quad \varepsilon = z \left[u_{AB} - \frac{1}{2}(u_{AA} + u_{BB}) \right]. \end{aligned} \quad (6.24)$$

The single parameter ε compares an unlike neighbour with the average of two like ones. When ε is positive, unlike neighbours are bound less strongly than like ones and mixing raises the energy; when it is negative, mixing lowers the energy. There is an approximation hidden in this calculation. The two species were taken to

be arranged at random when the pairs were counted, although the pair energies themselves make some arrangements more likely than others. A random arrangement is a valid approximation as long as $|\varepsilon|$ is small compared with $k_B T$.

The Free Energy of Mixing

At fixed temperature and pressure the equilibrium state is the one of lowest Gibbs function, so we combine the energy of mixing with the entropy of mixing. In the lattice model the mixture occupies the same N sites as the two separated pieces together, so the volume does not change on mixing and the pV term in $G = U + pV - TS$ contributes nothing to the change. Writing Δg_{mix} for the change in the Gibbs function per molecule on mixing,

$$\Delta g_{\text{mix}} = \varepsilon x(1-x) + k_B T [x \ln x + (1-x) \ln(1-x)]. \quad (6.25)$$

Figure 44 shows this function against composition at three temperatures, for a positive ε . At the highest temperature the entropy term dominates and the curve has a single minimum at $x = \frac{1}{2}$, so the uniform mixture is stable at every composition. As the temperature falls the energy term becomes relatively more important, and below a certain temperature the curve develops a hump in the middle with a minimum on either side of it. To find the temperature at which the shape of the curve changes in this way, we differentiate Δg_{mix} twice. The energy term differentiates to $\varepsilon(1-2x)$. For the entropy term we use the product rule, which gives $d(x \ln x)/dx = \ln x + 1$ and $d[(1-x) \ln(1-x)]/dx = -\ln(1-x) - 1$, whose sum is $\ln x - \ln(1-x)$, so that

$$\frac{d \Delta g_{\text{mix}}}{dx} = \varepsilon(1-2x) + k_B T \ln \frac{x}{1-x}. \quad (6.26)$$

Differentiating a second time, with $d \ln x/dx = 1/x$ and $d \ln(1-x)/dx = -1/(1-x)$,

$$\frac{d^2 \Delta g_{\text{mix}}}{dx^2} = -2\varepsilon + k_B T \left(\frac{1}{x} + \frac{1}{1-x} \right). \quad (6.27)$$

At the midpoint $x = \frac{1}{2}$ the bracket is $2 + 2 = 4$, so the curvature there is $-2\varepsilon + 4k_B T$, which vanishes at $k_B T = \varepsilon/2$. Above that temperature the two species mix in all proportions, and below it there is a range of compositions in the middle at which they do not.

The random-arrangement approximation, made above when the energy of mixing was worked out, does not hold well at the temperature just found. It was said to hold while $|\varepsilon|$ is small compared with $k_B T$, and at $k_B T = \varepsilon/2$ the parameter ε is twice $k_B T$, so the two species will in fact be somewhat clustered rather than arranged at random. The temperature $\varepsilon/2k_B$ should therefore be interpreted as an estimate of where separation sets in.

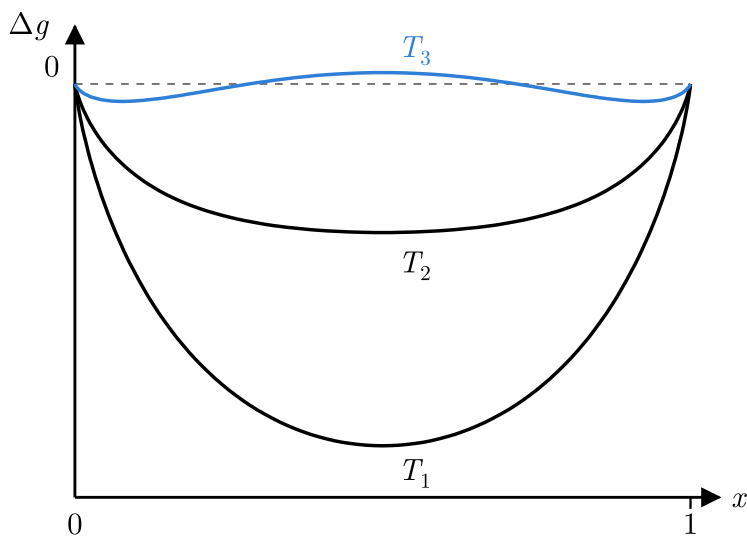


Figure 44. The Gibbs function of mixing per molecule against composition, for a positive ε , at three temperatures with $T_1 > T_2 > T_3$. At T_1 there is a single minimum at $x = \frac{1}{2}$; at T_3 , drawn in blue, there is a maximum at the middle and a minimum on either side. Every curve leaves the ends vertically.

Why a Trace of Anything Dissolves in Anything

Each curve in Figure 44 leaves the point $x = 0$ vertically downwards, and the derivative worked out above,

$$\frac{d \Delta g_{\text{mix}}}{dx} = \varepsilon(1 - 2x) + k_B T \ln \frac{x}{1 - x}, \quad (6.28)$$

shows why. As $x \rightarrow 0$ the logarithm of $x/(1-x)$ tends to $-\infty$, while the energy term $\varepsilon(1 - 2x)$ tends to the finite value ε . The slope is therefore negative for small enough x , however large and positive ε may be. Starting from pure A, adding a trace of B always lowers the Gibbs function, so a pure substance in equilibrium is never quite pure: a little of anything dissolves in anything. The first few B molecules gain a large amount of entropy, since there are a great many sites available to them, while the energy they bring in is proportional to their number.

The Chemical Potentials of a Mixture

To find what the two coexisting phases are, we need the chemical potential of each species. Write $g(x)$ for the Gibbs function per molecule of a uniform mixture at composition x , so that a sample of N molecules has $G = Ng(x)$, with $N_A = (1 - x)N$ molecules of A and $N_B = xN$ of B. We hold N_A fixed and add one molecule of B. The total number $N = N_A + N_B$ and the fraction $x = N_B/N$ both change, the first of them with $(\partial N / \partial N_B)_{N_A} = 1$ and the second by the quotient rule,

$$\left(\frac{\partial x}{\partial N_B} \right)_{N_A} = \frac{1}{N} - \frac{N_B}{N^2} \left(\frac{\partial N}{\partial N_B} \right)_{N_A} = \frac{N - N_B}{N^2} = \frac{1 - x}{N}, \quad (6.29)$$

where the last step uses $N - N_B = N_A = (1 - x)N$. Differentiating $G = Ng(x)$ by the product rule, and using the two derivatives just found,

$$\begin{aligned} \mu_B &= \left(\frac{\partial G}{\partial N_B} \right)_{N_A} = g(x) \left(\frac{\partial N}{\partial N_B} \right)_{N_A} + N \frac{dg}{dx} \left(\frac{\partial x}{\partial N_B} \right)_{N_A} \\ &= g + N \frac{dg}{dx} \frac{1 - x}{N}, \end{aligned} \quad (6.30)$$

in which the two factors of N cancel, leaving

$$\mu_B = g + (1 - x) \frac{dg}{dx}. \quad (6.31)$$

The same calculation with the roles of A and B exchanged holds N_B fixed instead, so that $x = N_B/N$ falls as N grows and $(\partial x / \partial N_A)_{N_B} = -N_B/N^2 = -x/N$, and it gives

$$\mu_A = g - x \frac{dg}{dx}. \quad (6.32)$$

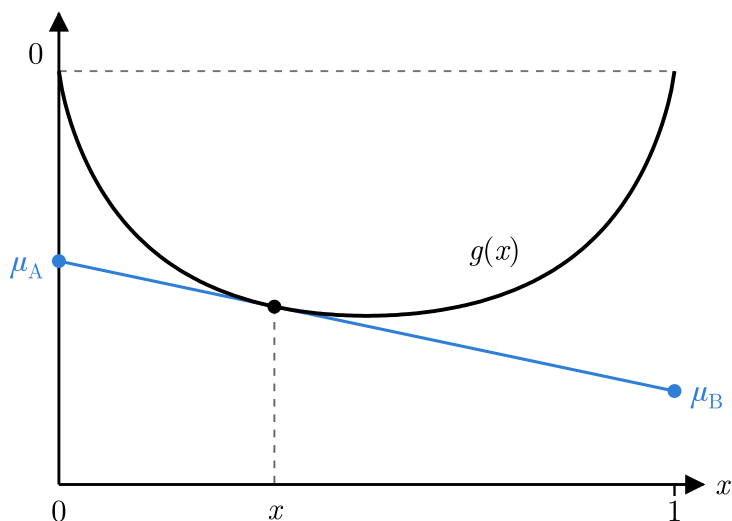


Figure 45. The chemical potentials read off the Gibbs function of mixing per molecule. The tangent to the curve at the composition x , drawn in blue, meets the left-hand edge of the diagram at the height μ_A and the right-hand edge at the height μ_B , both measured from the same zero as g itself. The construction is the same wherever along the curve the tangent is taken.

The chemical potential of A and the chemical potential of B are both read off the curve $g(x)$ at the composition in question, by the construction drawn in Figure 45. Writing g' for dg/dx and X for the composition running along the tangent line, the tangent to the curve at x is the line $y = g(x) + (X - x)g'(x)$. Setting $X = 0$ gives

the height $g - xg'$, which is μ_A , and setting $X = 1$ gives the height $g + (1 - x)g'$, which is μ_B . The tangent therefore meets the line $x = 0$ at the height μ_A and the line $x = 1$ at the height μ_B .

The Common Tangent

Suppose the sample separates into two phases, one at composition x_a and one at composition x_b . Molecules of A can pass between the two phases, and so can molecules of B, so the argument of Section 6.1, that the Gibbs function is at a minimum when a species has the same chemical potential in the two phases, applies to each species in turn and equilibrium requires

$$\mu_A(x_a) = \mu_A(x_b), \quad \mu_B(x_a) = \mu_B(x_b). \quad (6.33)$$

Subtracting μ_A from μ_B at any one composition, the terms in g cancel while the terms in g' add,

$$\mu_B - \mu_A = [g + (1 - x)g'] - [g - xg'] = g', \quad (6.34)$$

so the difference between the chemical potential of B and that of A at a given composition is simply the slope of the curve there. Applying this result at x_a and again at x_b , and using the two equilibrium conditions above, gives $g'(x_a) = g'(x_b)$: the curve has the same slope at the two compositions. Let m stand for that common slope, and put it back into the condition on μ_A , which becomes $g(x_a) - x_a m = g(x_b) - x_b m$. Collecting the terms in g on one side and the terms in m on the other gives $g(x_b) - g(x_a) = (x_b - x_a)m$, and dividing both sides by $x_b - x_a$,

$$\frac{g(x_b) - g(x_a)}{x_b - x_a} = g'(x_a) = g'(x_b). \quad (6.35)$$

A single straight line therefore touches the curve at x_a and again at x_b . This is called the common tangent construction, and it is drawn in Figure 46. The compositions of the two coexisting phases are the two points of contact.

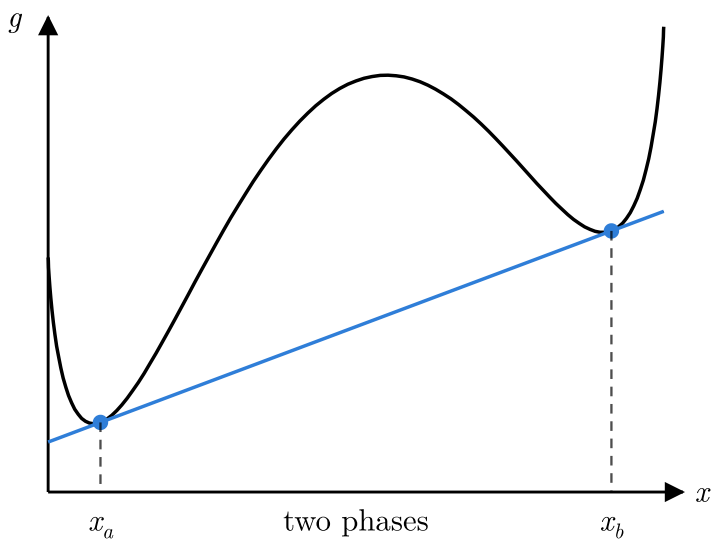


Figure 46. The common tangent construction. The straight line touching the curve $g(x)$ at x_a and at x_b lies below the curve between them, so a mixture whose overall composition lies in that range lowers its Gibbs function by separating into phases of composition x_a and x_b .

The construction also shows which compositions separate. A sample whose overall composition x lies between x_a and x_b can either stay uniform, with Gibbs function per molecule $g(x)$, or split into the two phases, in which case its Gibbs function per molecule is the weighted average of $g(x_a)$ and $g(x_b)$, with weights equal to the fractions of the sample in each phase. That average is the height at x of the chord joining the two contact points, which is the tangent line itself. The tangent lies below the curve everywhere between the two points of contact, so the separated state has the lower Gibbs function and the sample separates. Outside that range the uniform mixture is stable instead. A sample divided between phases of composition x_a and x_b has an overall composition lying somewhere between x_a and x_b , so a sample whose overall composition lies below x_a or above x_b cannot divide itself in that way; and the curve bends upwards between $x = 0$ and x_a , and again between x_b and $x = 1$, so no other pair of compositions serves either. The two coexisting phases have compositions x_a and x_b whatever the overall composition of the sample within that range, which changes only the amount of each phase present.

The Binary Phase Diagram

Repeating the construction at each temperature traces out the boundary of the two-phase region on a plot of temperature against composition, which is the phase diagram of the mixture. Figure 47 shows the form it takes for two substances that mix freely as liquids and hardly at all as solids, as tin and lead do. At each temperature there is a Gibbs function curve for the liquid and one for each of the two solid phases, the A-rich phase α and the B-rich phase β , and common tangents between the liquid curve and each solid curve give the compositions in equilibrium.

Pure A melts at T_A and pure B at T_B . Adding a little B to A lowers the temperature at which the solid first appears, for the same reason that a trace of anything dissolves in anything; the B molecules gain entropy by being dispersed in the liquid and are not taken into the solid, so the liquid is favoured. Each of the two descending curves marks that temperature at its composition, and the two meet at the eutectic composition, at the lowest temperature at which liquid exists anywhere on the diagram. Below the horizontal line through

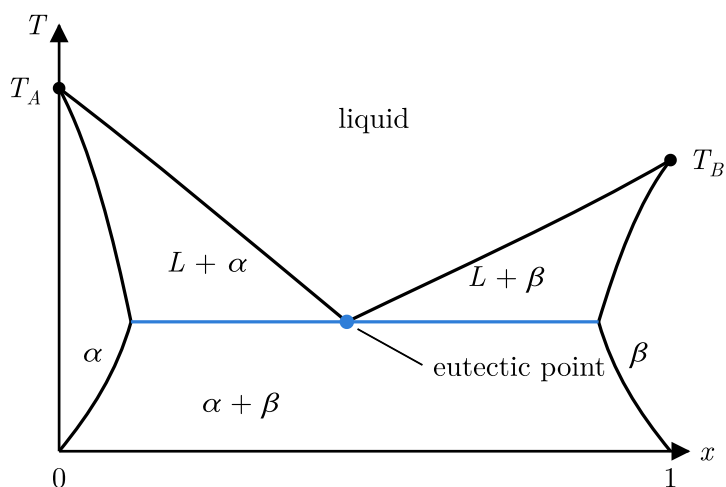


Figure 47. The phase diagram of a binary mixture whose two components mix in the liquid and separate in the solid. The upper pair of curves descending from the melting temperatures T_A and T_B of the pure components meet at the eutectic point, and below each of them runs a second curve, bounding the region in which a single solid phase is stable, and the horizontal line through it, drawn in blue, marks the lowest temperature at which any liquid exists.

the eutectic point the material is a mixture of the two solid phases α and β , except close to either end of the diagram, where a single solid phase holds a little of the other component in solution; between the descending curves and that line, a liquid coexists with one of the two solids. A mixture at the eutectic composition stays liquid all the way down to the eutectic temperature and then freezes at once into a fine interleaving of the two solids, which is why solders and casting alloys are often made at or near that composition.

Lowering the Freezing Point of a Solution

Let us consider an example of a dilute mixture, in which the phase diagram can be worked out near one end. Let A be a solvent in which a small amount of a solute B has been dissolved, and suppose the solute stays in the liquid when the solvent freezes. Writing μ_A^0 and μ_B^0 for the chemical potentials of the two pure liquids, the Gibbs function per molecule of the mixture is

$$g(x) = (1-x)\mu_A^0 + x\mu_B^0 + \varepsilon x(1-x) + k_B T [x \ln x + (1-x) \ln(1-x)], \quad (6.36)$$

whose derivative is

$$\frac{dg}{dx} = -\mu_A^0 + \mu_B^0 + \varepsilon(1-2x) + k_B T \ln \frac{x}{1-x}. \quad (6.37)$$

Substituting the expression for g and the expression for dg/dx into $\mu_A = g - x dg/dx$, the terms in μ_B^0 cancel between $x\mu_B^0$ and $-x\mu_B^0$; the terms in μ_A^0 combine as $(1-x)\mu_A^0 + x\mu_A^0 = \mu_A^0$; the energy terms give $\varepsilon[x(1-x) - x(1-2x)] = \varepsilon x^2$; and the entropy terms give $k_B T [x \ln x + (1-x) \ln(1-x) - x \ln x + x \ln(1-x)] = k_B T \ln(1-x)$. Adding these pieces together,

$$\mu_A = \mu_A^0 + \varepsilon x^2 + k_B T \ln(1-x). \quad (6.38)$$

For a dilute solution x is small, so that $\ln(1-x) \approx -x$ and the last term becomes $-k_B T x$. The term εx^2 is of second order in x and is negligible beside it, which leaves $\mu_A = \mu_A^0 - k_B T x$. Dissolving something in a liquid lowers the chemical potential of the liquid itself,

by an amount fixed by the fraction x of solute present, whatever the solute happens to be.

The solvent freezes when its chemical potential in the solution equals that of the pure solid, so the freezing temperature T of the solution satisfies

$$\mu_{A,\text{liq}}^0(T) - k_B T x = \mu_{A,\text{sol}}^0(T). \quad (6.39)$$

Let $\Delta\mu(T)$ be the difference $\mu_{A,\text{liq}}^0 - \mu_{A,\text{sol}}^0$ between the pure liquid and the pure solid, so that the condition above reads $\Delta\mu(T) = k_B T x$. This difference vanishes at the freezing temperature T_0 of the pure solvent, since the pure liquid and the pure solid are then in equilibrium with each other. Its derivative follows from $(\partial\mu/\partial T)_p = -s$ of Section 6.1 and is $-(s_{\text{liq}} - s_{\text{sol}}) = -\ell/T_0$, where ℓ is the latent heat of fusion per molecule and the last step uses the definition of the latent heat given in that section. A Taylor expansion about T_0 , kept to first order because the solution is dilute and its freezing temperature lies close to T_0 , therefore gives

$$\Delta\mu(T) \approx \Delta\mu(T_0) + (T - T_0) \left. \frac{d\Delta\mu}{dT} \right|_{T_0} = -\frac{\ell}{T_0} (T - T_0), \quad (6.40)$$

and equating this to $k_B T x$,

$$-\frac{\ell}{T_0} (T - T_0) = k_B T x, \quad (6.41)$$

and since T differs from T_0 only slightly we may put $T \approx T_0$ on the right, giving

$$T_0 - T = \frac{k_B T_0^2}{\ell} x. \quad (6.42)$$

The freezing temperature falls in proportion to the concentration of the solute, with a constant of proportionality that depends on the solvent alone, as water with salt dissolved in it freezes below the temperature at which pure water does. This is the descent of

the upper pair of curves of Figure 47 away from T_A and T_B , worked out for small x .

Problems

§6.1 Phase Coexistence

1. At the triple point all three coexistence curves meet. Using $L = \Delta H$ to show first that the latent heat of sublimation is the sum of the latent heats of fusion and vaporisation, show that the sublimation curve is steeper there than the vaporisation curve.
2. The volume per molecule of ice is larger than that of liquid water. Show that the melting curve of water has negative slope, and find the shift in the melting temperature produced by a small increase Δp in the pressure, in terms of the latent heat of fusion L and the volume change ΔV .
3. Suppose the latent heat per molecule of a liquid varies slowly with temperature as $\ell = \ell_0 + ak_B T$, with a a dimensionless constant. Show that the vapour pressure is then $p = C T^a e^{-\ell_0/k_B T}$, with C a constant.
4. A liquid boils at the temperature T_b at which its vapour pressure equals the ambient pressure p . Using $p = p_0 e^{-\ell/k_B T}$, find dT_b/dp , and show that a small fractional fall $\delta p/p$ in the ambient pressure lowers the boiling temperature by $\delta T_b = (k_B T_b^2/\ell)(\delta p/p)$.

§6.2 Mixtures and Phase Separation

1. Show that the entropy of mixing $\Delta S_{\text{mix}} = -Nk_B[x \ln x + (1-x) \ln(1-x)]$ is largest at $x = \frac{1}{2}$, where it equals $Nk_B \ln 2$. Show also that for small x it is approximately $Nk_B x(1 - \ln x)$, and hence that the entropy gained per solute molecule grows without limit as the solution is made more dilute.
2. Write $x = \frac{1}{2} + u$ in the expression for Δg_{mix} and expand in powers of u as far as u^4 , using $x \ln x + (1-x) \ln(1-x) = -\ln 2 + 2u^2 + \frac{4}{3}u^4 + \dots$. Find the two minima for a temperature just below $k_B T = \varepsilon/2$, and show that the two coexisting compositions move apart as the square root of $\varepsilon - 2k_B T$.

3. A sample of overall composition x separates into phases of composition x_a and x_b , with $x_a < x < x_b$. Let f be the fraction of the molecules in the phase of composition x_b . By counting B molecules, show that $f = (x - x_a)/(x_b - x_a)$, and hence that the Gibbs function per molecule of the separated sample is the height of the common tangent of Figure 46 at x .
4. A solution of a solute at small fraction x is separated from the pure solvent, at the same temperature, by a membrane through which the solvent passes and the solute does not. Equilibrium requires the chemical potential of the solvent to be the same on the two sides. Using $\mu_A = \mu_A^0(T, p) - k_B T x$ and $(\partial\mu/\partial p)_T = v$, show that the pressure on the solution exceeds that on the pure solvent by the osmotic pressure $\Pi = k_B T x/v$, where v is the volume per molecule of the solvent.

Solutions

Chapter 1. Kinetic Theory and Transport in Gases

§1.1 Thermal Physics at School

1. The mass of ν moles is $M = \nu M_{\text{m}}$, so $\rho = \nu M_{\text{m}}/V$. From $pV = \nu RT$ we have $\nu/V = p/RT$, and therefore $\rho = pM_{\text{m}}/RT$.
2. Each gas separately satisfies $p_i V = N_i k_B T$, so $p_i = N_i k_B T/V$. Adding the two partial pressures gives $p = (N_1 + N_2) k_B T/V$, which is the ideal gas equation for the mixture. Dividing p_1 by p gives $N_1/(N_1 + N_2)$.
3. Equating the two expressions for pV gives $\overline{c^2} = 3k_B T/m$ for each gas. At the same temperature $\overline{c^2}$ is inversely proportional to the mass of the molecule, so the root mean square speeds are in the ratio $\sqrt{m_2/m_1}$.
4. The new volume is $\ell^3(1 + \alpha \Delta T)^3$. Expanding the cube by the binomial theorem gives $\ell^3(1 + 3\alpha \Delta T + \dots)$, where the terms omitted are of second order and higher in $\alpha \Delta T$. The fractional change in the volume is therefore $3\alpha \Delta T$.

§1.2 Pressure and the Distribution of Molecular Speeds

1. With $a = m/2k_B T$ the integral required is $\int_0^\infty v^4 e^{-av^2} dv$. Integrating by parts, with $u = v^3$ and $dw = ve^{-av^2} dv$,

gives $\int_0^\infty v^4 e^{-av^2} dv = (3/2a) \int_0^\infty v^2 e^{-av^2} dv$, and a second integration by parts, with $u = v$, gives $\int_0^\infty v^2 e^{-av^2} dv = (1/2a) \int_0^\infty e^{-av^2} dv = \frac{1}{4} \sqrt{\pi/a^3}$. Hence $\int_0^\infty v^4 e^{-av^2} dv = \frac{3}{8} \sqrt{\pi/a^5}$, and multiplying by $4\pi(a/\pi)^{3/2}$ gives $\bar{v}^2 = 3/2a = 3k_B T/m$.

2. Since g is even, $\overline{|v_x|} = 2 \int_0^\infty v_x g(v_x) dv_x$. With $a = m/2k_B T$ the integral $\int_0^\infty v_x e^{-av_x^2} dv_x = 1/2a$, so $\overline{|v_x|} = 2\sqrt{a/\pi} \cdot (1/2a) = 1/\sqrt{\pi a} = \sqrt{2k_B T/\pi m}$. Comparing this with $\bar{v} = \sqrt{8k_B T/\pi m} = 2\sqrt{2k_B T/\pi m}$ shows that $\overline{|v_x|} = \frac{1}{2}\bar{v}$.
3. The fraction of molecules between z and $z+dz$ is proportional to $e^{-\gamma z} dz$ with $\gamma = mg/k_B T$, so $\bar{z} = \int_0^\infty z e^{-\gamma z} dz / \int_0^\infty e^{-\gamma z} dz$. Integrating the numerator by parts gives $1/\gamma^2$, and the denominator is $1/\gamma$, so $\bar{z} = 1/\gamma = k_B T/mg$ and $\overline{mgz} = k_B T$. Equipartition applies to terms in the energy that are quadratic in their variable, and mgz is linear in z , so the principle as stated does not cover this case.
4. There are six quadratic terms in the energy of an atom, three in the velocity components and three in the displacements, so the mean energy of an atom is $6 \times \frac{1}{2} k_B T = 3k_B T$. For one mole, $U = 3N_A k_B T = 3RT$, and differentiating with respect to T gives $C = 3R$.

§1.3 Effusion, Collisions, and the Mean Free Path

1. The mean speed in the beam is $\int_0^\infty v^4 e^{-av^2} dv / \int_0^\infty v^3 e^{-av^2} dv$, since the normalising constants cancel. Using the quoted integral and $\int_0^\infty v^3 e^{-av^2} dv = 1/2a^2$ from Section 1.2, the ratio is $\frac{3}{8} \sqrt{\pi/a^5} \cdot 2a^2 = \frac{3}{4} \sqrt{\pi/a}$. With $\bar{v} = 2/\sqrt{\pi a}$ the ratio of the mean speed in the beam to $\bar{v}$ is $\frac{3}{4} \sqrt{\pi/a} \cdot \frac{1}{2} \sqrt{\pi a} = \frac{3}{8}\pi$.
2. The number of molecules inside is $N = nV$, and molecules leave at the rate $\frac{1}{4} n \bar{v} A$, so $V dn/dt = -\frac{1}{4} n \bar{v} A$. This equation is separable, and integrating it gives $n = n_0 e^{-t/t_0}$ with $t_0 = 4V/A\bar{v}$. The temperature is fixed, so $\bar{v}$ does not change as the gas leaves.
3. Each species effuses at the rate $\frac{1}{4} n_i \bar{v}_i A$, and $\bar{v}_i \propto 1/\sqrt{m_i}$, so

the ratio of the rates is $(n_1/n_2)\sqrt{m_2/m_1}$. The ratio in the beam therefore exceeds the ratio n_1/n_2 inside by $\sqrt{m_2/m_1}$. Each stage multiplies the ratio by the same factor, so after s stages the ratio has been multiplied by $(m_2/m_1)^{s/2}$.

4. The collision rate for one molecule is $1/\tau = \sqrt{2}n\sigma\bar{v}$. With $n = p/k_B T$ and $\bar{v} \propto \sqrt{T}$, this is proportional to $(p/T)\sqrt{T} = p/\sqrt{T}$. The number of collisions per unit time in unit volume is $\frac{1}{2}n/\tau$, the half being there because each collision involves two molecules and would otherwise be counted twice, and this is proportional to $n \cdot p/\sqrt{T} \propto p^2/T^{3/2}$.

§1.4 Transport in Gases

1. In $\kappa = \frac{1}{3}nc\bar{v}\lambda$ substitute $\lambda = 1/\sqrt{2}n\sigma$, and the density cancels to leave $\kappa = c\bar{v}/3\sqrt{2}\sigma$, which depends on the temperature through $\bar{v}$ alone. In $D = \frac{1}{3}\bar{v}\lambda$ only one factor of the density appears, so $D = \bar{v}/3\sqrt{2}n\sigma$. Using $n = p/k_B T$ and $\bar{v} \propto \sqrt{T}$ gives $D \propto \sqrt{T} \cdot T/p = T^{3/2}/p$.
2. A monatomic molecule has three quadratic terms in its energy, so its mean energy is $\frac{3}{2}k_B T$ and $c = \frac{3}{2}k_B$. Putting this into $\kappa = c\eta/m$ gives $\kappa = 3k_B\eta/2m$.
3. With $J_{p_y} = -\eta du_y/dx$ constant across the gap, du_y/dx is constant, so u_y is linear in x . Taking $u_y = 0$ at one plate and u_0 at the other gives $du_y/dx = u_0/L$, and the magnitude of the momentum flux, which is the force per unit area on either plate, is $\eta u_0/L$.
4. Molecules strike each plate at the rate $\frac{1}{4}n\bar{v}$ per unit area. A molecule arriving at the stationary plate brings with it the y -momentum mu_0 it acquired at the moving plate, and leaves with none, so the momentum delivered per unit area per unit time is of order $\frac{1}{4}n\bar{v}mu_0$. This expression contains no mean free path, and it is proportional to n and therefore to the pressure. Comparing it with $\eta u_0/L$, the gas behaves as though it had a viscosity of order $\frac{1}{4}n\bar{v}mL$, which falls with the pressure as the vessel is pumped out.

Chapter 2. The Laws of Thermodynamics

§2.1 Thermodynamics at School

1. The work done by the gas is $p(V_2 - V_1)$, so $\Delta U = Q - p(V_2 - V_1)$.
2. Melting all the ice requires $m_2 L$, and the water can give up at most $m_1 c(T_1 - T_0)$ before reaching T_0 . All the ice melts provided $m_1 c(T_1 - T_0) \geq m_2 L$. The final temperature T_f then satisfies $m_1 c(T_1 - T_f) = m_2 L + m_2 c(T_f - T_0)$, so $T_f = (m_1 c T_1 + m_2 c T_0 - m_2 L) / [(m_1 + m_2) c]$.
3. From pV^γ constant, $p_2 = p_1(V_1/V_2)^\gamma$. From $TV^{\gamma-1}$ constant, $T_2 = T_1(V_1/V_2)^{\gamma-1}$.
4. For either device $Q_h = Q_c + W$, so $Q_h/W = Q_c/W + 1$. The heat delivered to the warm body is the heat taken from the cold one together with the work supplied, and the work supplied ends up in the warm body as well.

§2.2 Functions of State and the First Law

1. With $p = \nu RT/V$ the expression becomes $\delta W = 0 \cdot dT - (\nu RT/V) dV$, so $A = 0$ and $B = -\nu RT/V$ in the notation of the text, with T playing the part of x and V the part of y . Then $(\partial A/\partial V)_T = 0$ while $(\partial B/\partial T)_V = -\nu R/V$, which is not zero, so the differential is inexact.
2. From $p = \nu RT/V$, $(\partial p/\partial V)_T = -\nu RT/V^2$. From $V = \nu RT/p$, $(\partial V/\partial T)_p = \nu R/p$. From $T = pV/\nu R$, $(\partial T/\partial p)_V = V/\nu R$. The product is $-\nu RT/(pV) = -1$.
3. Here $p = \nu RT/(V - \nu b)$, so the work done on the gas is $-\int_{V_1}^{V_2} p dV = -\nu RT \ln[(V_2 - \nu b)/(V_1 - \nu b)]$. The ideal-gas answer is recovered when $b = 0$, and for a compression the magnitude is larger than the ideal-gas value because the molecules occupy a volume that cannot be squeezed away.
4. No heat enters, so the work done on the gas equals $\Delta U = C_V(T_2 - T_1)$. Using $C_V = \nu R/(\gamma - 1)$ and $\nu RT = pV$, this is $(p_2 V_2 - p_1 V_1)/(\gamma - 1)$.

§2.3 Heat Engines and the Carnot Cycle

1. On an adiabat, $C_V dT = -p dV = -\nu RT dV/(V - \nu b)$, which can be rearranged as $dT/T = -(\nu R/C_V) dV/(V - \nu b)$ and integrated to give $T(V - \nu b)^{\nu R/C_V}$ constant. For this gas $(\partial V/\partial T)_p = \nu R/p$, just as for an ideal gas, and the internal energy depends on T alone, so the argument of Section 2.2 again gives $C_p - C_V = \nu R$ and $\nu R/C_V = \gamma - 1$. On an isotherm the heat exchanged is $\nu RT \ln[(V_2 - \nu b)/(V_1 - \nu b)]$. The two adiabatic legs then give $(V_B - \nu b)/(V_A - \nu b) = (V_C - \nu b)/(V_D - \nu b)$, the logarithms cancel in the ratio Q_c/Q_h , and $\eta = 1 - T_c/T_h$.
2. The first engine takes Q_h and rejects $Q_m = Q_h T_m/T_h$, doing work $Q_h(1 - T_m/T_h)$. The second takes Q_m and does work $Q_m(1 - T_c/T_m)$. Adding the two amounts of work and dividing by Q_h gives $1 - T_m/T_h + (T_m/T_h)(1 - T_c/T_m) = 1 - T_c/T_h$.
3. Carnot's theorem gives $1 - Q_c/Q_h \leq 1 - T_c/T_h$, which rearranges to the stated inequality. Equality holds when the engine is reversible.
4. When the body is at temperature T , removing heat $\delta Q_c = -C dT$ reversibly requires work $\delta W = \delta Q_c(T_h/T - 1)$, since the coefficient of performance at that moment is $T/(T_h - T)$. Integrating from T_1 down to T_2 , $W = C [T_h \ln(T_1/T_2) - (T_1 - T_2)]$.

§2.4 Engine Cycles

1. From $T_2 = T_1 r^{\gamma-1}$ we have $r^{1-\gamma} = T_1/T_2$, so $\eta = 1 - r^{1-\gamma} = 1 - T_1/T_2$. The same relation applied to $T_3 = T_4 r^{\gamma-1}$ gives $r^{1-\gamma} = T_4/T_3$, so the efficiency equals $1 - T_4/T_3$ as well.
2. Write $x = r_c$. The claim is that $x^\gamma - 1 > \gamma(x - 1)$ for $x > 1$. Let $g(x) = x^\gamma - 1 - \gamma(x - 1)$. Then $g(1) = 0$ and $g'(x) = \gamma(x^{\gamma-1} - 1)$, which is positive for $x > 1$ because $\gamma > 1$. So $g(x) > 0$ for $x > 1$, and the factor multiplying $r^{1-\gamma}$ in the Diesel efficiency exceeds one, making the subtracted term larger.
3. Label the states as in the Otto cycle, so that $1 \rightarrow 2$ is the adiabatic compression, $2 \rightarrow 3$ the heating at p_2 , $3 \rightarrow 4$ the adiabatic expansion, and $4 \rightarrow 1$ the cooling at p_1 . Heat is supplied at

constant pressure p_2 and rejected at constant pressure p_1 , so $\eta = 1 - (T_4 - T_1)/(T_3 - T_2)$ with the same C_p on both legs. On an adiabat $Tp^{(1-\gamma)/\gamma}$ is constant, which follows from eliminating V between pV^γ and $pV = \nu RT$. Hence $T_2 = T_1(p_2/p_1)^{(\gamma-1)/\gamma}$ and $T_3 = T_4(p_2/p_1)^{(\gamma-1)/\gamma}$, and the ratio of the two temperature differences is $(p_1/p_2)^{(\gamma-1)/\gamma}$.

4. The indicated power is $NWf/2$. A four-stroke engine takes two revolutions to complete the sequence of induction, compression, power, and exhaust strokes, so each cylinder completes $f/2$ cycles per second.

§2.5 The Second Law

1. Let the two adiabats cross at a point P . Choose a point A on one adiabat and a point B on the other, both on the same isotherm, and run the cycle $P \rightarrow A$ along the first adiabat, $A \rightarrow B$ along the isotherm, and $B \rightarrow P$ along the second adiabat. Every leg is reversible, so we may traverse the cycle in whichever sense makes the net work positive, and then, since no heat is exchanged on the two adiabatic legs, the only heat entering comes from the single reservoir at the temperature of the isotherm, and over the cycle it is converted entirely into work. The Kelvin statement forbids this, so the adiabats cannot cross.
2. Write $\phi'(\theta) = f(\theta'_*, \theta)$ for the function built from the new reference reading. Putting $\theta_1 = \theta'_*$, $\theta_2 = \theta_*$, and $\theta_3 = \theta$ in the relation $f(\theta_1, \theta_3) = f(\theta_1, \theta_2)f(\theta_2, \theta_3)$ gives $\phi'(\theta) = f(\theta'_*, \theta_*)\phi(\theta)$, and the factor $f(\theta'_*, \theta_*)$ is a number that does not depend on θ . It therefore cancels from the ratio $\phi(\theta_b)/\phi(\theta_a)$, and the ratio of any two thermodynamic temperatures is the same whichever reference reading was used. The overall factor is fixed by assigning the value 273.16 K to the triple point of water.
3. The Clausius inequality gives $\oint \delta Q/T \leq 0$, and here the integral is a sum of two terms because the temperature of the supplier takes only two values. Reversibility turns the inequality into an equality by the argument of the text in which the cycle is run backwards.

4. The isothermal legs contribute $\nu R \ln(V_2/V_1)$ and $\nu R \ln(V_1/V_2)$, which cancel. The constant-volume legs contribute $\int C_V dT/T$ between the same two temperatures in opposite directions, namely $C_V \ln(T_2/T_1)$ and $C_V \ln(T_1/T_2)$, which also cancel. The sum is zero.

§2.6 Entropy

1. The body goes from T_1 to T_2 , so $\Delta S_{\text{body}} = C \ln(T_2/T_1)$. The reservoir absorbs $C(T_1 - T_2)$ at the fixed temperature T_2 , so $\Delta S_{\text{res}} = C(T_1 - T_2)/T_2$. The total is $C[\ln(T_2/T_1) + T_1/T_2 - 1]$. Writing $x = T_2/T_1$, this is $C[\ln x + 1/x - 1]$, which is non-negative because $\ln x \geq 1 - 1/x$, with equality only at $x = 1$.
2. We start from $dS = C_V dT/T + \nu R dV/V$ and eliminate V by logarithmic differentiation of $V = \nu RT/p$, whose logarithm is $\ln V = \ln(\nu R) + \ln T - \ln p$, so that differentiating gives $dV/V = dT/T - dp/p$. Then $dS = (C_V + \nu R) dT/T - \nu R dp/p = C_p dT/T - \nu R dp/p$, and integrating gives the stated result. Putting $p_2/p_1 = (T_2 V_1)/(T_1 V_2)$ recovers the T and V form.
3. Placed in contact, energy conservation gives $T_f = (T_1 + T_2)/2$. With a reversible engine between them the total entropy is unchanged, so $C \ln(T_f/T_1) + C \ln(T_f/T_2) = 0$, giving $T_f = \sqrt{T_1 T_2}$. The work is the energy the two bodies lose, $W = C(T_1 + T_2) - 2C\sqrt{T_1 T_2} = C(\sqrt{T_1} - \sqrt{T_2})^2$.
4. Each gas doubles its volume without changing its temperature, so with $\nu = pV/RT$ moles of each, $\Delta S = 2\nu R \ln 2$. If the two gases are the same, removing the partition produces no measurable change, and $\Delta S = 0$.

Chapter 3. Thermodynamic Potentials

§3.1 The Thermodynamic Potentials

1. Here $s = e^x$, so $x = \ln s$ and $g = e^x - sx = s - s \ln s$. Then $-dg/ds = -(1 - \ln s - 1) = \ln s = x$.
2. At constant volume the work vanishes, so $\delta Q = dU$ and $C_V = (\partial U/\partial T)_V$. At constant pressure $\delta Q = dH$, as shown in the text,

so $C_p = (\partial H/\partial T)_p$. For an ideal gas $H = U + pV = U + Nk_B T$. Differentiating with respect to T at constant pressure gives $C_p = (\partial U/\partial T)_p + Nk_B$, and since the internal energy of an ideal gas depends on its temperature alone, $(\partial U/\partial T)_p = (\partial U/\partial T)_V = C_V$, so that $C_p = C_V + Nk_B$.

3. Substituting $S = -(\partial F/\partial T)_V$ into $U = F + TS$ gives $U = F - T(\partial F/\partial T)_V$. Since $dG = -S dT + V dp$ gives $S = -(\partial G/\partial T)_p$, the same substitution in $H = G + TS$ gives $H = G - T(\partial G/\partial T)_p$.
4. Let the system take in heat Q during the change. The reservoir then absorbs heat $-Q$ at temperature T , so its entropy changes by $-Q/T$, and the system's entropy changes by ΔS . With no work done, $Q = \Delta U$, so the total change is $\Delta S - \Delta U/T = -(\Delta U - T\Delta S)/T = -\Delta F/T$. The second law requires this to be non-negative, so $\Delta F \leq 0$.

§3.2 Maxwell Relations and Their Uses

1. Setting $dS = 0$ in $dS = (C_p/T) dT - V\alpha dp$ gives $(C_p/T) dT = V\alpha dp$, and hence $(\partial T/\partial p)_S = TV\alpha/C_p$. For an ideal gas $\alpha = 1/T$ and $V = Nk_B T/p$, so the right-hand side is $Nk_B T/(pC_p)$ and $dT/T = (Nk_B/C_p) dp/p$. Since $Nk_B = C_p - C_V$, the exponent is $(\gamma - 1)/\gamma$, and integrating, with C_p taken as constant, gives $T \propto p^{(\gamma-1)/\gamma}$.
2. Differentiating the equation of state, $(\partial p/\partial T)_V = Nk_B/(V - Nb)$, so $T(\partial p/\partial T)_V = p + aN^2/V^2$ and the energy equation gives $(\partial U/\partial V)_T = aN^2/V^2$. The internal energy rises as the gas expands at fixed temperature. In a free expansion no work is done and no heat enters, so U is unchanged; the rise associated with the larger volume must be balanced by a fall in temperature, and the gas cools.
3. Using $C_V = T(\partial S/\partial T)_V$, we have $(\partial C_V/\partial V)_T = T \partial^2 S/\partial V \partial T$. Exchanging the order of differentiation and using the Maxwell relation $(\partial S/\partial V)_T = (\partial p/\partial T)_V$ gives $T(\partial^2 p/\partial T^2)_V$. For an ideal gas p is linear in T at fixed V , so the second derivative vanishes.

4. At constant temperature the heat taken in is $T\Delta S$, and $dS = -V\alpha dp$ along an isotherm, so the heat taken in is $-T \int V\alpha dp'$ and the heat given out is $T \int V\alpha dp'$. For a substance with $\alpha > 0$ this is positive, so compressing it releases heat.

§3.3 The Chemical Potential

1. Since $F = U - TS$, its differential is $dF = -S dT - p dV + \mu dN$, and the coefficient of dN at fixed T and V is μ . Solving the combined law for dS gives $dS = dU/T + p dV/T - \mu dN/T$, whose coefficient of dN at fixed U and V is $-\mu/T$.
2. At the common temperature $\mu = \mu_0(T) + k_B T \ln(p/p_0)$ in each vessel, and the logarithm increases with pressure, so $\mu_1 > \mu_2$. Particles therefore leave the first vessel, which lowers p_1 and raises p_2 , and this continues until the pressures and hence the chemical potentials are equal.
3. The condition that $\mu(T, p(z)) + mgz$ is independent of z gives $\mu_0(T) + k_B T \ln(p(z)/p_0) + mgz = \mu_0(T) + k_B T \ln(p(0)/p_0)$. Cancelling the reference terms gives $k_B T \ln(p(z)/p(0)) = -mgz$, so $p(z) = p(0)e^{-mgz/k_B T}$.
4. Taking the differential of $G = \mu N$ gives $dG = \mu dN + N d\mu$. Comparing with $dG = -S dT + V dp + \mu dN$ and cancelling the μdN terms gives $N d\mu = -S dT + V dp$, which is the stated relation.

§3.4 Chemical Equilibrium

1. The partial pressures are $p_i = x_i p$, where $x_i = N_i/N$ is the fraction of molecules of species i . Then $\prod_i (p_i/p_0)^{\nu_i} = (p/p_0)^{\sum_i \nu_i} \prod_i x_i^{\nu_i}$, and with $\sum_i \nu_i = 0$ the first factor is unity, so $\prod_i x_i^{\nu_i} = K(T)$ with no reference to p . The composition therefore depends on temperature alone.
2. Integrating $d \ln K / dT = \Delta h / k_B T^2$ with Δh constant gives $\ln[K(T_2)/K(T_1)] = -\frac{\Delta h}{k_B} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$. For $T_2 > T_1$ the bracket is negative, so the right-hand side is positive when $\Delta h > 0$ and K increases.

3. As $\xi \rightarrow 0$ the partial pressure of every product tends to zero, and each of those species contributes $\nu_i k_B T \ln(p_i/p_0)$ with $\nu_i > 0$ and the logarithm tending to $-\infty$. The reactant terms stay finite, so the whole sum tends to $-\infty$. The slope of G is then steeply downwards, so the equilibrium state of a mixture of pure reactants always lies a little way into the reaction, and a reaction of this kind never has an equilibrium with no products at all, though the equilibrium extent may be too small to detect. The argument says nothing about how quickly that equilibrium is reached, and a mixture may remain effectively unreacted for a very long time.
4. Writing the equilibrium condition as $\alpha^{-2} = 1 + 4p/(p_0 K)$ and differentiating with respect to T at fixed p gives $-2\alpha^{-3} d\alpha/dT = -(\alpha^{-2} - 1) d \ln K/dT$. Rearranging and using the van 't Hoff relation gives $d \ln \alpha/dT = \frac{1}{2}(1 - \alpha^2)\Delta h/k_B T^2$. A dissociation absorbs heat, so $\Delta h > 0$ and α increases with temperature, with the effect dying away as α approaches unity and there is little left to dissociate.

Chapter 4. Statistical Foundations and the Equilibrium Ensembles

§4.1 Probability for Many Particles

1. The first application gives $\bar{n} = Np(p+q)^{N-1}$. Applying $p \partial/\partial p$ again gives $\overline{n^2} = Np(p+q)^{N-1} + N(N-1)p^2(p+q)^{N-2}$. Setting $p+q=1$ gives $\bar{n} = Np$ and $\overline{n^2} = Np + N(N-1)p^2$, so $\sigma_n^2 = Np + N^2p^2 - Np^2 - N^2p^2 = Np(1-p) = Npq$.
2. With n steps to the right, $x = (2n - N)\ell$, so $\bar{x} = (2Np - N)\ell = (p - q)N\ell$ and the variance is $4\ell^2 Npq$. The ratio of the mean to the standard deviation is $(p - q)\sqrt{N}/(2\sqrt{pq})$, which is small when $|p - q| \ll 1/\sqrt{N}$.
3. The peak value is $P(\frac{1}{2}N) = N! 2^{-N} / [(\frac{1}{2}N)!]^2$. The more precise form gives $N! \approx \sqrt{2\pi N} (N/e)^N$ and $[(\frac{1}{2}N)!]^2 \approx \pi N (N/2e)^N$, so $P(\frac{1}{2}N) \approx \sqrt{2\pi N} (N/e)^N / [\pi N (N/2e)^N 2^N] = \sqrt{2\pi N}/\pi N = \sqrt{2/\pi N}$.

For the check, $\int_{-\infty}^{\infty} \sqrt{2/\pi N} e^{-m^2/N} dm = \sqrt{2/\pi N} \sqrt{\pi N/2} = 1$.

- Write $x = X_1 + X_2 + \cdots + X_N$, where X_i is the length of the i th step. Then $\bar{x} = \sum_i \bar{X}_i = 0$, because each step has mean zero, and $\overline{x^2} = \sum_i \sum_j \overline{X_i X_j}$. For $i \neq j$ the two steps are independent, so the mean of the product is the product of the means, $\overline{X_i X_j} = \bar{X}_i \bar{X}_j = 0$, and the N terms with $i = j$ each give $\overline{X_i^2} = \ell^2$. So $\overline{x^2} = N\ell^2$. The central limit theorem says that for large N the displacement is distributed as a Gaussian of mean zero and variance $N\ell^2$, independently of the shape of the single-step distribution.

§4.2 Microstates, Multiplicity, and Entropy

- $S = \alpha N k_B \ln U + \text{constant}$, so $1/T = \partial S / \partial U = \alpha N k_B / U$, giving $U = \alpha N k_B T$ and $C_V = \alpha N k_B$.
- $S/k_B = N \ln N - n \ln n - (N - n) \ln(N - n)$, using Stirling for all three factorials. Then $\partial S / \partial n = k_B \ln[(N - n)/n]$, and since $U = n\varepsilon$, $1/T = (k_B/\varepsilon) \ln[(N - n)/n]$. Solving, $(N - n)/n = e^{\varepsilon/k_B T}$, so $n/N = 1/(e^{\varepsilon/k_B T} + 1)$.
- Setting $\partial[\alpha N_1 \ln U_1 + \alpha N_2 \ln(U - U_1)] / \partial U_1 = 0$ gives $N_1/U_1 = N_2/U_2$, so $U_1^* = U N_1 / (N_1 + N_2)$. The second derivative is $-\alpha N_1/U_1^2 - \alpha N_2/U_2^2$, of order $-N/U^2$, so the Gaussian width is $\Delta U_1 \sim U/\sqrt{\alpha N}$ and $\Delta U_1/U_1^* \sim 1/\sqrt{N_1}$.
- With $dU_2 = -dU_1$ and $dV_2 = -dV_1$, the change in total entropy is $d(S_1 + S_2) = (1/T_1 - 1/T_2)dU_1 + (p_1/T_1 - p_2/T_2)dV_1$. This vanishes for arbitrary independent dU_1 and dV_1 only if $T_1 = T_2$ and then $p_1 = p_2$.

§4.3 Counting States of an Ideal Gas

- Write the entropy as $S = N k_B [\ln(V/N) + \frac{3}{2} \ln(U/N) + c_0 + \frac{5}{2}]$ with $c_0 = \frac{3}{2} \ln(4\pi m/3h^2)$. Then $\partial S / \partial N = k_B [\ln(V/N) + \frac{3}{2} \ln(U/N) + c_0 + \frac{5}{2}] - k_B(1 + \frac{3}{2})$, and $-\mu/T$ equals this, so $\mu = -k_B T \ln[(V/N)(4\pi m U/3N h^2)^{3/2}]$. Substituting $U = \frac{3}{2} N k_B T$ makes the bracket $(2\pi m k_B T/h^2)^{3/2}/n$, giving $\mu = k_B T \ln(n/n_Q)$.

2. At the same temperature and pressure the densities are equal, so $V_A/V = x_A$ and $V_B/V = x_B$. Then $\Delta S = -N_A k_B \ln x_A - N_B k_B \ln x_B = -N k_B (x_A \ln x_A + x_B \ln x_B)$, which is positive since the logarithms of fractions are negative.
3. $S/k_B = -p \ln p - (1-p) \ln(1-p)$, so $dS/dp = k_B \ln[(1-p)/p]$, which vanishes at $p = \frac{1}{2}$. The second derivative is negative there, so it is a maximum, and $S = k_B \ln 2$.
4. Replacing h^{3N} by $c^N h^{3N}$ divides Ω by c^N and so subtracts $N k_B \ln c$ from S . The added term depends on neither U nor V , so $\partial S/\partial U$ and $\partial S/\partial V$ are unchanged and T and p are the same. It does depend on N ; $-\mu/T$ decreases by $k_B \ln c$, so μ increases by $k_B T \ln c$.

§4.4 The Boltzmann Factor and the Partition Function

1. $Z = 1 + e^{-\beta\varepsilon} + e^{-2\beta\varepsilon}$ and $\bar{E} = \varepsilon(e^{-\beta\varepsilon} + 2e^{-2\beta\varepsilon})/(1 + e^{-\beta\varepsilon} + e^{-2\beta\varepsilon})$. As $T \rightarrow 0$ the numerator is dominated by $\varepsilon e^{-\beta\varepsilon}$ and $\bar{E} \rightarrow \varepsilon e^{-\beta\varepsilon} \rightarrow 0$. As $T \rightarrow \infty$ all three exponentials tend to one and $\bar{E} \rightarrow \varepsilon$, the average of the three level energies.
2. Each term of the sum acquires the same factor $e^{-\beta E_0}$, so $Z \rightarrow e^{-\beta E_0} Z$, and in $p_i = e^{-\beta(E_i + E_0)}/(e^{-\beta E_0} Z)$ the new factors cancel. Then $\ln Z \rightarrow \ln Z - \beta E_0$ and $\bar{E} = -\partial \ln Z / \partial \beta \rightarrow \bar{E} + E_0$.
3. For small $x = \beta \hbar \omega$, $e^x - 1 = x(1 + x/2 + \dots)$, so $\hbar \omega / (e^x - 1) \approx k_B T - \frac{1}{2} \hbar \omega$ and $\bar{E} \approx k_B T$. For large x , $e^x - 1 \approx e^x$ and $\bar{E} \approx \frac{1}{2} \hbar \omega + \hbar \omega e^{-\hbar \omega / k_B T}$.
4. The probability of finding a molecule at height z carries the factor $e^{-mgz/k_B T}$, and the number density is proportional to that probability, which gives the stated result. The mean height is $\int_0^\infty z e^{-mgz/k_B T} dz / \int_0^\infty e^{-mgz/k_B T} dz = k_B T / mg$, the integral in the numerator being done by parts.

§4.5 Free Energy and Fluctuations

1. $\sigma_E = \sqrt{k_B T^2 \cdot \frac{3}{2} N k_B} = k_B T \sqrt{3N/2}$, and dividing by $\bar{E} = \frac{3}{2} N k_B T$ gives $\sqrt{3N/2} / (\frac{3}{2} N) = \sqrt{2/3N}$.

2. $Z = z^N$, so $F = -Nk_B T \ln z$ and $\ln Z = N \ln z$. Then $\bar{E} = -\partial \ln Z / \partial \beta = -N \partial \ln z / \partial \beta$, and $S = (\bar{E} - F)/T = Nk_B (\ln z - \beta \partial \ln z / \partial \beta)$.
3. By symmetry the integrals over all q are twice those over positive q . Using $\int_0^\infty e^{-\beta b q} dq = 1/\beta b$ and $\int_0^\infty b q e^{-\beta b q} dq = 1/\beta^2 b$, the ratio is $1/\beta = k_B T$. A linear dependence on $|q|$ gives twice as much mean energy as a quadratic one, so the factor $\frac{1}{2}$ in equipartition comes from the energy being quadratic in the coordinate.
4. Differentiating $F = -Nk_B T \ln z + Nk_B T \ln N - Nk_B T$ with respect to N at fixed T and V gives $\mu = -k_B T \ln z + k_B T \ln N + k_B T - k_B T = -k_B T \ln(z/N)$. Since the number of single-particle states in any range of momentum is proportional to V , by the counting of Section 4.3, z is proportional to V for a free particle, so z/N depends on V/N alone and μ is intensive. Without the $N!$, $\mu = -k_B T \ln z$ changes with the size of the vessel at fixed density, which cannot be right for an intensive quantity.

§4.6 The Grand Canonical Ensemble

1. With $a = \beta(\mu + \varepsilon)$ and $f = 1/(e^{-a} + 1)$, $\partial f / \partial \mu = \beta f(1 - f)$, so $\sigma_N^2 = k_B T M \beta f(1 - f) = M f(1 - f)$. Each site is independently occupied with probability f , so the number occupied is binomially distributed with $N \rightarrow M$ and $p \rightarrow f$, whose variance is $M f(1 - f)$.
2. Substituting $e^{\beta \mu} = n/n_Q$ into $f = e^{\beta(\mu + \varepsilon)} / (1 + e^{\beta(\mu + \varepsilon)})$ gives $f = (n/n_Q) e^{\beta \varepsilon} / (1 + (n/n_Q) e^{\beta \varepsilon})$, which rearranges to the stated form. Since $p = nk_B T$ at fixed T , raising the pressure raises n and f tends to one.
3. Write $a = (\mu + \varepsilon)/k_B T$, so $\partial a / \partial T = -a/T$. Then $\partial \Phi / \partial T = -Mk_B \ln(1 + e^a) + Mk_B a f$, and $S = Mk_B [\ln(1 + e^a) - a f]$. Using $\ln f = a - \ln(1 + e^a)$ and $\ln(1 - f) = -\ln(1 + e^a)$, the expression $-Mk_B [f \ln f + (1 - f) \ln(1 - f)]$ reduces to the same thing.
4. With $dN_2 = -dN_1$, the change in the total entropy is $d(S_1 + S_2) = (-\mu_1/T + \mu_2/T) dN_1$, using $(\partial S / \partial N)_{U,V} = -\mu/T$ from

Section 4.2. This vanishes for arbitrary dN_1 only if $\mu_1 = \mu_2$. If $\mu_1 > \mu_2$, the total entropy increases when dN_1 is negative, so particles leave the first system, which is the one of higher chemical potential.

Chapter 5. Statistical Mechanics of Ideal Classical Systems

§5.1 The Ideal Gas from the Partition Function

1. The states with modulus between n and $n + dn$ occupy a quarter of an annulus, so their number is $\frac{1}{4} \cdot 2\pi n dn$. Then $Z_1 = \int_0^\infty \frac{1}{2}\pi n e^{-an^2} dn$ with $a = \beta h^2/8mL^2$, and $\int_0^\infty n e^{-an^2} dn = 1/2a$, giving $Z_1 = 2\pi m k_B T L^2/h^2 = A/\lambda_{\text{th}}^2$. Since $\ln Z_N$ now contains $N \ln T$ rather than $\frac{3}{2}N \ln T$, the energy is $U = Nk_B T$ and $C = Nk_B$. The entropy is $S = Nk_B [\ln(A/N\lambda_{\text{th}}^2) + 2]$, and $-(\partial F/\partial A)_T = Nk_B T/A$, which is a force per unit length.
2. With $b = \beta hc/2L$, $Z_1 = \frac{1}{2}\pi \cdot 2/b^3 = \pi(2L/\beta hc)^3 = 8\pi V(k_B T/hc)^3$. Then $\ln Z_N = N \ln V + 3N \ln T + \text{constant} - \ln N!$, so $U = -\partial \ln Z_N / \partial \beta = 3Nk_B T$ and $p = -(\partial F/\partial V)_T = Nk_B T/V$. Dividing, $pV = U/3$.
3. Without the factor, $S = Nk_B [\ln(V/\lambda_{\text{th}}^3) + \frac{3}{2}]$. Doubling N and V together then gives $2S + 2Nk_B \ln 2$ rather than $2S$, so the two samples, already at the same temperature and density, would seem to gain entropy on being joined. With the factor $1/N!$ the logarithm contains V/N , which the doubling leaves unchanged.
4. We have $-pV + \mu N = -Nk_B T + Nk_B T \ln(N\lambda_{\text{th}}^3/V)$. Since $N\lambda_{\text{th}}^3/V = (V/N\lambda_{\text{th}}^3)^{-1}$, the logarithm changes sign when the argument is inverted, and the right-hand side becomes $-Nk_B T [\ln(V/N\lambda_{\text{th}}^3) + 1]$, which is F .

§5.2 Internal Degrees of Freedom

1. The probability of the level n is $e^{-n\beta\hbar\omega} / \sum_m e^{-m\beta\hbar\omega}$, so $\bar{n} = \sum_n n e^{-nx} / \sum_n e^{-nx}$ with $x = \beta\hbar\omega$. The numerator is $-d/dx$ of the denominator, so $\bar{n} = -d \ln[(1 - e^{-x})^{-1}] / dx = 1/(e^x - 1)$. Comparing with the expression for $\bar{\epsilon}_{\text{vib}}$ gives $\bar{\epsilon}_{\text{vib}} = (\bar{n} + \frac{1}{2})\hbar\omega$.

For $T \gg \theta_v$, $e^x - 1 \approx x$ and $\bar{n} \approx k_B T / \hbar \omega$, which is large.

2. From $S = Nk_B(\ln Z_{\text{vib}} + \beta \bar{\epsilon}_{\text{vib}})$, and using $\ln Z_{\text{vib}} = -x/2 - \ln(1 - e^{-x})$ together with $\beta \bar{\epsilon}_{\text{vib}} = x/2 + x/(e^x - 1)$, the terms in $x/2$ cancel and $S/Nk_B = x/(e^x - 1) - \ln(1 - e^{-x})$. As $T \rightarrow 0$, $x \rightarrow \infty$ and both terms vanish exponentially. As $x \rightarrow 0$, $x/(e^x - 1) \rightarrow 1$ and $-\ln(1 - e^{-x}) \approx -\ln x$, giving $S/Nk_B \approx 1 + \ln(T/\theta_v)$.
3. With $\theta_r = \hbar^2/2Ik_B$ the sum is $\sum_m e^{-\theta_r m^2/T}$, and for $T \gg \theta_r$ the terms change slowly, so it may be replaced by $\int_{-\infty}^{\infty} e^{-\theta_r m^2/T} dm = \sqrt{\pi T/\theta_r}$ by the Gaussian integral of Section 1.2. Then $\ln Z = \frac{1}{2} \ln T + \text{constant}$, so $\bar{\epsilon} = \frac{1}{2} k_B T$ and $C = \frac{1}{2} Nk_B$, which is equipartition for one axis of rotation.
4. In the three ranges C_V/Nk_B is $\frac{3}{2}$, $\frac{5}{2}$, and $\frac{7}{2}$, so C_p/Nk_B is $\frac{5}{2}$, $\frac{7}{2}$, and $\frac{9}{2}$, and γ is $5/3$, $7/5$, and $9/7$. The ratio falls as more degrees of freedom become available.

§5.3 Two-Level Systems and Paramagnetism

1. Write $C/Nk_B = f(x) = x^2 e^x / (e^x + 1)^2 = x^2 / [4 \cosh^2(x/2)]$. By logarithmic differentiation, $\ln f = 2 \ln x - 2 \ln [2 \cosh(x/2)]$, so that $f'/f = 2/x - \tanh(x/2)$, which vanishes when $x \tanh(x/2) = 2$. The root is $x \approx 2.40$, so the peak lies at $k_B T / \Delta = 1/x \approx 0.42$, as in the figure.
2. Here $Z = 1 + 2 \cosh y$ with $y = \mu B / k_B T$, and $F = -Nk_B T \ln Z$. Then $M = -(\partial F / \partial B)_T = Nk_B T (\partial \ln Z / \partial B)$, and since $\partial y / \partial B = \mu / k_B T$, the result stated follows. For $y \ll 1$, $\sinh y \approx y$ and $\cosh y \approx 1$, so $M \approx 2N\mu y/3 = 2N\mu^2 B / 3k_B T$, which is Curie's law with a different constant.
3. Both $F/Nk_B T = -\ln[2 \cosh(\mu_B B / k_B T)]$ and S/Nk_B , which is obtained from it by differentiation, contain B and T only in the combination B/T . If no heat is exchanged and the change is slow, S is constant by the second law of Section 2.5, so B/T is constant and T falls in proportion to B . This is the method of adiabatic demagnetisation.
4. With $x = \Delta / k_B T$ small, $e^x \approx 1 + x + x^2/2$, so $\bar{\epsilon} = \Delta / (e^x + 1) \approx$

$\frac{1}{2}\Delta(1-x/2)$ and $U \approx \frac{1}{2}N\Delta - N\Delta^2/4k_B T$. Differentiating, $C \approx Nk_B(\Delta/k_B T)^2/4$, which goes as T^{-2} . The same limit follows from expanding the expression for C directly.

§5.4 The Einstein Solid

1. Dropping the 1 and writing $\ln \Omega = (q + \mathcal{N}) \ln(q + \mathcal{N}) - q \ln q - \mathcal{N} \ln \mathcal{N}$, we use $\ln(q + \mathcal{N}) = \ln q + \ln(1 + \mathcal{N}/q) \approx \ln q + \mathcal{N}/q$. The terms in $q \ln q$ cancel and, keeping the leading terms, $\ln \Omega \approx \mathcal{N} \ln(q/\mathcal{N}) + \mathcal{N}$. Then $1/T = (k_B/\hbar\omega) \partial \ln \Omega / \partial q = \mathcal{N} k_B / q \hbar \omega$, so $E = q \hbar \omega = \mathcal{N} k_B T$. Putting $\mathcal{N} = 3N$ gives $U = 3Nk_B T$ and $C = 3Nk_B$.
2. With $x = \theta_E/T$, $F = -3Nk_B T \ln Z_{\text{vib}} = 3Nk_B T [x/2 + \ln(1 - e^{-x})]$, and since $\partial x / \partial T = -x/T$, differentiating gives $S = -(\partial F / \partial T)_V = 3Nk_B [x/(e^x - 1) - \ln(1 - e^{-x})]$, the terms in $x/2$ cancelling; this is $3N$ times the entropy of one oscillator found in the problems of Section 5.2. As $T \rightarrow 0$ both terms vanish exponentially. As $x \rightarrow 0$ the first term tends to 1 and the second to $-\ln x$, giving $3Nk_B [1 + \ln(T/\theta_E)]$.
3. Keeping only the levels 0 and $\hbar\omega$, the partition function of one oscillator is $1 + e^{-x}$, which is that of a two-level system with $\Delta = \hbar\omega$. Each such oscillator then contributes $k_B x^2 e^x / (e^x + 1)^2$ to the heat capacity, and for large x that contribution is $k_B x^2 e^{-x}$. The full oscillator gives $k_B x^2 e^x / (e^x - 1)^2$, which for large x is also $k_B x^2 e^{-x}$. The two-level result and the full oscillator result agree because at low temperature the levels above the first make no appreciable contribution.
4. The mean number of quanta in an oscillator is $1/(e^x - 1)$ from Section 5.2, and for $x \gg 1$ this is e^{-x} . Multiplying by $3N$ oscillators and $\hbar\omega$ per quantum gives $U = 3N\hbar\omega e^{-\theta_E/T}$ above the lowest state. Since almost every excited oscillator holds exactly one quantum, $3N e^{-\theta_E/T}$ is the number of oscillators that are excited, and $e^{-\theta_E/T}$ is the Boltzmann factor for reaching the first excited level.

§5.5 Phonons and the Debye Model

1. With $x_D = \theta_D/T$, the integral becomes $\int_0^{x_D} x^2 (1 - x/2 + x^2/12) dx = x_D^3/3 - x_D^4/8 + x_D^5/60$. Multiplying by $9Nk_B T (T/\theta_D)^3$ gives $U = 3Nk_B T - \frac{9}{8}Nk_B \theta_D + 3Nk_B \theta_D^2/20T$. Differentiating with respect to T , the constant drops out and $C = 3Nk_B - 3Nk_B \theta_D^2/20T^2$, which is the result stated.
2. The lowest state of each mode has energy $\frac{1}{2}\hbar\omega$, so the total is $\int_0^{\omega_D} (9N/\omega_D^3)\omega^2 \cdot \frac{1}{2}\hbar\omega d\omega = \frac{9}{8}N\hbar\omega_D$. It does not depend on the temperature, so it drops out on differentiating. It is the constant that appeared in the high-temperature expansion of the first problem, with the opposite sign, so that when the energy of the lowest state is added back the energy at high temperature is $3Nk_B T$ with no constant term.
3. In two dimensions the modes with wavevector of modulus below k occupy a quarter-disc, so their number is proportional to k^2 and the number per unit frequency is proportional to ω . Let A be the area of the film. Counting as in the text, with two polarisations and a quarter of an annulus in place of an octant of a shell, the number of modes whose mode number lies between n and $n+dn$ is $2 \times \frac{1}{4} \cdot 2\pi n dn = \pi n dn$, and substituting $n = L\omega/\pi c_s$ gives $g(\omega) = A\omega/\pi c_s^2$. Requiring $\int_0^{\omega_D} g d\omega = 2N$ then gives $\omega_D = 2c_s \sqrt{\pi N/A}$, with N/A the number of atoms per unit area. The energy at low temperature is $\int_0^\infty a\omega \cdot \hbar\omega d\omega / (e^{\beta\hbar\omega} - 1) \propto T^3$, so $C \propto T^2$.
4. Integrating $g(\omega)$ from zero to ω gives $V\omega^3/2\pi^2 c_s^3$. Modes with $\hbar\omega$ much greater than $k_B T$ are almost always in their lowest state, so the modes that carry energy are those below $\omega \sim k_B T/\hbar$, and their number is proportional to T^3 . Each carries an energy of order $k_B T$, so $U \propto T^4$ and $C \propto T^3$.

§5.6 Photons and Blackbody Radiation

1. With $f(x) = x^3/(e^x - 1)$, $f'(x) = 3x^2/(e^x - 1) - x^3 e^x/(e^x - 1)^2$. Setting this to zero and multiplying through by $(e^x - 1)^2/x^2$ gives $3(e^x - 1) = x e^x$, and dividing by e^x gives $3(1 - e^{-x}) = x$.

Since x turns out to be near 3, the term e^{-x} is only about 0.05, so the equation is close to $x = 3$; the root is 2.82.

2. The number density is $\int g(\omega)\bar{n}(\omega) d\omega/V = \frac{1}{\pi^2 c^3} \int \omega^2 d\omega / (e^{\beta\hbar\omega} - 1)$. Substituting $x = \beta\hbar\omega$ gives $(k_B T / \hbar c)^3 \times 2.40/\pi^2$, which is proportional to T^3 . Dividing the energy density by the number density, the powers of T leave one factor of $k_B T$ and the numerical factor is $(\pi^4/15)/2.40 \approx 2.7$.
3. From $F = -\pi^2 V (k_B T)^4 / 45 c^3 \hbar^3$, $S = -(\partial F / \partial T)_V = 4\pi^2 V k_B^4 T^3 / 45 c^3 \hbar^3$, which is $4U/3T$. A slow expansion with no heat exchanged keeps S constant, so VT^3 is constant. Since $p = u/3 \propto T^4$, we have $T \propto p^{1/4}$ and $Vp^{3/4}$ is constant, which on raising to the power $4/3$ gives $pV^{4/3}$ constant.
4. At low frequency $u_\omega \approx \omega^2 k_B T / \pi^2 c^3$, in which $\hbar$ has cancelled between the number of modes and the mean energy $k_B T$ of each. Integrating this form to infinity diverges because the integrand grows as ω^2 , so the divergence comes from the high-frequency modes, of which there are ever more and each of which would hold $k_B T$. The quantisation of the mode energies removes the difficulty, since a mode with $\hbar\omega \gg k_B T$ holds almost nothing.

Chapter 6. Interacting Systems, Real Gases, and Phase Equilibria

§6.1 Phase Coexistence

1. For the sublimation curve and for the vaporisation curve, the vapour is far less dense than the condensed phase, so $\Delta V \approx V_{\text{gas}}$ is the same for both at the triple point. The slopes are therefore in the ratio of the latent heats, $L_{\text{sub}}/L_{\text{vap}}$. Since $L = \Delta H$ and the enthalpy is a function of state, the enthalpy change from solid to gas at the triple point is the sum of the changes from solid to liquid and from liquid to gas, so $L_{\text{sub}} = L_{\text{fus}} + L_{\text{vap}}$, and since $L_{\text{fus}} > 0$ the sublimation curve has the larger slope.
2. Melting absorbs heat, so $L > 0$, while $\Delta V = V_{\text{liquid}} - V_{\text{ice}} < 0$. Hence $dp/dT = L/(T\Delta V) < 0$. Inverting, $\Delta T = T\Delta V \Delta p/L$, which is negative for positive Δp : pressing on ice lowers the

temperature at which it melts.

3. With $\ell = \ell_0 + ak_B T$ the equation becomes $dp/p = (\ell_0/k_B T^2 + a/T) dT$. Integrating, $\ln p = -\ell_0/k_B T + a \ln T + \text{constant}$, which exponentiates to $p = C T^a e^{-\ell_0/k_B T}$.
4. Differentiating $p = p_0 e^{-\ell/k_B T_b}$ with respect to T_b gives $dp/dT_b = p\ell/(k_B T_b^2)$, so $dT_b/dp = k_B T_b^2/(\ell p)$. Multiplying by δp gives $\delta T_b = (k_B T_b^2/\ell)(\delta p/p)$.

§6.2 Mixtures and Phase Separation

1. Differentiating, $d\Delta S_{\text{mix}}/dx = -Nk_B \ln[x/(1-x)]$, which vanishes at $x = \frac{1}{2}$; the second derivative is negative there, and the value is $-Nk_B \ln \frac{1}{2} = Nk_B \ln 2$. For small x , $\ln(1-x) \approx -x - \frac{1}{2}x^2$, so $(1-x) \ln(1-x) \approx -x + \frac{1}{2}x^2$ and $\Delta S_{\text{mix}} \approx Nk_B x(1 - \ln x)$ to leading order. Dividing by the number of solute molecules Nx leaves $k_B(1 - \ln x)$, which grows without limit as $x \rightarrow 0$.
2. With $\varepsilon x(1-x) = \varepsilon(\frac{1}{4} - u^2)$, the expansion gives $\Delta g_{\text{mix}} = \frac{1}{4}\varepsilon - k_B T \ln 2 + (2k_B T - \varepsilon)u^2 + \frac{4}{3}k_B T u^4$. Differentiating with respect to u and setting the result to zero, $2(2k_B T - \varepsilon)u + \frac{16}{3}k_B T u^3 = 0$, so either $u = 0$ or $u^2 = 3(\varepsilon - 2k_B T)/(8k_B T)$. The second solution exists only for $k_B T < \varepsilon/2$, and then $u = \pm \sqrt{3(\varepsilon - 2k_B T)/(8k_B T)}$, so the separation of the two compositions grows as the square root of $\varepsilon - 2k_B T$.
3. Of N molecules in all, fN are in the phase of composition x_b and $(1-f)N$ in the other, so the number of B molecules is $fNx_b + (1-f)Nx_a$, which must equal Nx . Solving for f gives $f = (x - x_a)/(x_b - x_a)$. The Gibbs function per molecule is then $fg(x_b) + (1-f)g(x_a)$, which is the height at x of the line joining the two contact points.
4. Let the pure solvent be at pressure p and the solution at $p + \Pi$. Equality of the chemical potential of the solvent gives $\mu_A^0(T, p + \Pi) - k_B T x = \mu_A^0(T, p)$, and for small Π the left-hand side is $\mu_A^0(T, p) + v\Pi - k_B T x$, so $\Pi = k_B T x/v$. Since x/v is the number of solute molecules per unit volume in a dilute solution, the solute exerts the pressure of an ideal gas at the same number

density.